Root-n estimation of a parametric component in a panel NPIV model with a discrete instrument

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Abstract

We estimate the slope vector of a partially linear panel model whose nonparametric argument is a continuous endogenous regressor and whose only instrument is discrete, with $\mathsf{L} \geq 2$ cells. First differencing removes the individual effect. A conditional-mean restriction cannot then identify the unknown function, so we impose conditional independence between the differenced error and the instrument. That makes the function the solution of an ill-posed nonlinear operator equation, which a Bierens family turns into unconditional moments and Landweber–Fridman regularisation solves. A cross-fitted one-step on a Neyman-orthogonal moment estimates the slope vector, $\sqrt{n}$-consistently and asymptotically normally and with a consistent variance estimator, whenever the first-stage rate is faster than $n^{-1/4}$. A hedonic application on a county panel illustrates the estimator at $\mathsf{L} = 4$, reading a regulatory designation as the pair of triggers that produces it.

JEL codes: C14, C23, C26

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1 Introduction

Fixed effects and instrumental variables are the workhorse answers to endogeneity in panel data, and applied work uses them together. A first-difference transformation removes whatever is fixed about the unit, and an instrument handles what is left. Both steps are almost always taken inside a linear specification.

We relax linearity in one regressor and keep it in the others. Over two periods the model is

$$Y_{ij} \;=\; \varphi_\dagger(X_{ij}) \;+\; Z_{ij}'\theta_\dagger \;+\; \eta_i \;+\; U_{ij}, \qquad j \in \{1,2\},$$

with X_{ij} a scalar continuous regressor correlated with U_{ij}, $Z_{ij} \in \mathbb{R}^p$, and η_i an individual effect that may depend on everything observed. Differencing the two periods removes η_i and leaves

$$\Delta Y_i \;=\; \varphi_\dagger(X_{i2}) - \varphi_\dagger(X_{i1}) \;+\; \Delta Z_i'\theta_\dagger \;+\; \Delta U_i.$$

The instrument W_i takes finitely many values, $W_i \in \{0, 1, \ldots, \mathsf{L}-1\}$ with $\mathsf{L} \geq 2$ fixed and known to the analyst, and that single fact is what the rest of the paper is about. Sans-serif L is the number of instrument cells throughout, and $\mathsf{L} = 2$ is the binary case rather than the paper's scope.

The parameter of interest is $\theta_\dagger$. It is what a first-differenced linear panel regression already reports, and the point here is to report it again without having forced X into a line. A researcher who suspects one regressor enters nonlinearly, and who has a discrete instrument for it, can run the estimator of Section 3 and read a slope vector and a standard error in the usual way.

That estimator has two steps and only the second concerns $\theta_\dagger$. The first solves for $\varphi_\dagger$ by a regularised iteration at a rate slower than $n^{-1/2}$; the second is a one-step correction on a moment built to be insensitive to that error. The propensity score $\varrho = (\varrho_1, \ldots, \varrho_{\mathsf{L}-1})'$,

whose l-th coordinate is $\varrho_l(z) = P(W = l \mid \Delta Z = z)$, is the only conditional expectation the procedure estimates nonparametrically.

Take the exogeneity condition to be a conditional mean, $E[\Delta U \mid W, \Delta Z] = 0$. What it delivers about $\varphi_\dagger$ is the $\mathsf{L} - 1$ contrasts between the W-cells at each value of ΔZ, and identification asks those contrasts to separate every candidate from $\varphi_\dagger$. That is a completeness condition on the instrument, and a W with finitely many values does not supply one (see Domínguez and Lobato, 2004; Severini and Tripathi, 2006). The obstruction is a property of the instrument rather than of any estimator, and neither a larger sample nor a larger L removes it.

Instruments of this kind are common in applied panel work, and the two-cell case is the canonical one. A randomised offer, an eligibility rule crossed once, a programme a unit is or is not exposed to: each supplies $W \in \{0, 1\}$ and nothing more. Others supply a few cells and no continuum: a randomisation into a small number of arms, a rule with several thresholds, a designation that records which of two triggers fired.

Discrete instruments can be accommodated either by weakening what is asked of the data, reporting a pseudo-solution or a set rather than a function (see Babii and Florens, 2025; Chen, 2021); or by strengthening the exogeneity condition from mean independence to independence (see Centorrino, Fève, and Florens, 2025; Dunker et al., 2014; Dunker, 2021, among others).

The latter route is the one we undertake in this paper, in the conditional form $\Delta U \perp\!\!\!\perp W \mid \Delta Z$.

What independence buys is a family of restrictions where the mean condition gave $\mathsf{L} - 1$. Every indicator of the residual is an admissible test function, so the contrasts $\widetilde{W}_l = \mathbb{1}\{W = l\} - \varrho_l(\Delta Z)$ are asked to be uncorrelated with all of them at once, and the system that results has enough content to pin down a function on a continuum. Assumption A1b(iii) is that it does, and the completeness condition that reaches its linearised form is stated given the residual as well as the controls.

Those $\mathsf{L} - 1$ contrasts are the whole of what the instrument supplies, and nothing in the

W direction is approximated. Every conditionally centred function of $(W, \Delta Z)$ is a ΔZ-measurable linear combination of them, so a Bierens-type device is needed only in the ΔZ direction. That is why A1 itself is unchanged at every L: it is the statement $\Delta U \perp\!\!\!\perp W \mid \Delta Z$ about σ-fields, and what was binary in a previous draft of this paper was the representation and never the assumption.

What it costs is nonlinearity, and the cost is not cosmetic. The equation the family defines is nonlinear in φ, so its linearisation $\mathcal{A}_{\text{cond}}$ is a Fréchet derivative taken at the unknown truth and not a fixed conditional-expectation operator. The sample version is then noisy as an operator, and not merely in its right-hand side.

Nonlinearity also reaches $\theta_\dagger$. A debiasing weight that ignored the residual could be orthogonal only where the moment is already free of φ, and by Proposition 13 and Corollary 14 that is a design with nothing to debias. So the weight varies with the residual, and the moment carries a quadratic term in φ. That term is why the first stage faces a rate condition at all, and why the condition is $\delta_n = o(n^{-1/4})$ rather than a product of two estimation errors.

This is why the result is not a corollary of the linear theory.

On the discrete-instrument side the closest work is the cross-section version of the φ-problem, which we solve in a previous paper (Centorrino, Fève, and Florens, 2025). It takes the instrument independent of the structural error, writes the restriction as an operator equation, and estimates the structural function by an iterative Landweber–Fridman scheme, with an upper bound on the mean integrated squared error and explicit guidance on the initial condition. There is no parametric component in that paper, no panel, and no $\sqrt{n}$ statement about anything.

The machinery carries over and the question does not. Adding $Z'\theta$ changes little about how $\varphi_\dagger$ is computed and much about what has to be proved. The parameter $\theta_\dagger$ must be identified without recourse to a source condition, and the moment used to estimate it must be made insensitive to a first stage converging at a nonparametric rate.

On the panel side the closest prior work is Fève and Florens (2014). They analyse first-differenced panel models with endogenous regressors as a linear inverse problem solved by Tikhonov regularisation, and their adjoint is an integral against the same difference of joint densities that runs through this paper. Their operator is linear, there is no parametric component anywhere in it, and nothing requires the instrument to be discrete.

Their non-identification condition and their Proposition 2.2 are used here rather than distinguished. The first, that the operator annihilates every φ when the two periods are exchangeable given the instrument, is the linear-case form of Proposition 13; the second, of which they write that its assumption is too strong, does not transplant at all. Their completeness hypothesis is posed on $g \mapsto E[g \mid W]$; the conditional-independence analogue would be posed on $\Lambda \mapsto E[\widetilde{W}\Lambda \mid \Delta U, \Delta Z]$, and that operator annihilates the constants, because $\widetilde{W}$ is conditionally centred. The analogue is therefore not merely stronger than theirs but unsatisfiable, and we record it as a route that is closed rather than as one we take.

On the third side, $\sqrt{n}$ estimation of a finite-dimensional object whose nuisance solves an ill-posed equation is settled for linear operators. Severini and Tripathi (2006) give the condition, that the relevant representer lie in the range of the adjoint. Florens, Johannes, and Van Bellegem (2012) put it to work in the partially linear model, obtaining $\sqrt{n}$ normality of the slope under Tikhonov regularisation. Florens and Lapenta (2024) reach the same model by Landweber–Fridman, keeping the instrument out of the smoothing by fixing its bandwidth.

That literature now runs without completeness, reaches strongly identified functionals of weakly identified functions, and delivers one-step estimators of average marginal effects (see Bennett et al., 2023; Bennett et al., 2026; Chen, 2021; Girard and Lapenta, 2026, among others). Every operator in that list is linear.

A nonlinear framework does cover this model in principle. Chen and Pouzo (2012, 2015) treat conditional moment models whose generalized residual is nonsmooth and nonlinear in

the unknown function, and deliver $\sqrt{n}$ normality for regular functionals under a sieve Riesz representer condition. To the best of our understanding, instantiating it here would call for a continuum of generalized residuals, nonparametric smoothing in a three-to-five dimensional conditioning variable, and a Donsker property for a regularised inverse with a data-driven stopping rule. The estimator of Section 3 needs none of the three.

On the independence side the nearest work is Poirier (2017), who computes the semiparametric efficiency bound for a finite-dimensional parameter under independence restrictions. His examples are a linear instrumental-variables model and a partially linear model whose nonparametric argument is exogenous, so no structural inverse appears in either.

He then names the step he does not take: an independence restriction whose residual depends on an unknown function as well as on a finite-dimensional parameter (Poirier, 2017, §5).

That is the case we take up, and the differences should be recorded. He writes the restriction unconditionally and we maintain the conditional form, and the nuisance here additionally solves an ill-posed nonlinear operator equation. His sentence is therefore the closest available statement of the gap and not a description of what we do.

The combination is new to the best of our knowledge: an ill-posed *nonlinear* operator equation for the nuisance, a discrete instrument, a differenced panel, and a $\sqrt{n}$ statement about the parametric component.

We contribute to the existing literature in several directions.

First, a negative result. Proposition 13 shows that a moment with weight g is free of φ exactly when the two periods' g-tilted marginals of the endogenous regressor coincide. At $g \equiv 1$ that is marginal stationarity of X. So the partialling-out argument behind the partially linear instrumental-variables literature is unavailable here, and its unavailability is a property of the differenced design and not of our estimator.

Second, an orthogonal moment and the condition it needs. We add to the differenced least-

squares moment a term $\widetilde{W}$ times a weight evaluated at the residual, and Proposition 20 shows that the moment's derivative in φ vanishes in every direction if and only if the weight's derivative solves $\mathcal{A}^{*}_{\mathrm{cond}}\ell^{\Pi} = -\nu/\pi$. That is the range condition of Severini and Tripathi (2006), read at the Fréchet derivative of a nonlinear operator instead of at a fixed one. We prove it sufficient and do not show it necessary.

Third, a device that keeps the conditioning variables out of the nonparametric machinery. Lemma 11 converts the conditional restriction into an unconditional family indexed by a scalar residual threshold and a Bierens index, in the manner of Bierens (1990) and at the generality of Stinchcombe and White (1998) (see also Bierens and Ploberger, 1997; Escanciano, 2006; Fan and Li, 2000). The conversion is an equivalence and not an approximation, so no conditional law given ΔZ is ever estimated, and the Landweber–Fridman machinery carries over with a larger index set and no new curse of dimensionality.

Finally, the theorem. Theorem 29 states that the cross-fitted one-step estimator is $\sqrt{n}$-consistent and asymptotically normal with influence function $J^{-1}\psi_{\star}$, that $J = E[\Delta Z \Delta Z']$ exactly, and that the sample sandwich is consistent for the limiting variance. The studentised statement holds in every direction, and the positive definiteness it needs is implied by the theorem's own hypotheses rather than added to them.

The rate the theorem asks of the first stage is stated in primitives. Proposition 28 delivers the first-stage rate δ_n, and the frontier (7) asks that it be faster than $n^{-1/4}$.

Read in the primitives, the frontier is a smoothness requirement on the propensity score: $r > d_Z(2\mu+1)/\{2(2\mu-1)\}$, with r its Hölder smoothness, d_Z its number of non-degenerate arguments and μ the source index. At $d_Z = 1$ it approaches $r > 2$ as $\mu \downarrow \frac{5}{6}$ and weakens at every larger μ, so a scalar conditioning variable and five derivatives clear it throughout.

What we do not do is worth stating here rather than leaving to be found. The estimator has two ingredients that we assume to exist and build nowhere, and it rests on three conditions we take as primitive rather than derive. Section 3 lists both sets, Section 7 repeats them,

and the supplement's remark on open verification obligations inventories a separate matter: the four forward obligations of the block, of which two remain open.

We assume global identification of $\varphi_\dagger$ and do not derive it, and we offer no primitive for the linearised condition either: the instrument-completeness route that would have reached it is unsatisfiable in this design, for the reason given in Section 2. Nor would that condition have delivered a neighbourhood of uniqueness had it been reachable: $\mathcal{A}_{\mathrm{cond}}$ is compact by Proposition 9, hence not bounded below, so by Proposition 8(i) a competitor becomes admissible at a radius shrinking to zero along the singular system.

The theorem covers mildly ill-posed problems. Under exponentially decaying singular values the rate attainable from any source condition is logarithmic, so (7) fails at every r.

Near the frontier the asymptotics do not bind at any sample size anyone will run, and that is quantified rather than gestured at. At $\mu = 2$, with a scalar conditioning variable and five derivatives in the first stage, the remainder rises in n before it falls and the turn comes at $n^* \approx 1{,}140$. At the configuration the supplement's first Monte Carlo experiment runs, which sits just inside the frontier, it comes at $n^* \approx 10^{58}$.

The paper is organised as follows. Section 2 sets out the model, the operator equation and the identification results, including the negative result on φ-free moments and the localisation result. Section 3 defines the estimator, the stopping rule and the cross-fitting scheme. Section 4 contains the asymptotic theory and the frontier condition. Section 5 reports a Monte Carlo experiment that runs the shipped estimator on estimated nuisances at three instrument cardinalities and at sample sizes that bracket the application's; the supplement carries it in full, together with three further experiments, one at the configuration the theory names, one at a configuration with room to spare at the frontier, and one that varies the instrument's cardinality on the first of those two designs. Section 6 illustrates the estimator on a first-differenced county panel, estimating a hedonic price function for air quality with a regulatory designation read as four instrument cells, and the last section concludes. Proofs of all results

are in the online supplement.

2 Local identification under independence

Assumptions

2.0.0.1 The instrument, its contrasts, and the range reference measure. The instrument W takes finitely many values, $W \in \{0, 1, \ldots, \mathsf{L}-1\}$, with $\mathsf{L} \geq 2$ an integer that is *fixed and known to the analyst. Sans-serif* L *is the number of instrument cells and nothing else*: italic L remains the debiasing weight of Definition 19, L_x and L_u remain the sieve dimensions, and $L^2(\cdot)$ remain the function spaces. Write, for $l \in \{0, 1, \ldots, \mathsf{L}-1\}$, $\varrho_l(z) := P(W = l \mid \Delta Z = z)$, and collect the *non-base* cells into

$$\varrho \;:=\; \big(\varrho_1, \ldots, \varrho_{\mathsf{L}-1}\big)', \qquad \widetilde{W}_l \;:=\; \mathbb{1}\{W = l\} - \varrho_l(\Delta Z), \qquad \widetilde{W} \;:=\; \big(\widetilde{W}_1, \ldots, \widetilde{W}_{\mathsf{L}-1}\big)' \in \mathbb{R}^{\mathsf{L}-1}, \tag{1}$$

so that $\varrho_0 = 1 - \mathbf{1}'\varrho$ with $\mathbf{1}$ the vector of ones in $\mathbb{R}^{\mathsf{L}-1}$. Cell 0 is the base cell. The *conditional contrast covariance* is

$$\Omega(z) \;:=\; \mathrm{Var}\big(\widetilde{W} \mid \Delta Z = z\big) \;=\; \mathrm{diag}\big(\varrho(z)\big) - \varrho(z)\varrho(z)', \qquad z \in \mathbb{R}^p. \tag{2}$$

At $\mathsf{L} = 2$ these are the scalar objects: $\varrho = \varrho_1 = P(W = 1 \mid \Delta Z)$ is the propensity, $\widetilde{W} = W - \varrho(\Delta Z)$, and $\Omega = \varrho(1 - \varrho)$.

The contrasts span, and nothing in the W direction is approximated: every measurable $a(W, \Delta Z)$ with $E[|a| \mid \Delta Z] < \infty$ and $E[a \mid \Delta Z] = 0$ almost surely is $a = c(\Delta Z)'\widetilde{W}$ for a ΔZ-measurable c, uniquely so at a $\Delta Z = z$ at which every cell is non-null, where the space has dimension exactly $\mathsf{L} - 1$ and (1) is a basis for it. *The spanning statement needs no overlap condition*; at a z with null cells the dimension is one less than the number of non-null cells.

Consequently the Bierens device is needed only in the ΔZ direction, where it always was, and B1 is unchanged apart from B1(iv.d). The derivation, the exact statement at a degenerate z, and the fact that neither depends on which cell is taken as the base are in the Discussion paragraph *A1*.

Type declaration, stated once and binding on every statement in this corpus. The debiasing weight and the objects built from it carry one column per contrast: L, $\ell = \partial_u L$, the anchor b and their starred and hatted variants are $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued, one *row* per coordinate of θ and one *column* per contrast, and the range-space representer ℓ^{Π} carries the contrast index as a third argument, $\ell^{\Pi}(u,t,l)$. *Norms carried by these objects are Frobenius*; matrix norms elsewhere in this corpus are spectral unless the site says otherwise, and the two are named at every site at which both occur. The moment function is $\psi = \Delta Z\,\Delta U + L(\Delta U, \Delta Z)\,\widetilde{W}$ of (38), the contrast standing to the *right* of the $p \times (\mathsf{L}-1)$ weight. *At* $\mathsf{L} = 2$ *each of these is the printed object*: a $p \times 1$ matrix is a vector in $\mathbb{R}^p$, the Frobenius norm is the Euclidean norm, $\ell^{\Pi}(u,t,1) = \ell^{\Pi}(u,t)$, and $L\widetilde{W} = \widetilde{W}L$ because $\widetilde{W}$ is then a scalar.

The estimator carries the **raw** contrast (1), on both sides: the operator equation is written at $\widetilde{W}$ and so is ψ, while Ω enters as the conditional covariance (2) and as the prefactor (S:omegaprefac), and *never as a weight*. The normalised contrast $\widetilde{W}^{\Omega} := \Omega(\Delta Z)^{-1}\widetilde{W}$, what it makes explicit, and the reason the paper does **not** estimate at it are Remark S:htread.

The range reference measure gains a third factor: $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ with Π_{w} a probability measure on $\{1,\dots,\mathsf{L}-1\}$, masses $\varpi_{\mathsf{w}}(l)$, required strictly positive by B1(iv.d). *At* $\mathsf{L} = 2$, Π_{w} *is the point mass at* 1 *and* $\Pi = \Pi_u \otimes \Pi_t$, under the canonical identification of $\mathbb{R} \times \mathcal{T} \times \{1\}$ with $\mathbb{R} \times \mathcal{T}$; c_{Π} of B1(iv.c) is unchanged at every L, Π_{w} being a probability measure and contributing the factor 1.

2.0.0.2 How a citation of this block is read. *On a result's hypothesis line* — inside a "Suppose … hold" — a citation is a **carrier**: naming a tag carries every condition under it, and naming a clause carries that clause and no other. *Inside a definition* a citation is a

pointer when it merely locates the object it names, as Definition 5's "$\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ as in Assumption B1(iv)" does, and imposes nothing; it is **not** a pointer when it is constitutive, as A2b's "admits a version satisfying A2′(i.a)–(i.b)" and Definition 2's use of the constant A4(i) declares are. *And several conditions below are written in two halves*: a **declarative** half, which introduces a symbol, a type or a carrier set and is *standing* — in force at every statement in this corpus, named or not, since without it the statement has no objects to be about, so it is never listed as a hypothesis — and a **clause**, which restricts, and which is carried on the hypothesis line of every result whose proof reads it. A0, A4(i), A8, B1(iii) and B1(iv) are split that way. Where the split falls, why it replaced an earlier test, and the register of the three constants the analyst chooses are in the Discussion of the assumptions, under the heading *The three registers*.

Assumption (A0: Sampling)**.** $\{(Y_{i1}, Y_{i2}, X_{i1}, X_{i2}, Z_{i1}, Z_{i2}, W_i)\}_{i=1}^{n}$ are independent and identically distributed draws from a distribution P on $\mathbb{R}^2 \times \mathcal{X}^2 \times \mathbb{R}^{2p} \times \{0, 1, \ldots, \mathsf{L}-1\}$, where $\mathcal{X} \subseteq \mathbb{R}$ is the carrier of the regressor, $\underline{x} := \inf \mathcal{X}$ and $\bar{x} := \sup \mathcal{X}$ — *that much is A0's* ***declaration*** — and $\mathsf{L} \geq 2$ is a finite integer that does not depend on n and is known to the analyst. **The condition on** $\mathcal{X}$ is that it is a *compact interval*, so that $\mathcal{X} = [\underline{x}, \bar{x}]$ with $\underline{x} < \bar{x}$; *that is a clause and is carried by the results that read it.*

Assumption (A1: Conditional exogeneity)**.** $\Delta U \perp\!\!\!\perp W \mid \Delta Z$ and $E[\Delta U \mid \Delta Z] = 0$.

Note on A1. A1 is a statement about σ-fields and carries no reference to the cardinality of W's support. Its representation is (1), and that representation is exact: restriction (A) of Definition 7 written at the $\mathsf{L}-1$ contrasts is *equivalent* to $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, at every ΔZ, at every L, and under **no** overlap condition — *A4(i) is not a hypothesis of that equivalence and must not be listed as one.* The equivalence, the base-cell invariance, and the sense in which extra cells are *surplus* rather than necessary are in the Discussion of the assumptions, under the heading *A1 (conditional exogeneity).*

Assumption (A1b: Identification)**.**

(i) *Normalisation.* $\varphi_\dagger \in \mathcal{V}$, that is, $\int_{\mathcal{X}} \varphi_\dagger \, d\pi = 0$.

(ii) *Measure-separability.* Every function of X_1 that is almost surely equal to a function of X_2 is almost surely equal to a constant.

(iii) *Global identification of* $\varphi_\dagger$. $\mathcal{V}_{CI} = \{\varphi_\dagger\}$, where $\mathcal{V}_{CI} := \{\varphi \in \mathcal{V} : \ \Delta Y - \Delta\varphi \perp\!\!\!\perp W \mid \Delta Z\}$.

(iv) *Identification of* $\theta_\dagger$. $E[\Delta Z \, \Delta Z']$ is nonsingular, and $E\|\Delta Z\|^2 < \infty$, $E[\Delta U^2] < \infty$, $E\|\Delta Z \, \Delta U\|^2 < \infty$.

Assumption (A1c: First-order local identification)**.** The linearisation is injective: $\ker \mathcal{A}_{\text{cond}} = \{0\}$, where $\mathcal{A}_{\text{cond}} : \mathcal{V} \to L^2(\Pi)$ is the Fréchet derivative of Proposition 8 evaluated at $(\varphi_\dagger, \theta_\dagger)$.

Assumption (B1: Bierens family and range space)**.**

(i) *Transform.* $\Phi : \mathbb{R}^p \to \mathbb{R}^p$ is Borel, one-to-one, and bounded: $c_\Phi := \sup_z \|\Phi(z)\|_\infty < \infty$. The maintained choice is Φ componentwise arctan, for which $c_\Phi = \pi/2$. *The value of* c_Φ *is not pinned by this block and no result reads it*: only $c_\Phi < \infty$ is consumed, at Lemma S:bierenscore (compactness of the image cube) and through $\bar{w}$ at (ii). c_Φ *is not separately identified from the scale of* $\mathcal{T}$ — see Lemma S:scaleinv(vii) — *and enters the programme only through* c_Π *of* (iv.c).

(ii) *Weight.* $w = \exp$. Write $\bar{w}(t) := e^{c_\Phi \|t\|_1}$, so that $|w(t'\Phi(z))| \le \bar{w}(t)$ for every z.

(iii) *Index set.* $\mathcal{T} \subset \mathbb{R}^p$ has strictly positive Lebesgue measure.

(iv) *Range reference measure.* $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$, where

(iv.a) Π_u is a finite measure on $\mathbb{R}$ that is absolutely continuous with respect to Lebesgue measure, with a density ϖ_u that is bounded, strictly positive at every point of $\mathbb{R}$, and continuously differentiable with $\|\varpi_u'\|_\infty < \infty$; in particular Π_u has full topological support, $\Pi_u(O) > 0$ for every nonempty open $O \subseteq \mathbb{R}$;

(iv.b) Π_t is a finite measure on $\mathcal{T}$ for which there exists a Borel set $\mathcal{T}_0 \subseteq \mathcal{T}$ such that $\text{Leb}|_{\mathcal{T}_0}$ and $\Pi_t|_{\mathcal{T}_0}$ are *not mutually singular*;

(iv.c) $c_\Pi^2 := \Pi_u(\mathbb{R}) \int_{\mathcal{T}} \bar{w}(t)^2 \, \Pi_t(dt) < \infty$;

(iv.d) Π_{w} is a *probability* measure on $\{1, \dots, \mathsf{L}-1\}$ whose masses $\varpi_{\mathsf{w}}(l) := \Pi_{\mathsf{w}}(\{l\})$ are strictly positive at every $l \in \{1, \dots, \mathsf{L}-1\}$.

(v) *Correction measure.* γ is a finite signed measure on $\mathcal{T}$ with $\int_{\mathcal{T}} \bar{w}(t) \, |\gamma|(dt) < \infty$; and, where $\partial_u L$ is required, $|\partial_u \lambda(u,t)| \le \Lambda_1(t)$ for all u, with $\int_{\mathcal{T}} \Lambda_1(t) \, \bar{w}(t) \, |\gamma|(dt) < \infty$.

Assumption (A2: Source condition for θ)**.** Each coordinate of ν/π lies in $\mathcal{R}(\mathcal{A}_{\text{cond}}^*)$, the range of the adjoint of $\mathcal{A}_{\text{cond}} : \mathcal{V} \to L^2(\Pi)$, where $\nu := \nu_{\Delta Z}$ and $\nu_g(x) := E[g \mid X_2 = x] f_{X_2}(x) - E[g \mid X_1 = x] f_{X_1}(x)$ as in (20). Equivalently (Picard): $\nu/\pi \perp \ker \mathcal{A}_{\text{cond}}$ in $L^2(\pi)$ and $\sum_{j \ge 1} \sigma_j(\mathcal{A}_{\text{cond}})^{-2} \, \langle \nu/\pi, \, e_j \rangle_{L^2(\pi)}^2 < \infty$.

Assumption (A2′: Source condition with usable representer)**.** A2 holds, and the selected solution $\ell_\star^\Pi \in \ell_\dagger^\Pi + \ker \mathcal{A}_{\text{cond}}^*$ of $\mathcal{A}_{\text{cond}}^* \ell^\Pi = -\nu/\pi$, where $\ell_\dagger^\Pi$ denotes any one particular solution, so that the displayed set is the whole solution coset and $\ell_\star^\Pi$ is a selection from it, *admits a version* satisfying

(i) *Range-space regularity.*

(i.a) $\Lambda_\star^{(0)} := \max_{1 \le l \le \mathsf{L}-1} \int_{\mathcal{T}} \left(\sup_{u \in \mathbb{R}} \left\| \ell_\star^\Pi(u,t,l) \right\| \right) \bar{w}(t) \, \Pi_t(dt) < \infty$;

(i.b) for every $l \in \{1, \dots, \mathsf{L}-1\}$ and Π_t-almost every t the map $u \mapsto \ell_\star^\Pi(u,t,l)$ is continuously differentiable, and $\Lambda_\star^{(1)} := \max_{1 \le l \le \mathsf{L}-1} \int_{\mathcal{T}} \left(\sup_{u \in \mathbb{R}} \left\| \partial_u \ell_\star^\Pi(u,t,l) \right\| \right) \bar{w}(t) \, \Pi_t(dt) < \infty$;

and $\ell_\star := \mathcal{W} \ell_\star^\Pi$ is defined *pointwise* from that version, through (S:Wdef), columnwise in l;

(iii) *Anchor.* The debiasing weight attached to $\ell_\star$ is $L_\star := L_{\ell_\star}^{b_\star}$ of Definition 19 for a measurable $b_\star : \mathbb{R}^p \to \mathbb{R}^{p \times (\mathsf{L}-1)}$ with $E \left\| b_\star(\Delta Z) \right\|^2 < \infty$, the norm being Frobenius.

The two budgets aggregate the $p(\mathsf{L}-1)$ *slicewise quantities in* ***two different ways***: the $\|\cdot\|$ inside each supremum is the **Euclidean** norm of the vector $\ell_\star^\Pi(u,t,l) \in \mathbb{R}^p$, and the aggregation over the contrast index l is by the **maximum**. *The asymmetry is deliberate*, and

why each half is what it is is in the Discussion of the assumptions, under $A2'$. At $p = 1$ the Euclidean norm on $\mathbb{R}$ is the modulus, and at $\mathsf{L} = 2$ the maximum is over a one-element set.

Assumption (A2b: Admissible representer (a *property*, not maintained))**.** A function $\ell^{\Pi} \in L^2(\Pi)$ is an *admissible representer* if it admits a version satisfying A2′(i.a)–(i.b) with ℓ^{Π} in place of $\ell^{\Pi}_{\star}$, that is, with

$$\Lambda^{(0)} := \max_{1 \le l \le \mathsf{L}-1} \int_{\mathcal{T}} \Big(\sup_{u \in \mathbb{R}} \big\| \ell^{\Pi}(u,t,l) \big\| \Big) \bar{w}(t)\, \Pi_t(dt) < \infty,$$
$$\Lambda^{(1)} := \max_{1 \le l \le \mathsf{L}-1} \int_{\mathcal{T}} \Big(\sup_{u \in \mathbb{R}} \big\| \partial_u \ell^{\Pi}(u,t,l) \big\| \Big) \bar{w}(t)\, \Pi_t(dt) < \infty,$$

the norms and the aggregation over l being A2′(i.a)–(i.b)'s, and $u \mapsto \ell^{\Pi}(u,t,l)$ continuously differentiable for every $l \in \{1, \dots, \mathsf{L}-1\}$ and Π_t-almost every t, and if $\ell := \mathcal{W}\ell^{\Pi}$ is defined *pointwise* from that version through (S:Wdef), columnwise in l. *A2b is NOT MAINTAINED*: it is a hypothesis a result may impose on a representer it quantifies over. Under A2′(i) the selected solution $\ell^{\Pi}_{\star}$ satisfies A2b, with $\Lambda^{(0)} = \Lambda^{(0)}_{\star}$ and $\Lambda^{(1)} = \Lambda^{(1)}_{\star}$; that implication is the *only* relation between the two tags.

Assumption (A2c: Source condition of index μ'' on the representer (a *property*, not maintained))**.** Fix $\mu'' > \frac{3}{2}$. For $g \in L^2(\mathbb{R}, du)$ write $(\mathcal{F}g)(\xi) := \int_{\mathbb{R}} g(u) e^{-\mathrm{i}u\xi} du$ for the Fourier transform in u, and let

$$H^{\mu''}(\mathbb{R}) \;:=\; \Big\{ g \in L^2(\mathbb{R}, du) : \; \big\| g \big\|^2_{H^{\mu''}(\mathbb{R})} := \tfrac{1}{2\pi} \int_{\mathbb{R}} \big(1 + \xi^2\big)^{\mu''} \big| (\mathcal{F}g)(\xi) \big|^2 d\xi < \infty \Big\} \qquad (3)$$

be the u-Sobolev scale of order μ'' on $\mathbb{R}$, *taken with respect to Lebesgue measure du and not with respect to* Π_u. For an $\mathbb{R}^p$-valued g on $\mathbb{R}$ write $H^{\mu''}(\mathbb{R}; \mathbb{R}^p)$ for the same scale with $|\cdot|$ replaced inside (3) by the Euclidean norm $\|z\| := \big(\sum_{j=1}^p |z_j|^2 \big)^{1/2}$ on $\mathbb{C}^p$, which is where $(\mathcal{F}g)(\xi)$ takes its values, so that $\|g\|^2_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)} = \sum_{j=1}^p \|g_j\|^2_{H^{\mu''}(\mathbb{R})}$ and $g \in H^{\mu''}(\mathbb{R}; \mathbb{R}^p)$ if and only if every coordinate lies in $H^{\mu''}(\mathbb{R})$; *at* $p = 1$ *they are the same object.* A function $\ell^{\Pi} \in L^2(\Pi)$ *satisfies the source condition of index* μ'' if it admits a jointly measurable

representative, *fix one; everything below is read at it*, whose $\mathsf{L}-1$ slices satisfy: there is a Borel set $\mathcal{T}_* \subseteq \mathcal{T}$, *one set, serving every* l, with $\Pi_t(\mathcal{T}\setminus\mathcal{T}_*)=0$ such that the slice $\ell^{\Pi}(\cdot,t,l)$ lies in $H^{\mu''}(\mathbb{R};\mathbb{R}^p)$ for *every* $t\in\mathcal{T}_*$ and every $l\in\{1,\dots,\mathsf{L}-1\}$, and

$$\Lambda^{(\mu'')} \;:=\; \max_{1\le l\le \mathsf{L}-1}\; \int_{\mathcal{T}_*} \left\|\ell^{\Pi}(\cdot,t,l)\right\|_{H^{\mu''}(\mathbb{R};\mathbb{R}^p)}\, \bar{w}(t)\,\Pi_t(dt) \;<\; \infty, \tag{4}$$

the budget being aggregated as in A2′(i).

Notes on A2c. That it is a property and not a maintained condition, whose sole role is to imply A2′(i) at the starred representer, and the $\bar{w}$-weighted geometry it is stated in, are Remark S:reprsourceread.

Assumption (H1: Hilbert-scale link)**.** $\mathcal{A}_{\mathrm{cond}}:\mathcal{V}\to L^2(\Pi)$, read into the *whole* three-factor range space of Definition 5, not contrast by contrast, has infinite rank and its singular values satisfy $\sigma_j(\mathcal{A}_{\mathrm{cond}})\asymp j^{-a}$ for some $a>0$; the source set for $\varphi_\dagger$ is a ball of the *classical* Sobolev space $H^s(\mathcal{X})$, for non-integer s, the interpolation space between the neighbouring integer orders, **with no boundary condition imposed at either endpoint of** $\mathcal{X}$; and the operator is *matched* to that scale in the sense of Engl, Hanke, and Neubauer (2000, §8.5), which is here the following stated comparison: there is a finite c, not depending on ζ, such that every $\zeta\in\mathcal{V}\cap H^s(\mathcal{X})$ admits

$$\zeta \;=\; \left(\mathcal{A}^*_{\mathrm{cond}}\mathcal{A}_{\mathrm{cond}}\right)^{s/(2a)} h \qquad \text{with}\quad \|h\|_{L^2(\pi)} \;\le\; \mathsf{c}\,\|\zeta\|_{H^s}. \tag{5}$$

Notes on H1. That a is an exponent of the **stacked** operator and not of any one contrast, that it does not depend on the range weights Π_{w}, what a per-contrast reading of it does and does not license at $\mathsf{L}\ge 3$, and why $\mathsf{L}=2$ reduces with no exception, are the supplement's Discussion paragraph *H1*. The disposition of the two numbers the glyph μ names is Remark S:pasrc.

Assumption (A4: Overlap and propensity regularity)**.**

(i) *Overlap.* **Fix** $\varepsilon_\varrho \in \left(0, \frac{1}{\mathsf{L}}\right]$, once and for all in this block — *that much is A4(i)'s* ***declaration****, and it is what Definition 2 builds* (11) *from.* **The condition** is that $\varrho_l(\Delta Z) \geq \varepsilon_\varrho$ almost surely, *for every* $l \in \{0, 1, \ldots, \mathsf{L}-1\}$: *every cell of the instrument is reached, at every value of the controls, with probability bounded away from zero*, which rules out a design in which some cell empties out over part of the support of ΔZ. Consequently

$$\lambda_{\min}\Big(\Omega(\Delta Z)\Big) \;\geq\; \varepsilon_\Omega \;:=\; \frac{\varepsilon_\varrho}{\mathsf{L}} \;>\; 0 \qquad \text{almost surely.} \tag{6}$$

(ii) *Smoothness.* Each of $\varrho_1, \ldots, \varrho_{\mathsf{L}-1}$ belongs to a Hölder ball of order $r > 0$, the *same* r for every l, on the support of the non-degenerate coordinates of ΔZ, whose number is d_Z with $1 \leq d_Z \leq p-1$.

(iii) Estimation, and the measurability that makes the display a probability statement. $\hat{\varrho} = (\hat{\varrho}_1, \ldots, \hat{\varrho}_{\mathsf{L}-1})'$ is computed on an auxiliary fold; its *own pointwise function* $(\omega, z) \mapsto \hat{\varrho}(\omega)(z)$, the one Definition 2 evaluates at each ΔZ_i, is **jointly measurable**, with respect to the product of the σ-algebra generated by the auxiliary fold with $\mathfrak{B}(\mathbb{R}^p)$; and

$$\big\|\hat{\varrho} - \varrho\big\|_{L^2} \;:=\; \Big(\int \big\|\hat{\varrho}(z) - \varrho(z)\big\|^2 P_{\Delta Z}(dz)\Big)^{1/2} = O_P\Big(n^{-r/(2r+d_Z)}\Big) =: O_P(r_n),$$

the inner $\|\cdot\|$ being Euclidean on $\mathbb{R}^{\mathsf{L}-1}$ and the integral running over z alone, so that the left side is a function of the auxiliary fold.

Assumption (N1: Conditional centring of a weight)**.** A debiasing weight L *satisfies N1* if $E\Big[L(\Delta U, \Delta Z) \mid \Delta Z\Big] = 0$ almost surely, *the zero on the right being the zero matrix of* $\mathbb{R}^{p\times(\mathsf{L}-1)}$: N1 is $p(\mathsf{L}-1)$ scalar conditions, one per pair (j, l), and it is **not** a condition on any contraction of the $\mathsf{L}-1$ columns. For a fixed $\mathsf{v} \in \mathbb{R}^{\mathsf{L}-1}$ with $\mathsf{v} \neq 0$ the contracted condition $E\Big[L \mid \Delta Z\Big]\,\mathsf{v} = 0$ is p conditions; it is implied by N1 and implies N1 *only* at $\mathsf{L} = 2$, where the columns number one and the two statements are the same p conditions. Corollary 21 makes

the gap explicit and prices it. N1 is a property that a weight may or may not have; it is not maintained, and a result that needs it says so.

Assumption (N1′: Conditional centring of a direction)**.** A perturbation direction D in the weight *satisfies N1′* if $E\big[D(\Delta U, \Delta Z) \mid \Delta Z\big] = 0$ almost surely.

Assumption (A5: Nuisance rates)**.** Let $h := \hat{\varphi} - \varphi_\dagger$, let $\hat{\delta}_n$, δ_n and ϵ_n be the rates the clauses below define, and let r_n be the propensity rate of A4(iii).

(i) Operator noise, at a declared level with two branches. The level is the *deterministic* sequence

$$\hat{\delta}_n \;:=\; \max\Big\{\hat{\delta}_n^{(T)}, \hat{\delta}_n^{(\varrho)}\Big\}, \qquad \hat{\delta}_n^{(T)} \;:=\; n^{-1/2}\sqrt{\ln n}, \qquad \hat{\delta}_n^{(\varrho)} \;:=\; r_n,$$

so that $\hat{\delta}_n \geq n^{-1/2}\sqrt{\ln n}$ at every n; and the operator error is $O_P(\hat{\delta}_n)$.

(ii) Structural rate, with the logarithm of (56) carried **in the symbol**. $\|h\|_{L^2(\pi)} = O_P(\delta_n)$ with

$$\delta_n \;:=\; \Big(\hat{\delta}_n \ln n\Big)^{2\mu/(2\mu+1)},$$

and $\delta_n = o_P(n^{-1/4})$. *The first half is* ***assumed*** *and is not delivered by Proposition 28*: $h = \hat{\varphi} - \varphi_\dagger$ is formed at the estimator Definition 22(b) instructs, while that proposition concludes at the infeasible reading $\hat{\varphi}^\pi$, and the passage between them is Remark S:phirscope(7), two of whose five parts are open. The second half is a statement about the deterministic δ_n alone and *is* delivered, under (7).

(iii) *Propensity frontier.* Writing $\hat{\delta}_n = n^{-r/(2r+d_Z)}$, which is (i)'s maximum for all n large enough, A4(ii)'s $d_Z \geq 1$ giving $r/(2r+d_Z) < \frac{1}{2}$, condition (ii) holds if and only if

$$\frac{2\mu}{2\mu+1} \cdot \frac{r}{2r+d_Z} \;>\; \frac{1}{4} \qquad\Longleftrightarrow\qquad r \;>\; \frac{d_Z\,(2\mu+1)}{2\,(2\mu-1)}, \tag{7}$$

which in particular requires $\mu > \frac{1}{2}$, whatever the value of r.

(iv) *Debiasing weight, on the derivative.* The estimator $\hat{\ell}$ of the representer, a random element of $L^2(P_{\Delta U,\Delta Z})$ measurable with respect to the σ-algebra generated by the auxiliary fold and read at one fixed *jointly* measurable representative, satisfies

$$\left\|\hat{\ell} - \ell_\star\right\|_{L^2(P)} = O_P(\epsilon_n), \qquad \delta_n\,\epsilon_n = o_P\Big(n^{-1/2}\Big).$$

(v) *Plug-in.* $\sqrt{\hat{m}}\,n^{-1/2} = o_P(\delta_n)$.

(vi) Sobolev stability of the first-stage error, at the final iterate. $\|h\|_{H^s} = O_P(1)$, with s as in H1.

(vi′) Sobolev stability along the linearisation path, as a w.p.$\to 1$ envelope at a **fixed** radius. With $\hat{\varphi}^{(0)}, \ldots, \hat{\varphi}^{(\bar{k})}$ the stages of Definition 22 and $\bar{k}$ finite, there is one finite R, not depending on n, such that

$$P\Big(\max_{0\le k\le \bar{k}} \left\|\hat{\varphi}^{(k)} - \varphi_\dagger\right\|_{H^s} \;\le\; \mathsf{R}\Big) \;\longrightarrow\; 1.$$

Assumption (A8: Sieve window)**.** The balancing sieve $b_1, \ldots, b_{L_x}$ is a *B-spline* basis of *degree exceeding* $s - \frac{1}{2}$ on a quasi-uniform mesh over $\mathcal{X}$ *with simple interior knots*, with

$$n^{1/(2s)} \;\ll\; L_x \;\ll\; n^{1/2},$$

and $\beta_1, \ldots, \beta_{L_u}$ is a basis on the residual range with $L_u \le L_x$. *Consequently the sieve is contained in the Sobolev space the source conditions are written in*:

$$\mathcal{S}_n \;=\; \operatorname{span}(b_1, \ldots, b_{L_x}) \cap \mathcal{V} \;\subset\; H^s(\mathcal{X}), \tag{8}$$

so $\|\zeta\|_{H^s} < \infty$ at every $\zeta \in \mathcal{S}_n$, with $\mathcal{S}_n$ the sieve of Definition 22 and $H^s(\mathcal{X})$ the *classical* Sobolev space of H1, *the one that carries no boundary condition*, which is what makes the

containment available to a basis that satisfies none.

Definition 1 (The integrated kernel)**.** Let C_{h_u} be the kernel of C3 and, for $I \in \mathfrak{I}$, define

$$\mathsf{C}_{h_u}(v; I) \;:=\; \int_{-\infty}^{v/h_u} \Theta_I(\varsigma)\, d\varsigma \;=\; \int_{-\infty}^{v/h_u} C_{h_u}\Big(h_u\varsigma, I\Big)\, d\varsigma \;=\; \frac{1}{h_u}\int_{-\infty}^{v} C_{h_u}(\varsigma, I)\, d\varsigma, \qquad v \in \mathbb{R}, \tag{9}$$

with Θ_I of (14), and write $\mathsf{C}_{h_u}(v) := \mathsf{C}_{h_u}(v; \mathbb{R})$. The same definition with h_x in place of h_u defines C_{h_x}. *The boundary index is carried only in the x-direction*: in the residual direction the argument u ranges over all of $\mathbb{R}$ and there is no boundary, so (12) uses $\mathsf{C}_{h_u}(\cdot) = \mathsf{C}_{h_u}(\cdot; \mathbb{R})$ throughout — *in u the interval is always* $\mathbb{R}$; in the x-direction $\mathcal{X}$ is a compact interval by A0 and the index is used. Its properties are Lemma S:intker.

Definition 2 (The sample operator $\hat{T}$)**.** Adopt the sample-splitting convention of Lemma S:balance: $\{1, \ldots, n\} = \mathcal{I}_a \cup \mathcal{I}_m$, $n_{\mathrm{a}} := \#\mathcal{I}_a$, $n_{\mathrm{m}} := \#\mathcal{I}_m$, the two folds independent by A0, with $\hat{\varrho}$ computed on $\mathcal{I}_a$ as A4(iii) requires. *The split is* ***balanced***: n_{a} and n_{m} are non-random with

$$n_{\mathrm{m}} \asymp n \qquad (\text{equivalently, } \liminf_n n_{\mathrm{m}}/n > 0). \tag{10}$$

Construction convention, and it costs nothing: $\hat{\varrho}$ is taken to have values in the ε_ϱ-interior of the simplex,

$$\Big\{q \in \mathbb{R}^{\mathsf{L}-1} :\; q_l \geq \varepsilon_\varrho \;\text{ for every } l \in \{1, \ldots, \mathsf{L}-1\},\;\; \textstyle\sum_{l=1}^{\mathsf{L}-1} q_l \leq 1 - \varepsilon_\varrho\Big\}, \tag{11}$$

with ε_ϱ the overlap constant of A4(i); if a first-stage estimator does not, replace it by its metric projection onto that set, which is well defined because (11) is closed, convex and, since A4(i) *declares* $\varepsilon_\varrho \leq 1/\mathsf{L}$, non-empty — *the declaration alone and not the condition.* Writing

$\widehat{\widetilde{W}}_i := \big(\mathbb{1}\{W_i = 1\} - \hat{\varrho}_1(\Delta Z_i), \ldots, \mathbb{1}\{W_i = \mathsf{L}-1\} - \hat{\varrho}_{\mathsf{L}-1}(\Delta Z_i)\big)'$, the *sample operator* is

$$\hat{T}(\varphi,\theta)(u,t,l) \;:=\; \frac{1}{n_{\mathrm{m}}} \sum_{i \in \mathcal{I}_m} \widehat{\widetilde{W}}_{i,l}\, \mathsf{C}_{h_u}\Big(u - \Delta U_i(\theta,\varphi)\Big)\, w\Big(t'\Phi(\Delta Z_i)\Big), \qquad (u,t,l) \in \mathbb{R}\times\mathcal{T}\times\{1,\ldots,\mathsf{L}-1\}, \tag{12}$$

with $\Delta U_i(\theta,\varphi)$ the trial residual of (19). Its x-smoothed variant $\hat{T}_{h_x}$, which is what Definition 22 iterates on and what $\hat{\mathcal{A}}_\varphi$ differentiates, is (S:hatTx).

Notes on Definition 2. What the balanced split (10) costs and where it is consumed, what the truncation buys, the scope note for a smaller known lower bound, and which assumption names (12) are Remark S:hatTread.

Definition 3 (The smoothed sample linearisation $\hat{\mathcal{A}}$)**.** For $\varphi \in \mathcal{V}$ let $\mathcal{A}_\varphi : \mathcal{V} \to L^2(\Pi)$ be the Fréchet derivative of $T(\cdot,\theta)$ at φ, so that $\mathcal{A}_{\mathrm{cond}} = \mathcal{A}_{\varphi_\dagger}$, with kernel $\mathsf{k}(u,t,l;x) = \nu_{g_{u,t,l}}(x)/\pi(x)$ as in Proposition 9, where $g_{u,t,l}$ is that proposition's $g_{u,t}$ read at the l-th contrast. Write $\hat{\mathcal{A}}_\varphi$ for the *kernel-smoothed sample analogue* of $\mathcal{A}_\varphi$: the integral operator whose kernel is obtained from k by replacing the indicator $\mathbb{1}\{\Delta U(\theta,\varphi) \le u\}$ inside $g_{u,t,l}$ by its C_{h_u}-smoothed version, and each of the two products $E[\,\cdot \mid X_j = x]f_{X_j}(x)$ constituting $\nu_{g_{u,t,l}}$ by its C_{h_x} kernel estimate. Write $\hat{\mathcal{A}}^*_\varphi$ for its adjoint in the $L^2(\pi)$–$L^2(\Pi)$ pairing, and

$$\hat{\mathcal{A}}_{\mathrm{cond}} := \hat{\mathcal{A}}_{\varphi_\dagger}, \qquad \hat{\mathcal{A}}^*_{\mathrm{cond}} := \hat{\mathcal{A}}^*_{\varphi_\dagger},$$

the running Landweber–Fridman iterate being written $\hat{\mathcal{A}}_{\hat{\varphi}_m}$ wherever the base point matters.

Assumption (A7: Stopping feasibility, at the amended threshold)**.** The level used by the discrepancy principle dominates the true operator noise level with probability approaching one: for some $\tau_{\mathrm{DP}} \ge 1$,

$$P\Big(\big\|\hat{T}(\varphi_\dagger,\theta)\big\|_{L^2(\Pi)} \le \tau_{\mathrm{DP}}\, \hat{\delta}_n\, \ln n\Big) \;\longrightarrow\; 1, \tag{13}$$

with $\hat{T}$ as in (12). *Any divergent factor serves in place of* $\ln n$; $\ln n$ is fixed here so that one

symbol, not a family, propagates downstream.

Assumption (C1: Densities)**.** Write $x = (x_1, x_2) \in \mathcal{X}^2$.

(i) The conditional density $f_{\Delta Y, X_1, X_2 | \Delta Z, W}$ and the joint density $f_{\Delta Y, X_1, X_2 | \Delta Z}$ are p_K times continuously differentiable and uniformly bounded above.

(ii) f_{X_1, X_2}, f_{X_1}, f_{X_2} and π are uniformly bounded above and below away from 0 on $\mathcal{X}^2$ and $\mathcal{X}$ respectively; consequently $\bar{c}_\pi := \max_{j \in \{1,2\}} \|f_{X_j}/\pi\|_\infty < \infty$.

Assumption (C2′: Smoothness of the residual law)**.**

(i) The conditional density $f_{\Delta Y | X_1, X_2, \Delta Z, W}(\cdot \mid \cdot)$ exists, is $p_K - 1$ times continuously differentiable in its first argument, and admits envelopes $\bar{f}_k(x_1, x_2, z, \mathsf{w}) := \sup_u \left| \partial_u^k f_{\Delta Y | X_1, X_2, \Delta Z, W}(u \mid x_1, x_2, z, \mathsf{w}) \right|$ with $E[\bar{f}_k] < \infty$ for $k = 0, 1, \ldots, p_K - 1$ and $\|\bar{f}_0\|_\infty + \|\bar{f}_1\|_\infty < \infty$.

(ii) For every $t \in \mathcal{T}$ the map $u \mapsto \mathcal{M}(u, t)$ is p_K times continuously differentiable, with $p_K = 8$ and

$$\max_{0 \le k \le p_K} \sup_{t \in \mathcal{T}} \bar{w}(t)^{-1} \sup_{u \in \mathbb{R}} \left| \partial_u^k \mathcal{M}(u, t) \right| < \infty.$$

Assumption (C3: Kernel, and the defining property of the boundary adjustment)**.** Fix $p_K = 8$. *Write* $\mathfrak{I}$ *for the set of closed intervals* $I \subseteq [-\infty, +\infty]$ *with* $\inf I \in [-\infty, 0]$, $\sup I \in [0, +\infty]$ *and* $\sup I - \inf I \geq 2$ *— half-lines and* $\mathbb{R}$ *included — and, for a bandwidth* $h > 0$ *and* $I \in \mathfrak{I}$*,*

$$\Theta_I(\varsigma) := C_h\Big(h\varsigma,\, I\Big), \qquad \varsigma \in \mathbb{R}, \tag{14}$$

for the kernel read as a function of its scaled argument. The smoothing kernel used in the residual direction is a univariate generalized kernel C_{h_u} with:

(i) Order p_K at the interior index. Writing $\Theta_1 := \Theta_{\mathbb{R}}$ *— the subscript* 1 *is an abbreviation for the untruncated index* $I = \mathbb{R}$*, and it is the* ***digit*** *that is not an interval and so not a member of* $\mathfrak{I}$*; the index it abbreviates,* $I = \mathbb{R}$*, is a member,* the notation being retained

because it is the interior kernel every bound below names —

$$\int_{\mathbb{R}} \Theta_1(\varsigma)\, d\varsigma \;=\; 1, \qquad \int_{\mathbb{R}} \varsigma^{j}\, \Theta_1(\varsigma)\, d\varsigma \;=\; 0 \quad (1 \le j \le p_K{-}1), \qquad \int_{\mathbb{R}} |\varsigma|^{p_K} \big|\Theta_1(\varsigma)\big|\, d\varsigma \;<\; \infty.$$

(ii) *Boundary adjustment.* There is a measurable **envelope** $\bar{\Theta} : \mathbb{R} \to [0, \infty)$ with $\|\bar{\Theta}\|_\infty < \infty$, $\int_{\mathbb{R}} \bar{\Theta} < \infty$ and $\int_{\mathbb{R}} |\varsigma|^{p_K} \bar{\Theta} < \infty$, such that for every $I \in \mathfrak{I}$ the function Θ_I vanishes off I, satisfies $|\Theta_I| \le \bar{\Theta}$, and

$$\int_I \varsigma^{j}\, \Theta_I(\varsigma)\, d\varsigma \;=\; \delta_{j0} \qquad (0 \le j \le p_K - 1), \tag{15}$$

with δ_{j0} the Kronecker symbol. Write

$$\mathsf{c}_2 \;:=\; \|\bar{\Theta}\|_\infty, \qquad c_C \;:=\; \frac{\|\bar{\Theta}\|_{L^1}}{2\,\mathsf{c}_2}, \qquad \text{so that} \qquad \int_{\mathbb{R}} \bar{\Theta} \;=\; 2\, c_C\, \mathsf{c}_2. \tag{16}$$

(iii) *Regularity at the interior index.* Θ_1 is symmetric and Lipschitz, with Lipschitz constant c_4.

The same three conditions with the same p_K apply to the kernel C_{h_x} used in the x-direction inside $\hat{\mathcal{A}}_{\mathrm{cond}}$ and $\hat{\mathcal{A}}^{*}_{\mathrm{cond}}$, both as defined in Definition 3. *C3 is not vacuous at* $p_K = 8$: Lemma S:bdryfam exhibits a family satisfying (i)–(iii) whose interior member is *exactly* the order-p_K Gaussian-based kernel the application computes.

Notes on C3. What (14) is for, and how clause (ii)'s boundary index and defining property are read, are Remark S:app3read. *Clause* (ii) *asks for* ***an*** *envelope, so* (16) *returns a different pair for each admissible* $\bar{\Theta}$; every result below holds at whichever pair the analyst can verify, and none is sharper for a larger one. Lemma S:bdryfam discharges (ii) at the **smallest** envelope the family admits, the pointwise supremum $\sup_{I \in \mathfrak{I}} |\Theta_I|$, and *its* constants at $p_K = 8$ are the ones reported here: $\mathsf{c}_2 = 34.0$ and $c_C = 0.155$, so that $2 c_C \mathsf{c}_2 = \|\bar{\Theta}\|_{L^1} = 10.56$. *Neither constant enters a rate*; they enter the relaxation parameter (35), with c_2 **squared**, and hence

the cap (36), where the combination is $c_C \mathsf{c}_2^4$. Which further bounds they enter, how a looser envelope moves each, and which of three admissible envelopes a given reading should use are in the Discussion of the assumptions, under *C3*.

Assumption (C5′: Smoothing parameters)**.**

(i) *Residual direction, operator component.* $p_K = 8$ and

$$h_u \asymp \mathrm{sd}(\Delta U)\,\hat{\delta}_n^{1/p_K} = \mathrm{sd}(\Delta U)\,n^{-\alpha/8}, \qquad \alpha := \frac{r}{2r + d_Z} \in \left(0, \tfrac{1}{2}\right),$$

(ii) *x-direction, derivative and adjoint components.* $h_x \le (\bar{x} - \underline{x})/2$ at every n, $h_x \to 0$, and

$$\left(n\, h_x\, h_u\right)^{-1} \ln n \longrightarrow 0.$$

(iii) *Propensity.* $\hat{\varrho}$ satisfies A4(iii) with r obeying the frontier (7).

(iv) *No further smoothing.* No bandwidth is used in t, in ΔZ, or in W.

Notes on C5′. What clause (i)'s display buys, and clause (ii)'s stronger inherited form together with the window on s it leaves against (P-g), are Remark S:app5read.

We ask two things of the differenced residual in A1, and the two halves do different work. The mean-zero half, $E[\Delta U \mid \Delta Z] = 0$, delivers restriction (B) of Definition 7, the unconditional orthogonality $E[\Delta Z\, \Delta U] = 0$; the conditional-independence half, $\Delta U \perp\!\!\!\perp W \mid \Delta Z$, delivers restriction (A). Each implication runs one way, A1 being the stronger statement in both. Only the first identifies $\theta_\dagger$.

We build everything else from the second: the operator equation, the debiasing weight, the Jacobian of the moment, and its orthogonality in two of the three nuisance directions.

Restriction (B) identifies $\theta_\dagger$ on its own. Under A1 and A1b(iv), $\theta_\dagger = \left(E[\Delta Z \Delta Z']\right)^{-1} E[\Delta Z(\Delta Y - \Delta\varphi_\dagger)]$, the inverse existing and the expectations being finite by that same clause. Neither the

source condition A2 nor a rank condition on any other matrix enters. That is worth stating plainly rather than leaving to be inferred: routing identification of the parametric component through a matrix built from the representer would make it contingent on a source condition, and it is not.

Restriction (A) carries no first-order information about $\theta_\dagger$, and that is a theorem rather than a shortcoming of the argument. By Lemma 10 the set of pairs admitted by (A) is a product set, so no functional of the conditional law of the residual given $(W, \Delta Z)$ distinguishes $\theta_\dagger$ from any other value.

It does not follow that $\theta_\dagger$ is unidentified, the display above having identified it, and it does not follow that (A) is idle. A Neyman correction is mean-zero at the truth *by construction*, which is what makes it a correction rather than a second restriction, and that is a property of every Riesz representer in the debiased-machine-learning construction (Chernozhukov, Chetverikov, et al., 2018; Chernozhukov, Newey, and Singh, 2022a; Chernozhukov, Newey, and Singh, 2022b) and not a defect peculiar to this model.

The comparison with Poirier (2017) runs the other way, and the direction is worth stating. He shows that replacing a conditional independence restriction by a conditional moment restriction loses identification; here identification of $\theta_\dagger$ comes from the mean restriction, and conditional independence supplies the debiasing channel instead.

What (A) costs is over-identification. Under an *unconditional* independence assumption the residual at a wrong θ would fail to be independent of W whenever ΔZ and W are dependent, so (A) would deliver a testable restriction on θ. Under A1 as stated, $\theta_\dagger$ is exactly identified by (B) and (A) is a pure debiasing device. Solvability of the operator equation remains testable in the usual nonparametric-instrumental-variables sense; what carries no over-identifying restriction is θ.

What we get from the conditional-independence half and not from the mean-zero half is one identity: $E[\widetilde{W}\,\Xi(\Delta U, \Delta Z)] = 0$ for every integrable Ξ (Lemma L-1(v) of the supplement).

From that identity, the θ-Jacobian of the debiased moment is $-J$ with $J = E[\Delta Z \Delta Z']$ *exactly*, the debiasing summand of (40) being the zero matrix (part (iv) of the moment-validity lemma of the supplement). We do not assume the local rank condition such a moment would ordinarily require, nonsingularity of a matrix built from $\ell_\star$ and therefore statable only under A2: it follows from A1 and A1b(iv).

Where that identity stops is where our difficulty lives, and it belongs beside it. Δh is $\sigma(X_1, X_2)$-measurable and not $\sigma(\Delta U, \Delta Z)$-measurable, so the same tower step does not run in the structural direction: A2 is precisely the assertion that the pairing there equals $-E[\Delta Z\, \Delta h]$ for every $h \in \mathcal{V}$, and were it zero for every h the instrument would not move the regressor at all.

We convert that conditional restriction rather than approximate it.

Lemma 11 carries (A) to $T(\varphi_\dagger, \theta_\dagger) = 0$ in $L^2(\Pi)$. The contrasts $\widetilde{W}$ replace the W-argument by the finite index l and the Bierens index t removes the z-argument, leaving a family indexed by (u, t, l) with u scalar and l running over $\{1, \ldots, \mathsf{L} - 1\}$; the problem is therefore still amenable to the existing Landweber–Fridman machinery, at an enlarged index set and no new curse of dimensionality. B1(iv.d)'s third factor on the reference measure is the whole of the cost: no Bierens-type device is needed in the W direction, because the space of conditionally centred functions of $(W, \Delta Z)$ is finite-dimensional and (1) is a basis for it.

It is also cheaper than it looks. It requires no moment condition whatsoever on ΔY, ΔU or ΔZ, and no Lebesgue density for ΔU or ΔZ. That is, ΔZ may be discrete, ΔU may have atoms, and neither need have a finite mean. That is exactly what boundedness of Φ in B1(i) purchases, and the smoothness and moment conditions of C1, C2′ and A1b(iv) enter only when the operator is differentiated or the parameter is identified. Compactness of $\mathcal{T}$ is sufficient and is not assumed; what the proofs consume is the integrability of B1(iv.c) and B1(v).

The weight is the exponential, and B1(ii) fixes it as a correction rather than a convenience.

Under the textbook condition, with w real-analytic and non-polynomial, the converse direction of the Bierens core lemma of the supplement fails against the maintained componentwise arctan, which is odd. For $w = \cos$ and a two-point V the family sees only the even part of the image measure, and the supplement's remark on the choice of weight exhibits the witness inside the paper's own transform.

That failure reaches Lemma 11 itself, condition (c) then holding whenever the *unconditional* restriction does while the ΔZ-conditional one fails through opposite-signed violations that cancel. Repair is not a matter of strengthening Φ but of adding an intercept and indexing the family over a subset of $\mathbb{R}^{1+p}$. It is Stinchcombe and White (1998, Thm. 3.1), and it is load-bearing rather than a *see also*. We do not adopt it.

Adopting it would change the range space and redo the envelope bookkeeping of B1(iv)–(v) throughout. We maintain $w = \exp$, which is what Bierens (1990, Thm. 1) characterises, and the additional generality is consumed nowhere in the programme.

Identification of $\varphi_\dagger$ separates into a linearised statement and a global one, and we offer a primitive for neither. Proposition 15 characterises the kernel of the linearisation. The obvious primitive would be the transplant of Fève and Florens (2014, Prop. 2.2) to the conditional-independence design, an instrument-completeness condition on $\Lambda \mapsto E[\widetilde{W}\Lambda \mid \Delta U, \Delta Z]$. *That transplant is unavailable, and not because it is strong.* Under conditional exogeneity $\widetilde{W}$ is centred given ΔZ, so the operator annihilates every function that is almost surely a function of $(\Delta U, \Delta Z)$, the constants included; the completeness condition fails at $\Lambda \equiv 1$ at every law the model admits. In Fève and Florens (2014) the restriction is a conditional mean and the operator is $g \mapsto E[g \mid W]$, which does not annihilate constants, and their operator is linear in φ, so injectivity *is* identification. Our design is the harder one twice over: the operator equation is nonlinear, so the linearised and global statements come apart, and the completeness hypothesis that would have delivered the linearised one cannot be posed.

The obstruction is nameable. Membership of $\mathcal{V}_{CI}$ is a family indexed by test functions of

the *observed* residual at the trial value, whereas running the linearised argument globally would need the strictly larger family indexed also by test functions of the *true* residual, about which the restriction says nothing at a trial φ. They coincide at $\varphi = \varphi_\dagger$ and nowhere else. Proposition 9 closes off the obvious repair: $\mathcal{A}_{\text{cond}}$ is compact, hence not bounded below, so A1c does not even deliver a neighbourhood of uniqueness.

We acknowledge that A1b(iii) is assumed rather than derived, and that no primitive for it is offered anywhere in this paper. An earlier draft carried one for the linearised condition and we have withdrawn it: the hypothesis it rested on is empty in this model, for the reason above. *We would rather record a route as closed than leave a corollary standing whose hypotheses nothing satisfies.*

Global identification has been treated directly in a nonlinear instrumental-variables problem with a duration outcome (Beyhum et al., 2023). Conditions of that kind are harder to interpret than the linearised one and are difficult to check on data.

What a finer instrument does to that hypothesis runs the opposite way to the reflex, and Definition 16 and Proposition 17 say what it does instead. A conflict between two cells refutes A1, and where a violation exists the cells in conflict with a maximal consistent set are where the failure sits. That route is available in a design where A1b(iii) is assumed rather than derived, because Proposition 17(ii) carries no identification hypothesis at all.

Two readings have to be blocked, and each is the one a reader arriving from the linear literature will supply. Extra cells do not buy identification: A1b(iii) at a coarsening of W, together with A1, *implies* A1b(iii) at the full instrument, so a finer instrument makes the identification hypothesis weaker and never stronger. Nor do they buy testability or instrument strength, for the reasons Remark 18(5) gives. And the mechanism is not the conditioning of the inverse problem: the degree of ill-posedness is a property of the singular values of $\mathcal{A}_{\text{cond}}$, and the cardinality of W does not act on it in any way this paper can state. What the extra cells add are moment directions, restrictions on the same φ, and the per-cell decomposition

of those directions is the localisation.

Proposition 17(iv) is where the design earns its keep and where the binary case is not merely weaker but empty. One limit belongs with it and is carried by neither statement: consistency of every cell set is necessary for A1 and not sufficient, since the diagnostic detects $\mathcal{V}_{CI} = \emptyset$ and a violation that some other $\varphi \in \mathcal{V}$ reconciles is invisible to it, so a cell showing no conflict is not exonerated.

Proposition 13 is our negative result and it is an equivalence, not an impossibility theorem. Read its condition $\nu_g \equiv 0$ at $g \equiv 1$ and it is marginal stationarity of X; read at $g = \Delta Z$ it is $\nu \equiv 0$, the design in which the debiasing weight degenerates and the moment collapses to the differenced least-squares one.

The condition was reached independently in the linear case. Fève and Florens (2014) observe that their operator annihilates every φ under exchangeability of the two periods given the instrument, which is (30) imposed simultaneously for every bounded $g = g(W)$. What Proposition 13 adds is the one-weight-at-a-time equivalence, theirs being the simultaneous condition and correspondingly stronger, and the enlargement of g from functions of W to functions of $(W, \Delta Z)$, which is the class the debiasing construction actually searches.

The proposition does not assert that (30) *holds or fails.* Whether it fails is a property of the application, testable through the Picard diagnostic and through $\hat{\nu}_g$ directly, and we report it rather than assert it.

A design in which it holds for the weights of interest is one in which there is nothing to debias.

Corollary 14 closes the chain from that equivalence to the shape of the remainder. If the debiasing weight did not vary with the residual, the moment would reduce to a $\sigma(W, \Delta Z)$-measurable weight times the residual, and orthogonality would then force (30). So in any design where (30) fails for such weights the selected $\ell_\star$ must have $\partial_u \ell_\star \not\equiv 0$.

That is what puts a quadratic term in φ into the moment, and it is why the second-order remainder of §4 is unavoidable within this moment class rather than an artefact of the expansion.

Necessity is not sufficiency here. The condition $\partial_u \ell_\star \not\equiv 0$ is necessary and not sufficient for the second-order form to be nonzero: the conditional expectation $q = E[\widetilde{W}\,\partial_u \ell_\star \mid X_1, X_2]$ can still vanish by cancellation, and excluding that is the separate hypothesis O-16 carried explicitly by Proposition 27(iii)–(iv). And we make no claim about whether the moment class is exhausted.

Definition 4 (Model and differencing)**.** The structural model is

$$Y_{ij} \;=\; \varphi_\dagger(X_{ij}) \;+\; Z_{ij}'\theta_\dagger \;+\; \eta_i \;+\; U_{ij}, \qquad j \in \{1,2\}, \tag{17}$$

with $X_{ij} \in \mathcal{X} \subset \mathbb{R}$ scalar and continuously distributed, $Z_{ij} \in \mathbb{R}^p$, $W_i \in \{0, 1, \ldots, \mathsf{L}-1\}$ with $\mathsf{L} \geq 2$ fixed and known, and η_i an individual effect that may depend on (X, Z, W). Differencing across the two periods,

$$\Delta Y_i \;=\; \Delta\varphi_{\dagger i} \;+\; \Delta Z_i'\theta_\dagger \;+\; \Delta U_i, \qquad \Delta\varphi_i := \varphi(X_{i2}) - \varphi(X_{i1}), \tag{18}$$

and, for a trial pair (φ, θ), the residual at trial values is

$$\Delta U_i(\theta, \varphi) \;:=\; \Delta Y_i - \Delta\varphi_i - \Delta Z_i'\theta, \qquad \Delta U_i \;=\; \Delta U_i(\theta_\dagger, \varphi_\dagger). \tag{19}$$

The subscript i is suppressed whenever a statement concerns the population distribution P.

Definition 5 (Spaces, and the domain of the operator)**.** Let π be the reference density on $\mathcal{X}$ of Assumption C1, namely the *pooled design density* of the regressor across the two periods, $\pi := \frac{1}{2}\big(f_{X_1} + f_{X_2}\big)$. *It is an* ***unknown*** *feature of P and not a measure the analyst selects*: the estimator of Definition 22 divides by an *estimate* $\hat{\pi}$ of it. *It is fixed at P once and is*

not re-indexed by a restriction of the sample, which is the reading at which Remark 18(2)'s sub-population transfer is stated. What that declaration costs, what it delivers for free and the one thing it does not settle are Remark S:spacesread(4). Set

$$\mathcal{V} \;:=\; \Big\{\varphi \in L^2(\mathcal{X},\pi) :\; \textstyle\int_{\mathcal{X}} \varphi \, d\pi = 0\Big\}, \qquad L^2(\Pi) \;:=\; L^2\Big(\mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\},\; \Pi\Big),$$

with $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ as in Assumption B1(iv), the base carrying $\mathfrak{B}(\mathbb{R})\otimes\mathfrak{B}(\mathcal{T})\otimes 2^{\{1,\dots,\mathsf{L}-1\}}$, *the third factor is a finite set with its power set*, which is what makes B1(iv.d)'s $\varpi_{\mathsf{w}}(l) := \Pi_{\mathsf{w}}(\{l\})$ well posed and what makes Π-measurability equivalent to measurability of the $\mathsf{L}-1$ sections. The elements of $L^2(\Pi)$ are *scalar* functions of (u,t,l); *and, for an integer* $q \geq 1$, $L^2(\Pi;\mathbb{R}^q)$ *denotes the same space with* $\mathbb{R}^q$*-valued elements*, normed by $\Big(\int \|\cdot\|^2 d\Pi\Big)^{1/2}$ with $\|\cdot\|$ the Euclidean norm on $\mathbb{R}^q$, so that $L^2(\Pi;\mathbb{R}^1) = L^2(\Pi)$. The *range* of the operator is the scalar one, which is why the pairing below is written on scalars; the *representer* ℓ^{Π} of A2′, A2b and A2c is $\mathbb{R}^p$-valued and lives in this space at $q = p$, with $\mathcal{A}^*_{\text{cond}}$ read coordinatewise, as Definition 19 spells out. At $p = 1$ the two spellings are one symbol. Inner products are $\langle h, h'\rangle_{L^2(\pi)} = \int hh'\,\pi\,dx$ and

$$\begin{aligned}\langle \Xi, \Xi'\rangle_{L^2(\Pi)} \;&=\; \int \Xi\,\Xi'\,d\Pi \\ &=\; \sum_{l=1}^{\mathsf{L}-1} \varpi_{\mathsf{w}}(l) \int_{\mathbb{R}\times\mathcal{T}} \Xi(u,t,l)\,\Xi'(u,t,l)\,\Pi_u(du)\,\Pi_t(dt),\end{aligned}$$

where h, h' denote elements of the domain and Ξ, Ξ' elements of the range. This pairing is where the masses $\varpi_{\mathsf{w}}(l)$ of B1(iv.d) act, and every adjoint taken in it carries them. (S:Wdef) *displays* them because $\mathcal{W}$ absorbs what the pairing *supplies*; they are not applied twice. Write $P_{\mathcal{V}}$ for the $L^2(\pi)$-orthogonal projection onto $\mathcal{V}$. *At* $\mathsf{L} = 2$ *the third factor is the one-point set* $\{1\}$ *and* Π_{w} *the point mass at it*, under the canonical identification of $\mathbb{R}\times\mathcal{T}\times\{1\}$ with $\mathbb{R}\times\mathcal{T}$. The period-difference operator is $\Delta : \mathcal{V} \to L^2(P)$, $\Delta h = h(X_2) - h(X_1)$. For an

integrable random variable g write

$$\nu_g(x) \;:=\; E[g \mid X_2 = x]\, f_{X_2}(x) \;-\; E[g \mid X_1 = x]\, f_{X_1}(x), \qquad \nu \;:=\; \nu_{\Delta Z} \in \mathbb{R}^p, \tag{20}$$

the latter understood coordinatewise.

Every operator in this paper has domain $\mathcal{V}$, with one declared exception named below, and the decomposition $L^2(\pi) = \mathcal{V} \oplus \{\text{constants}\}$ is orthogonal.

The exception is the second-order form operator $\mathcal{B}$ of (S:Bpi)*, which is read on all of $L^2(\pi)$.* No other object in the paper leaves $\mathcal{V}$.

Notes on Definition 5. Which letters are the bound variables on each side, why the restriction to $\mathcal{V}$ costs nothing, what governs the sharp constant of the form on the declared exception, and what the declaration of π as an unknown feature of P costs are Remark S:spacesread.

Definition 6 (The operator equation)**.** Let $\Delta Y_\theta := \Delta Y - \Delta Z'\theta$, so that $\Delta U(\theta, \varphi) = \Delta Y_\theta - \Delta\varphi$. For $(\varphi, \theta) \in \mathcal{V} \times \mathbb{R}^p$ define the *Bierens moment family*

$$\mathcal{M}(u, t, l; \varphi, \theta) \;:=\; E\Big[\widetilde{W}_l \,\mathbb{1}\big\{\Delta Y_\theta - \Delta\varphi \le u\big\}\, w\big(t'\Phi(\Delta Z)\big)\Big], \qquad (u, t, l) \in \mathbb{R} \times \mathcal{T} \times \{1, \dots, \mathsf{L}-1\}, \tag{21}$$

with $\widetilde{W}_l = \mathbb{1}\{W = l\} - \varrho_l(\Delta Z)$ the l-th contrast of (1), and let

$$T:\; \mathcal{V} \times \mathbb{R}^p \longrightarrow L^2(\Pi), \qquad T(\varphi, \theta) \;:=\; \big[(u, t, l) \mapsto \mathcal{M}(u, t, l; \varphi, \theta)\big], \tag{22}$$

the contrast index being an argument of $\mathcal{M}$ and not a coordinate of its value, so that $T(\varphi, \theta)$ is a scalar function on the base set of Π fixed in Definition 5. The *operator equation of the model* is

$$T(\varphi, \theta) \;=\; 0 \quad \text{in } L^2(\Pi). \tag{23}$$

We abbreviate $\mathcal{M}(u, t, l) := \mathcal{M}(u, t, l; \varphi_\dagger, \theta_\dagger)$ and $T(\varphi) := T(\varphi, \theta_\dagger)$ when no ambiguity arises.

At $\mathsf{L} = 2$ the index takes the single value 1 and $\widetilde{W}_1 = W - \varrho(\Delta Z)$.

Definition 7 (Admissible sets)**.** Restrictions (A) and (B) of plan §2.1 are

$$\text{(A)} \quad E\Big[\widetilde{W}\,\varsigma(\Delta U)\,\Big|\,\Delta Z\Big] = 0 \text{ a.s. for every bounded measurable } \varsigma, \qquad \text{(B)} \quad E\Big[\Delta Z\,\Delta U\Big] = 0.$$

Define

$$\mathcal{I} \;:=\; \Big\{(\varphi,\theta) \in \mathcal{V}\times\mathbb{R}^p :\; \Delta Y - \Delta\varphi - \Delta Z'\theta \;\perp\!\!\!\perp\; W \,\Big|\, \Delta Z\Big\}, \qquad \mathcal{V}_{CI} \;:=\; \Big\{\varphi \in \mathcal{V} :\; (\varphi,0) \in \mathcal{I}\Big\}.$$

Conditional independence $\xi \perp\!\!\!\perp W \mid \Delta Z$ is understood in the factorisation sense, $P(\xi \in B, W = j \mid \Delta Z) = P(\xi \in B \mid \Delta Z)\,P(W = j \mid \Delta Z)$ a.s. for all Borel B and $j \in \{0, 1, \ldots, \mathsf{L}-1\}$. Two features of this form are used below, and at $\mathsf{L} \geq 3$ they are features of two *different* sets of values of ΔZ. *First*, at a $\Delta Z = z$ at which every W-cell is non-null it is equivalent to the conditional-expectation form $P(\xi \in B \mid \Delta Z, W) = P(\xi \in B \mid \Delta Z)$ a.s., and where some cell is null it is the form that behaves correctly, the factorisation holding at a null cell with both sides zero whatever version is taken there. *Second*, it holds automatically, whatever the conditional law of ξ, exactly where W is *conditionally degenerate*, that is, where $\max_{0\leq l\leq \mathsf{L}-1} \varrho_l(z) = 1$. Both facts are used in the proof of Lemma 11. *Degeneracy is that condition, and a null cell is not by itself degeneracy*: at $\mathsf{L} \geq 3$ a z may have null cells and two or more surviving ones, and there the restriction is not vacuous, it still forces the surviving cells to share one conditional law of ξ. *At* $\mathsf{L} = 2$ *the two conditions coincide*, one null cell leaving one survivor.

Proposition 8 (Fréchet derivative in φ, with a Lipschitz derivative map)**.** *T is Fréchet differentiable in φ with a quadratic remainder, and the derivative map is Lipschitz. Suppose B1(i), (ii), (iv.c), (iv.d), C1(ii) and C2′(i) hold;* B1(iv) is cited by clause*, its factorisation $\Pi = \Pi_u \otimes \Pi_t \otimes \Pi_{\mathsf{w}}$ being a declaration and not a hypothesis. Then, for every $\theta \in \mathbb{R}^p$ and every $\varphi \in \mathcal{V}$, the linear operator $\mathcal{A}_\varphi : \mathcal{V} \to L^2(\Pi)$ defined, at each $(u,t,l) \in \mathbb{R}\times\mathcal{T}\times\{1,\ldots,\mathsf{L}-1\}$,*

by

$$\mathcal{A}_\varphi h(u,t,l) \;=\; E\Big[\widetilde{W}_l\, w\big(t'\Phi(\Delta Z)\big)\, f_{\Delta Y_\theta - \Delta\varphi|X_1,X_2,\Delta Z,W}(u)\,\big\{h(X_2)-h(X_1)\big\}\Big] \tag{24}$$

is bounded, with $\|\mathcal{A}_\varphi\| \le 2\,\bar{c}_\pi^{1/2}\|\bar{f}_0\|_\infty c_\Pi$*, and:*

(i) Fréchet differentiability with a quadratic remainder: *for all* $h \in \mathcal{V}$*,*

$$\Big\|T(\varphi+h,\theta) - T(\varphi,\theta) - \mathcal{A}_\varphi h\Big\|_{L^2(\Pi)} \;\le\; C_{\mathcal{A}}\,\|h\|^2_{L^2(\pi)}, \qquad C_{\mathcal{A}} := 2\,\bar{c}_\pi\,\|\bar{f}_1\|_\infty\, c_\Pi; \tag{25}$$

(ii) Lipschitz derivative map: *for all* $h \in \mathcal{V}$*,*

$$\Big\|\mathcal{A}_{\varphi+h} - \mathcal{A}_\varphi\Big\| \;\le\; 2\,C_{\mathcal{A}}\,\|h\|_{L^2(\pi)}. \tag{26}$$

At the truth we write $\mathcal{A}_{\mathrm{cond}} := \mathcal{A}_{\varphi_\dagger}$ *with* $\theta = \theta_\dagger$*, in which case the conditional density in* (24) *is* $f_{\Delta U|X_1,X_2,\Delta Z,W}$*. At* $\mathsf{L} = 2$ *the index takes the single value* 1 *and* $\widetilde{W}_1 = W - \varrho(\Delta Z)$*.*

Proposition 9 (Compactness)**.** $\mathcal{A}_{\mathrm{cond}}$ *is Hilbert–Schmidt, hence compact. Suppose B1(i), (ii), (iv.c), (iv.d), C1(ii) and C2′(i) hold;* B1(iv) is cited by clause*, its factorisation being a declaration and not a hypothesis. Then* $\mathcal{A}_{\mathrm{cond}}$ *is the integral operator on* $\mathcal{V}$ *with kernel*

$$\mathsf{k}(u,t,l;x) \;=\; \frac{\nu_{g_{u,t,l}}(x)}{\pi(x)}, \qquad g_{u,t,l} \;:=\; \widetilde{W}_l\, w\big(t'\Phi(\Delta Z)\big)\, f_{\Delta U|X_1,X_2,\Delta Z,W}(u), \tag{27}$$

and $\|\mathcal{A}_{\mathrm{cond}}\|_{\mathrm{HS}} \le 2\,\bar{c}_\pi\,\|\bar{f}_0\|_\infty\, c_\Pi < \infty$*. Moreover* $\mathsf{k}(u,t,l;\cdot) \in \mathcal{V}$ *for every* $(u,t,l) \in \mathbb{R}\times\mathcal{T}\times\{1,\dots,\mathsf{L}-1\}$*, so* $\mathcal{A}^*_{\mathrm{cond}}$ *maps* $L^2(\Pi)$ *into* $\mathcal{V}$*; and* $\mathcal{A}_{\mathrm{cond}}$ *admits a singular system* $(\sigma_j; e_j, \tilde{e}_j)_{j\in\mathcal{J}}$*,* $\mathcal{J} \subseteq \mathbb{N}$*, with* $\{e_j\}_{j\in\mathcal{J}}$ *an orthonormal basis of* $(\ker \mathcal{A}_{\mathrm{cond}})^\perp \cap \mathcal{V}$ *and, when* $\mathcal{J}$ *is infinite,* $\sigma_j \downarrow 0$*.* At $\mathsf{L} = 2$ the index takes the single value 1 and $\mathsf{k}(u,t,1;x) = \mathsf{k}(u,t;x)$*, the Hilbert–Schmidt constant included,* Π_{w} *being a probability measure at every* L*.*

Lemma 10 (L1: shift invariance)**.** *Restriction (A) is invariant to the value of* θ*, so it does*

not identify $\theta_\dagger$. The set $\mathcal{I}$ of Definition 7 satisfies, with no assumption beyond that definition,

$$\mathcal{I} \;=\; \mathcal{V}_{CI} \times \mathbb{R}^p; \tag{28}$$

and if in addition A1 holds, then $\varphi_\dagger \in \mathcal{V}_{CI}$, so $(\varphi_\dagger, \theta) \in \mathcal{I}$ for every $\theta \in \mathbb{R}^p$.

Lemma 11 (Bierens equivalence for the panel restriction)**.** *The ΔZ-conditional restriction (A) is equivalent to the vanishing of the unconditional family $\{\mathcal{M}(u,t,l)\}$, so no conditional distribution given ΔZ ever has to be estimated. Suppose B1(i)–(iv) hold. Then the following are equivalent:*

(a) $\Delta U \perp\!\!\!\perp W \mid \Delta Z$*;*

(b) $E[\widetilde{W}\,\varsigma(\Delta U) \mid \Delta Z] = 0$ *a.s. for every bounded measurable ς, that is, restriction* (A)*;*

(c) $\mathcal{M}(u,t,l) = 0$ *for every* $(u,t,l) \in \mathbb{R} \times \mathcal{T} \times \{1,\dots,\mathsf{L}-1\}$*;*

(d) $T(\varphi_\dagger, \theta_\dagger) = 0$ *in* $L^2(\Pi)$*.*

Note on Lemma 11. Clause (b) is the $\mathbb{R}^{\mathsf{L}-1}$-valued identity (1) types it as, hence the $\mathsf{L}-1$ scalar restrictions $E[\widetilde{W}_l\varsigma(\Delta U) \mid \Delta Z] = 0$; that (a) and (b) are *equivalent* at every L, and under **no** overlap condition, is established in the Discussion of the assumptions, under the heading *A1 (conditional exogeneity)*, and is cited and not reproved here.

Corollary 12 (to Lemma 10 and Lemma 11)**.** *The operator equation has the same solution set in φ at every θ, and that set is $\mathcal{V}_{CI}$. Suppose B1(i)–(iv) hold. Then for every $\theta \in \mathbb{R}^p$,*

$$\big\{\varphi \in \mathcal{V} :\; T(\varphi,\theta) = 0 \text{ in } L^2(\Pi)\big\} \;=\; \mathcal{V}_{CI},$$

and under A1b(iii) this set equals $\{\varphi_\dagger\}$.

Proposition 13 (R2: when a weight yields a φ-free moment)**.** *A moment is free of the structural function exactly when the two periods' g-tilted marginals of the endogenous regressor*

coincide. Suppose C1(ii) holds and let g be a random variable satisfying

$$E|g| < \infty, \qquad E\Big|g\,h(X_j)\Big| < \infty \;\; (j = 1, 2) \text{ for every } h \in \mathcal{V}, \qquad E\Big|g\,\Delta U(\theta, 0)\Big| < \infty. \quad (29)$$

Then the map $\varphi \mapsto E\Big[g\,\Delta U(\theta, \varphi)\Big]$ is constant on $\mathcal{V}$, for one and hence every $\theta \in \mathbb{R}^p$, if and only if

$$E[g \mid X_2 = x]\, f_{X_2}(x) \;=\; E[g \mid X_1 = x]\, f_{X_1}(x) \qquad \text{for almost every } x \in \mathcal{X}, \qquad (30)$$

that is, if and only if $\nu_g \equiv 0$.

Corollary 14 (to Proposition 13: the debiasing weight must vary with the residual)**.** *If the debiasing weight does not depend on the residual, the debiased moment is not merely first-order insensitive to the structural function but exactly free of it. Suppose A1, A1b(i), A1b(iv) and C1(ii) hold, let L be differentiable in u* at every *(u, z) with $\partial_u L(u, z) = c(z)$ for some Borel $c : \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$, and write*

$$c(z)\,\widetilde{W} \;:=\; \sum_{l=1}^{\mathsf{L}-1} \widetilde{W}_l\, c_{\cdot l}(z) \;\in\; \mathbb{R}^p$$

for the Euclidean contraction of the $\mathsf{L}-1$ columns of c against the contrast vector. Suppose $g := \Delta Z + c(\Delta Z)\widetilde{W}$ satisfies (29) coordinatewise, and suppose L satisfies N1. Then $L(u, z) = c(z)\,u$, the moment function is $\psi = g\,\Delta U(\theta, \varphi)$ with g measurable with respect to $\sigma(W, \Delta Z)$, and the following are equivalent:

(a) $E[\psi]$ is first-order insensitive to φ at $\varphi_\dagger$, i.e. $\nu_{\Delta Z + c\widetilde{W}} \equiv 0$;

(b) $\varphi \mapsto E[\psi]$ is constant on $\mathcal{V}$;

(c) $\nu_{c\widetilde{W}} = -\nu$ for the *c* of the hypothesis*; equivalently, membership $-\nu \in \mathcal{N}_0 := \{\nu_{\tilde{c}(\Delta Z)\widetilde{W}} : \tilde{c} \in \mathcal{C}_0\}$ is realised* at that *c, where $\mathcal{C}_0$ is the set of $\tilde{c}$ for which $\Delta Z + \tilde{c}(\Delta Z)\widetilde{W}$ satisfies (29) coordinatewise.*

Moreover $\varphi \mapsto E[\psi]$ is affine on $\mathcal{V}$, so every term of order two or higher in its expansion vanishes identically whether or not (a)–(c) hold*; what* (a)–(c) *add is that the first-order term vanishes as well. Contrapositively, if $-\nu \notin \mathcal{N}_0$, then every $\ell_\star$ satisfying $\nu_{\Delta Z + \ell_\star \widetilde{W}} \equiv 0$ has $\partial_u \ell_\star \not\equiv 0$. At $\mathsf{L} = 2$ there is one column, $c(z) \in \mathbb{R}^p$, $\widetilde{W}$ is the scalar $W - \varrho(\Delta Z)$, and $c\widetilde{W} = \widetilde{W}c$.*

Proposition 15 (Characterisation of the kernel of the linearisation)**.** *A direction is annihilated by the linearised operator exactly when the instrument fails to move its period difference, given the residual and the controls. Suppose B1(i)–(iv), C1(ii) and C2′(i) hold, and let $h \in \mathcal{V}$. Then*

(i) the following are equivalent: (a) $\mathcal{A}_{\mathrm{cond}} h = 0$ *in* $L^2(\Pi)$*;* (b) $E\big[\widetilde{W}\,\{h(X_2) - h(X_1)\} \,\big|\, \Delta U, \Delta Z\big] = 0$ *almost surely;*

(ii) if in addition A1 and A4(i) hold, (b) *is equivalent to* (c) $E\big[h(X_2) - h(X_1) \mid \Delta U, \Delta Z, W = l\big] = E\big[h(X_2) - h(X_1) \mid \Delta U, \Delta Z\big]$ *for every* $l \in \{0, 1, \ldots, \mathsf{L} - 1\}$*, almost surely on* $\{\varrho_l(\Delta Z) > 0\}$.

Notes on Proposition 15. Clause (b) is the $\mathbb{R}^{\mathsf{L}-1}$-valued identity (1) types it as, hence the $\mathsf{L}-1$ scalar restrictions $E[\widetilde{W}_l\{h(X_2) - h(X_1)\} \mid \Delta U, \Delta Z] = 0$. *Clause* (c) *compares each cell against the expectation given $(\Delta U, \Delta Z)$ alone, and not the cells against one another*, which is the comparison the companion treatment of discrete instruments, Centorrino, Fève, and Florens (2025), is stated at; it names no base cell and is well posed where a cell is null, on which it says nothing, exactly as the factorisation form of Definition 7 does. *The two readings coincide at every* L: under A1, $P(W = l \mid \Delta U, \Delta Z) = \varrho_l(\Delta Z)$, so $E[h(X_2) - h(X_1) \mid \Delta U, \Delta Z]$ is the ϱ-weighted average of the cell expectations over the non-null cells, and all cells agreeing with one another and all cells agreeing with that average are the same condition. The quantifier over l runs over the fixed set $\{0, \ldots, \mathsf{L}-1\}$ and the qualification is on the **event** $\{\varrho_l(\Delta Z) > 0\}$, so no index set depends on the data; *under* (ii)*'s own hypotheses that qualification is vacuous*, A4(i) making each of those events almost sure, and it is carried only so that (c) remains a

statement if A4(i) is relaxed.

Definition 16 (Per-subset restriction sets)**.** For a set of instrument cells $\mathsf{S} \subseteq \{0, 1, \ldots, \mathsf{L}-1\}$ put

$$\mathcal{V}_{\mathsf{S}} := \Big\{\varphi \in \mathcal{V} : \varrho_k(\Delta Z)\, P\big(\Delta Y - \Delta\varphi \in B,\ W = l \mid \Delta Z\big) = \varrho_l(\Delta Z)\, P\big(\Delta Y - \Delta\varphi \in B,\ W = k \mid \Delta Z\big)\Big\},$$

the identity holding, *for each Borel B and each pair $l, k \in \mathsf{S}$*, almost surely; by right-continuity in the endpoint of a half-line and a π–λ argument a single null set serves every B and every pair, so the stronger simultaneous reading defines the same set. Here $\varrho_0 = 1 - \mathbf{1}'\varrho$ as in (1). *The restriction is written in the factorisation sense of Definition 7 and in cross-multiplied form*, so that no conditional law at a null cell is ever named: where every cell of S is non-null it says that $\Delta Y - \Delta\varphi$ has *one* conditional law given ΔZ across the cells of S, and where a cell of S is null it says nothing about that cell. *The set is unchanged if $\Delta Y - \Delta\varphi$ is replaced by $\Delta U(\theta, \varphi)$ of (19) at any $\theta \in \mathbb{R}^p$*, the two differing by a ΔZ-measurable shift; this is the invariance Lemma 10 records for $\mathcal{I}$, read at a set of cells. Call S *consistent* if $\mathcal{V}_{\mathsf{S}} \neq \emptyset$; and, for a consistent S, call

$$\Big\{l \in \{0, 1, \ldots, \mathsf{L}-1\} \setminus \mathsf{S} :\ \mathcal{V}_{\mathsf{S}\cup\{l\}} = \emptyset\Big\}$$

the set of cells *in conflict with* S. *Sans-serif S is a set of instrument cells and nothing else*, on the pattern of L: the subscript of $\mathcal{V}_{\mathsf{S}}$ is always a set of cells and never an integer, which separates it from the source-condition ball $\mathcal{V}_n$ of (S:srcball) and from $\mathcal{V}_{CI}$, whose subscript is a word.

Proposition 17 (Localisation of a violation to a set of cells)**.** *Extra instrument cells localise a failure of conditional exogeneity to a set of cells, and do so without any identification hypothesis on any subset of cells. Suppose B1(i)–(iv) hold, and read $\mathcal{V}_{\mathsf{S}}$,* consistent *and* in conflict with *as in Definition 16. Then*

(i) Structure of the family. $\mathcal{V}_{\mathsf{S}'} \subseteq \mathcal{V}_{\mathsf{S}}$ *whenever* $\mathsf{S} \subseteq \mathsf{S}'$*;* $\mathcal{V}_{\mathsf{S}} = \mathcal{V}$ *whenever* $\#\mathsf{S} \leq 1$*; and*

$$\mathcal{V}_{\{0,1,\ldots,\mathsf{L}-1\}} \;=\; \mathcal{V}_{CI} \;=\; \left\{\varphi \in \mathcal{V} :\; T(\varphi,\theta) = 0 \text{ in } L^2(\Pi)\right\} \qquad \text{for every } \theta \in \mathbb{R}^p.$$

B1(i)–(iv) is consumed by the second equality of this clause, through Corollary 12, and by nothing else in this proposition*, so localisation does not rest on the Bierens apparatus.*

(ii) Joint consistency. *If in addition A1 holds, then* $\varphi_\dagger \in \mathcal{V}_{\mathsf{S}}$ *for every* S*, so every set of cells is consistent.* This clause uses no identification hypothesis*: A1b(iii) is a hypothesis of no part of this proposition. Contrapositively, if some* S *is not consistent then A1 fails.*

(iii) The conflict set, and what it cannot be freed from. Consistency is inherited by subsets, and every S *with* $\#\mathsf{S} \leq 1$ *is consistent. The cells in conflict with* S *are among those in conflict with* S' *whenever* $\mathsf{S} \subseteq \mathsf{S}'$ *are both consistent; at a maximal consistent* $\mathsf{S}^\star$ *they are exactly* $\{0, 1, \ldots, \mathsf{L}-1\} \setminus \mathsf{S}^\star$*; and some consistent* S *has a non-empty conflict set if and only if* $\mathcal{V}_{CI} = \emptyset$*. Moreover there is exactly one maximal consistent set of cells if and only if* $\mathcal{V}_{CI} \neq \emptyset$*, and no cell is in conflict with* every *consistent* S*. So wherever there is a violation to attribute, the attribution is relative to a reference set of cells. What* P *determines is the* family *of maximal consistent sets, which is a functional of* P *and is identified; what it does not determine is which member of that family is the reference set, and any rule for choosing one is a convention.*

(iv) Localisation. *Suppose* $\mathcal{V}_{CI} = \emptyset$ *while* $\mathcal{V}_{\mathsf{S}_0} \neq \emptyset$ *for some* S_0 *with* $\#\mathsf{S}_0 \geq 2$*. Then there is a maximal consistent* $\mathsf{S}^\star \supseteq \mathsf{S}_0$*, necessarily with* $\#\mathsf{S}^\star \geq 2$*, and the cells in conflict with it form a non-empty* proper *subset of* $\{0, 1, \ldots, \mathsf{L}-1\}$ *omitting at least two cells. The hypothesis of this clause is unsatisfiable at* $\mathsf{L} = 2$*.*

At $\mathsf{L} = 2$ clauses (i)–(iii) are true and say nothing beyond Corollary 12*: under a violation the maximal consistent sets are* $\{0\}$ *and* $\{1\}$*, the cells in conflict with them are* $\{1\}$ *and* $\{0\}$*, so*

the verdict is that the restriction fails and it names no channel. Clause (iv) is **vacuous** at $\mathsf{L} = 2$*, its hypothesis being unsatisfiable there, so the* $\mathsf{L} = 2$ *reading of this proposition is a vacuity and not the verbatim reduction the statements around it carry.*

Remark 18 (The slices of T, the sub-population reading, and the statistic).

(1) The slices of T carry joint consistency and not localisation. Write $\mathcal{V}_{CI}(g(W))$ for the set of Definition 7 computed at the instrument $g(W)$ in place of W, so that $\mathcal{V}_{CI} = \mathcal{V}_{CI}(W)$. Under B1(i)–(iii), for each $l \in \{1, \ldots, \mathsf{L}-1\}$ the set of $\varphi \in \mathcal{V}$ at which the l-th slice of $T(\cdot, \theta_\dagger)$ vanishes at every $(u,t) \in \mathbb{R} \times \mathcal{T}$ is exactly $\mathcal{V}_{CI}(\mathbb{1}\{W = l\})$, the solution set at the *binary coarsening* that separates cell l from the rest; and the intersection of those $\mathsf{L}-1$ sets is $\mathcal{V}_{CI}$. *So the slices carry Proposition 17*(ii) *exactly*, and what they do not carry is localisation: a coarsening imposes fewer restrictions, $\mathcal{V}_{CI}(W) \subseteq \mathcal{V}_{CI}(g(W))$, and slice l compares cell l against all remaining cells *pooled*, so one deviant cell moves the pooled benchmark and with it every slice. *The same two readings meet at Proposition 15*(ii)(c)*, where they coincide*, there the quantifier runs over all L cells at once, and the Notes on that proposition show cell-against-average and cell-against-cell are then the same condition. They part company only at a *proper* subset of cells, which is the object Definition 16 supplies and the slices do not.

(2) Sub-population reading, and exactly what transfers with it. Under A4(i), for $\#\mathsf{S} \geq 2$ the set $\mathcal{V}_{\mathsf{S}}$ is $\mathcal{V}_{CI}$ of Definition 7 computed under the law $Q := P(\cdot \mid W \in \mathsf{S})$, the same restriction, on the sub-population on which the instrument takes its values in S. *No new operator and no normalisation enters*, and the propensities it induces are $\varrho_l / \varrho_{\mathsf{S}}$ with $\varrho_{\mathsf{S}} := \sum_{k \in \mathsf{S}} \varrho_k$. Whether the maintained conditions survive the passage to Q is a separate question from the set identity, and it is answered here rather than assumed. A0 survives: $\{i : W_i \in \mathsf{S}\}$ is an i.i.d. sample from Q. A4(i) survives with the *same* ε_ϱ, since $\varrho_l / \varrho_{\mathsf{S}} \geq \varrho_l \geq \varepsilon_\varrho$ and $\varepsilon_\varrho < 1/\mathsf{L} \leq 1/\#\mathsf{S}$. B1 survives unchanged, being a set of choices made by the analyst and not a condition on P. C1(ii)'s *upper* bounds

survive with $\bar{c}_\pi$ inflated to $\bar{c}_\pi / P(W \in \mathsf{S})$. *Two things do not survive, and neither is repaired here*: C1(ii)'s *lower* bounds **on the X-densities** f_{X_1}, f_{X_2} and f_{X_1,X_2}, because A4(i) bounds ϱ_l given ΔZ and says nothing about $P(W \in \mathsf{S} \mid X_j)$, which is what a density of X_j under Q is bounded below by; and the source conditions A2, A2′ and H1, which are conditions on the linearisation the law induces, and at Q that is a different operator. *Only the second of the two binds anything*: those three lower bounds are consumed by no result in this paper, whereas A2, A2′ and H1 carry the whole rate theory, which therefore has no counterpart at Q until it is posed there. *The narrowing to the X-densities is deliberate*: C1(ii) also bounds π below, and *that* bound is consumed — at Proposition S:hatAprop(i), at (35) and hence at (36), at Lemma S:aerr and at Lemma S:levels(i). *Nothing is claimed here about π.*

(3) The statistic is apparatus and is subordinate. Proposition 17 is a statement about population sets. The corresponding procedure is a J-type one, the letter is used as a name and not as a symbol, italic J being the Jacobian of (40): for a candidate reference set S, solve for φ on the sub-population of clause (2), evaluate at that solution the criterion of the cells outside S, and read a value large relative to its own noise level as evidence that the cell is in conflict with S. *Its construction, its rate under the null and its limit law are not established here*, and clause (iii) of Proposition 17 bounds what any such procedure can deliver: the output is a family of maximal consistent sets of cells, and a selection among them, the largest, or the one containing a designated reference cell, is a convention and not an identified object. *And if S is itself chosen from the data, a conflict set read off it is a post-selection object*, whose coverage is not the coverage of the same set read off a reference set fixed in advance; nothing here prices that.

(4) *Base invariance.* $\mathcal{V}_\mathsf{S}$ is symmetric in the cells of S and names no base cell, so the family, the consistent sets and the conflict sets are all invariant to which cell (1) omits. This is the invariance established in the Discussion of the assumptions, under the heading *A1*

(conditional exogeneity), and it is cited and not reproved here.

(5) *What is **not** claimed.* Neither Proposition 17 nor this remark asserts that extra cells make A1 testable or that they make the instrument stronger. A1 is testable at $\mathsf{L} = 2$, the restriction is a continuum of equations against one unknown function, and splitting an instrument creates no variation. The separation of testability, instrument strength and localisation is argued in that same Discussion paragraph.

(6) *Two constructions establish that the hypothesis of Proposition 17*(iv) *is not empty*, and they are in the online supplement, under the heading *Two constructions for the hypothesis of Proposition 17*(iv). The first is a witness at $\mathsf{L} = 3$ with an *irrelevant* instrument: W independent of (X_1, X_2), so there is no first stage, $\mathcal{V}_{\{0,1\}} = \mathcal{V}$ precisely because nothing is identified, and $\mathcal{V}_{CI} = \emptyset$ all the same — which isolates the fact that emptiness of $\mathcal{V}_{CI}$ is settled at the law and not by an identification hypothesis. *The companion treatment of discrete instruments says what that regime is*: Centorrino, Fève, and Florens (2025, Remark 2) records that when the residual is independent of the regressor and the instrument, the relevance condition of that paper reverts to ordinary completeness, which a purely discrete instrument cannot satisfy, and the structural function is then only *partially* identified. The second is a worked example at $\mathsf{L} = 3$ with a **relevant** instrument — relevant in all three senses the question has — in which $\varphi_\dagger \in \mathcal{V}_{\{0,1\}}$ while $\mathcal{V}_{\{0,2\}} = \emptyset$, so that $\{0, 1\}$ is a maximal consistent set and the cells in conflict with it are exactly $\{2\}$; the same construction at $\mathsf{L} = 4$ puts them at exactly $\{2, 3\}$, which is the form the application's chronic-against-transient hypothesis takes. *The hypothesis of clause* (iv) *is maintained nowhere in this paper*: it is the configuration the procedure of clause (3) exists to detect, and its standing is that of the conditional-relevance condition A1b(iii), the conditional analogue of Centorrino, Fève, and Florens (2025, Assumption 2.2) — a condition on the law whose plausibility is discussed and whose scope is bounded, not a theorem. An example illustrates such a

hypothesis and is not asked to exhaust it.

Definition 19 (Representers, weights, and the anchor)**.** Let $\ell^{\Pi} \in L^2(\Pi; \mathbb{R}^p)$ and $\ell := \mathcal{W}\ell^{\Pi}$ with $\mathcal{W}$ as in (S:Wdef), applied coordinatewise in the p coordinates of θ and *columnwise in* l: column l of the $\mathbb{R}^{p\times(\mathsf{L}-1)}$-valued ℓ is generated by the slice $\ell^{\Pi}(\cdot,\cdot,l)$ and by no other slice, so the $\mathsf{L}-1$ contrasts carry $\mathsf{L}-1$ *separately choosable* generators — separately choosable, not linearly independent: nothing prevents two slices coinciding, and localisation needs only the former. *This is the reading A2′(i) and A2b fix, not a further restriction.*

(a) ℓ^{Π} is an *admissible representer* if it satisfies **A2b**. *That tag, and not A2′(i), is what this clause names*: A2′(i) is a *maintained assertion* about $\ell^{\Pi}_{\star}$; A2b is a *property* a candidate representer may or may not have, and it is the property every result quantifying over representers imposes. Under B1(i), (ii), (iv.a) and (iv.d), Corollary S:elldiff, itself stated at a generic admissible representer, applies to ℓ entry by entry: each entry is bounded and continuously differentiable in u, uniformly in z, with budgets $\Lambda^{(0)}, \Lambda^{(1)}$ aggregated as A2b aggregates them, which therefore bound the Euclidean norm of a whole column and hence *a fortiori* every entry.

(b) For a Borel $b : \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$, the *weight generated by* ℓ *and anchored at* b is

$$L^b_\ell(u,z) \;:=\; \int_0^u \ell(v,z)\,dv \;+\; b(z), \qquad (u,z) \in \mathbb{R}\times\mathbb{R}^p, \tag{31}$$

the integral taken entrywise and well defined because each entry of $\ell(\cdot,z)$ is continuous and bounded. It is the unique function that is differentiable in u at *every* point with $\partial_u L^b_\ell = \ell$ and $L^b_\ell(0,\cdot) = b$. The weight is *admissible* if $E\|b(\Delta Z)\|^2 < \infty$, the norm being Frobenius.

(c) ℓ^{Π} is an *admissible selection* if it is an admissible representer and solves $\mathcal{A}^*_{\mathrm{cond}}\ell^{\Pi} = -\nu/\pi$.

Under A2 and A2′(i), $\ell^{\Pi}_{\star}$ is an admissible selection, A2 gives that it solves the adjoint equation, A2′(i) that it satisfies A2b, with $\ell_\star = \mathcal{W}\ell^{\Pi}_{\star}$, and $L_\star := L^{b_\star}_{\ell_\star}$ is the admissible weight fixed by

A2′(iii). Write $\zeta_\star := (\varphi_\dagger, L_\star, \varrho)$ for the nuisance triple at the truth.

Proposition 20 (Neyman orthogonality)**.** *The debiased moment is orthogonal in all three nuisances: in the structural function exactly when the debiasing weight solves the adjoint equation, and in the weight and the propensity exactly, for every admissible direction. Suppose A1b(i), A1b(iv), B1(i), (ii), (iv), C1(ii) and C2′(i) hold. Then*

(i) Structural function, at an arbitrary admissible representer. Let ℓ^Π be an admissible representer in the sense of A2b, $\ell := \mathcal{W}\ell^\Pi$ with $\mathcal{W}$ as in (S:Wdef)*, and L an admissible weight generated by ℓ in the sense of Definition 19. Write $\ell\widetilde{W} := \sum_{l=1}^{\mathsf{L}-1} \widetilde{W}_l\, \ell_{\cdot l}(\Delta U, \Delta Z) \in \mathbb{R}^p$ for the contraction of the $\mathsf{L}-1$ columns of ℓ against the contrast vector. For $h \in \mathcal{V}$ define*

$$\begin{aligned} \Psi_L(\tau_1) \;&:=\; E\Big[\psi\Big(\theta_\dagger, (\varphi_\dagger + \tau_1 h,\; L,\; \varrho)\Big)\Big] \\ &=\; E\Big[\Delta Z\,(\Delta U - \tau_1 \Delta h) + L\Big(\Delta U - \tau_1 \Delta h, \Delta Z\Big)\,\widetilde{W}\Big], \qquad \tau_1 \in \mathbb{R}, \end{aligned} \tag{32}$$

which is finite for every $\tau_1 \in \mathbb{R}$*. At $L = L_\star$, $\Psi_{L_\star}(\tau_1) = \Psi(\tau_1, 0, 0)$ in the notation of Lemma* S:secondord*, so the two agree wherever both are defined. Then for every $h \in \mathcal{V}$, $\partial_{\tau_1}\Psi_L(0) = -\big\langle \nu_{\Delta Z + \ell\widetilde{W}}/\pi,\, h\big\rangle_{L^2(\pi)}$, and this vanishes for every $h \in \mathcal{V}$ if and only if*

$$\nu_{\Delta Z + \ell\widetilde{W}} \equiv 0 \qquad \Longleftrightarrow \qquad \mathcal{A}^*_{\mathrm{cond}}\ell^\Pi = -\nu/\pi. \tag{33}$$

The two sides of (33) are one condition written before and after routing through $\mathcal{W}$: *$\ell\widetilde{W}$ is the combination $\widetilde{W}'(\mathcal{W}\ell^\Pi)$ of* (S:adjclosed)*, read coordinatewise in the p coordinates of θ, so the masses $\varpi_{\mathsf{w}}(l)$ of B1(iv.d) enter only inside $\mathcal{W}$ and ψ carries the raw contrast on both sides.* In particular, under A2′ the left-hand condition holds at $\ell = \ell_\star$, *and $\partial_{\tau_1}\Psi(0) = 0$ for every $h \in \mathcal{V}$.*

(ii) Debiasing weight, exactly. *Assume in addition A1, A2, A2′(i) and A2′(iii). For every Borel $D : \mathbb{R} \times \mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$ with $E[\|D\| \mid \Delta Z] < \infty$ almost surely and $E\|D\widetilde{W}\| < \infty$,*

and every $\tau_2 \in \mathbb{R}$, $E[\psi(\theta_\dagger, (\varphi_\dagger, L_\star + \tau_2 D, \varrho))] = E[\psi(\theta_\dagger, \zeta_\star)]$.

(iii) Propensity: exact orthogonality is equivalent to N1. Assume in addition A2, A2′(i), A2′(iii) and A4(i), and do not assume *N1, nor A1. Then, for every bounded Borel* $\mathsf{r} : \mathbb{R}^p \to \mathbb{R}^{\mathsf{L}-1}$ *with* $\varrho + \tau_3 \mathsf{r}$ *taking values in the simplex* $\{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \geq 0$ *for every* l, $\mathbf{1}'q \leq 1\}$ *for all small* τ_3*, and every such* τ_3*,*

$$E\Big[\psi(\theta_\dagger, (\varphi_\dagger, L_\star, \varrho + \tau_3 \mathsf{r}))\Big] = E\Big[\psi(\theta_\dagger, \zeta_\star)\Big] \qquad \Longleftrightarrow \qquad E\Big[L_\star(\Delta U, \Delta Z) \mid \Delta Z\Big] = 0 \text{ a.s.},$$

the right-hand condition being exactly N1, both sides being well defined.

At $\mathsf{L} = 2$, L, ℓ *and* D *have one column,* r *is scalar, the three products against* $\widetilde{W}$ *commute, and the simplex of clause* (iii) *is the interval* $[0, 1]$.

Notes on Proposition 20. Why A1 is not in the global list, why N1 is assumed in no part, and what the finiteness of (32) and the well-posedness of clause (iii)'s two sides rest on are Remark S:orthoghyp.

Corollary 21 (to Proposition 20(iii): the arity of N1, and what a contracted centring leaves unpriced)**.** *N1 is a condition on a matrix and therefore* $p(\mathsf{L} - 1)$ *scalar conditions, not* p*: centring one contraction of the columns is strictly weaker at every* $\mathsf{L} \geq 3$*, it fails inside the class of weights this paper admits, and what it leaves uncentred enters the propensity direction at the full propensity rate. Suppose A1, A1b(iv), A2, A2′(i), A2′(iii), A4(i) and B1(i), (ii), (iv.a), (iv.d) hold, and fix* $\mathsf{v} \in \mathbb{R}^{\mathsf{L}-1}$ *with* $\mathsf{v} \neq 0$. Do not assume *N1.* The four B1 clauses are the $\|\ell_\star\|_\infty$ bridge*: the finiteness* (34) *requires runs through the sup-norm bound on* $\ell_\star$*, which is reached from A2′(i) under exactly those four. They are a proper part of Proposition 20's own line, so they add no restriction on* P *that this corollary's parent does not already carry. Then*

(i) Arity, and the failure of the converse inside the admissible class. *N1 at* $L_\star$*, namely* $E\Big[L_\star(\Delta U, \Delta Z) \mid \Delta Z\Big] = 0$ *almost surely, is* $p(\mathsf{L} - 1)$ *scalar conditions, one per entry*

$(j,l) \in \{1,\dots,p\} \times \{1,\dots,\mathsf{L}-1\}$, *the zero being the zero matrix of* $\mathbb{R}^{p\times(\mathsf{L}-1)}$. *It implies the* p *contracted conditions* $E\big[L_\star \mid \Delta Z\big]\,\mathsf{v} = 0$ *almost surely.* The converse fails at every $\mathsf{L} \geq 3$, and it fails at a weight the paper's own hypotheses admit. *Let* $\mathsf{v}^\perp \in \mathbb{R}^{\mathsf{L}-1}$ *satisfy* $\mathsf{v}^\perp \neq 0$ *and* $(\mathsf{v}^\perp)'\mathsf{v} = 0$, *which is possible if and only if* $\mathsf{L}-1 \geq 2$, *let* b_{N1} *be the N1 anchor of Remark* S:anchor *at* $\ell_\star$, *and let* $\mathsf{f} : \mathbb{R}^p \to \mathbb{R}^p$ *be Borel with* $E\|\mathsf{f}(\Delta Z)\|^2 < \infty$ *and* $P\big(\mathsf{f}(\Delta Z) \neq 0\big) > 0$. *Then* $L := L^b_{\ell_\star}$ *with* $b(z) := b_{\mathrm{N1}}(z) + \mathsf{f}(z)\,(\mathsf{v}^\perp)'$, *a Borel map* $\mathbb{R}^p \to \mathbb{R}^{p\times(\mathsf{L}-1)}$ — the **outer** product of the $\mathbb{R}^p$-valued $\mathsf{f}(z)$ with the $\mathbb{R}^{\mathsf{L}-1}$-valued $\mathsf{v}^\perp$, and not f evaluated at $\mathsf{v}^\perp$, which is not even typed — *is, in the sense of Definition 19*(b), *an admissible weight with* $\partial_u L = \ell_\star$, *it satisfies the contracted condition at* v, *and it violates N1 on an event of positive probability.*

(ii) What the gap costs at the population propensity. *For every bounded Borel* $\mathsf{r} : \mathbb{R}^p \to \mathbb{R}^{\mathsf{L}-1}$ *admissible in the sense of Proposition 20*(iii),

$$E\Big[L_\star(\Delta U, \Delta Z)\,\mathsf{r}(\Delta Z)\Big] \;=\; E\Big[E\Big[L_\star \mid \Delta Z\Big]\,\mathsf{r}(\Delta Z)\Big],$$

the $\sigma(\Delta Z)$-measurable factor standing on the right, *so the derivative in the propensity reads the* whole *matrix and not its contraction against* v. *If* $P\big(E[L_\star \mid \Delta Z] \neq 0\big) > 0$ *there is such an* r *making it non-zero, and it may be taken of the entrywise form* $\mathsf{r} = \varepsilon_\varrho \operatorname{sign}\Big(\big(E[L_\star \mid \Delta Z]\big)_{jl}\Big)\,e_l$ *with* $e_l \in \mathbb{R}^{\mathsf{L}-1}$ *the* l*-th standard basis vector, at any entry* (j,l) *on which* $E[L_\star \mid \Delta Z]$ *is non-zero with positive probability.*

(iii) What it costs at the estimated propensity, and why it does not vanish. *For every bounded Borel* r *as in* (ii),

$$\Big\|E\Big[L_\star(\Delta U, \Delta Z)\,\mathsf{r}(\Delta Z)\Big]\Big\| \;\leq\; \Big(E\Big\|E[L_\star \mid \Delta Z]\Big\|^2\Big)^{1/2}\,\Big(E\|\mathsf{r}(\Delta Z)\|^2\Big)^{1/2}, \tag{34}$$

the first norm Frobenius and the second Euclidean. If in addition $\liminf_n \Big(E\|\mathsf{r}\|^2\Big)^{-1/2}\Big\|E[L_\star \mathsf{r}]\Big\| > 0$ *along a sequence of such directions, the left side is of* exact *order* $\Big(E\|\mathsf{r}\|^2\Big)^{1/2}$. *Read*

at $\mathsf{r} = \hat{\varrho} - \varrho$ — conditionally on $\sigma(\mathcal{I}_a)$, on which $\hat{\varrho}$ is frozen, so that the clause is applied at a **bounded Borel** r as it is stated and not at a random one*; boundedness holds because* (11) *confines both* $\hat{\varrho}$ *and* ϱ *to the* ε_ϱ*-interior simplex, and the expectation in* (34) *is then over a* $(\Delta U, \Delta Z)$ *draw independent of* $\mathcal{I}_a$*, by Definition 2's sample split — A4(iii) makes that order* $r_n = n^{-\alpha}$ *with* $\alpha = r/(2r + d_Z)$*, so after the* $\sqrt{n}$ *scaling of Theorem 29 the contribution is of order* $n^{1/2-\alpha}$, which diverges*: A4(ii) gives* $r > 0$ *and* $d_Z \geq 1$*, so* $\alpha = r/(2r + d_Z) < \frac{1}{2}$*, the range C5′(i) also records.*

At $\mathsf{L} = 2$ the whole statement collapses to the printed one*: there is one column,* v *is a non-zero scalar, no* $\mathsf{v}^\perp \neq 0$ *exists, so clause* (i)*'s witness cannot be built, and N1 and the contracted condition are the same p conditions.*

3 Estimation

Recall that the differenced model is $\Delta Y = \Delta\varphi_\dagger + \Delta Z'\theta_\dagger + \Delta U$, with $\varphi_\dagger$ the unknown function and W taking values in $\{0, 1, \ldots, \mathsf{L} - 1\}$. We estimate $\varphi_\dagger$ in a φ-step, which solves an ill-posed inverse problem by regularisation and returns a rate slower than $n^{-1/2}$; the θ-step is a cross-fitted one-step on a moment built to be insensitive to that error to first order. The multinomial first stage fits the $\mathsf{L} - 1$ propensities $\varrho_1, \ldots, \varrho_{\mathsf{L}-1}$, which is why A0 asks that L be known: the analyst has to know how many functions to fit. A4(i)'s overlap floor ε_ϱ lies in $(0, 1/\mathsf{L})$ and is imposed at every cell including the base one, so it tightens mechanically as cells are added, and (6) converts it into $\lambda_{\min}(\Omega(\Delta Z)) \geq \varepsilon_\varrho/\mathsf{L}$.

One requirement of the assumption block is a restriction on how the analyst writes ΔZ, and it is stated here because it is invisible where it is imposed. A4(ii) asks that the number d_Z of non-degenerate coordinates of ΔZ satisfy $1 \leq d_Z \leq p - 1$, degenerate meaning almost surely equal to a constant. Read against A1b(iv) the upper bound is sharper than it looks. It says at least one coordinate of ΔZ is almost surely constant; that constant cannot be zero, since a zero coordinate makes a row and a column of $E[\Delta Z \Delta Z']$ vanish and A1b(iv) fails; and there

cannot be two of them, since two non-zero constant coordinates make two rows of that matrix proportional and A1b(iv) fails again. So the two clauses together force $d_Z = p - 1$ exactly, and force ΔZ to carry exactly one constant coordinate, non-zero: *the differenced equation must include a period effect.* What is not required is that the period effect be non-zero in the population, its coefficient being an ordinary component of $\theta_\dagger$ that may be zero; what is required is that the coordinate be present, so that ΔZ spans an intercept in differences. A specification in which every coordinate of ΔZ varies is excluded by A4(ii), and one in which two do not is excluded by A1b(iv).

The lower bound does different work and is not a modelling choice. At $d_Z = 0$ the propensity exponent $\alpha = r/(2r + d_Z)$ is $\frac{1}{2}$ exactly, A5(i)'s declared floor puts $\hat{\delta}_n$ at $n^{-1/2}\sqrt{\ln n}$, and the sieve-entropy condition $L_x \ln n = o(n\hat{\delta}_n^2)$ then reads $L_x = o(1)$, which no integer $L_x \geq 1$ satisfies. Proposition 28 has no admissible sieve dimension there and Theorem 29 is vacuous rather than false. That is not a pathology of the propensity, which at $d_Z = 0$ is a constant and estimable at the parametric rate; what fails is the φ-step's entropy budget, which needs room between the noise level and the sieve dimension.

The φ-step is the iterated linearisation of Definition 22, run on the spline sieve of A8 (see DeVore and Lorentz, 1993; Schumaker, 2007) at the relaxation (35) (see Dunker et al., 2014; Engl, Hanke, and Neubauer, 2000).

That every pass is applied to a linear equation is what makes a tangential cone condition (Hanke, Neubauer, and Scherzer, 1995) unnecessary, and the point is stronger than convenience. The cone condition is not merely strong here: the supplement refutes it, proving that its event has probability zero, and proving that for every bounded candidate linearisation and for the one unbounded candidate as well.

The refutation is bounded, and the three qualifiers are load-bearing: the condition is posed on $\mathcal{V}$, in the levels parameterisation, off the sieve. A cone condition restricted to $\mathcal{S}_n$ is untouched by the refutation and is also unproved, as part (3) of the supplement's remark on where the

failure stops records, and in a space of derivatives the proposition dissolves altogether at a cost in the source index (55) will not pay. Nothing in the argument above needs any of that: the φ-step is licensed by the linearity of each pass, not by the refutation.

The derivative parameterisation is available and we decline it. Estimating the derivative of the structural function is standard in the linear problem and carries into the nonlinear one, where it lifts a saturation ceiling on the source index (Florens, Racine, and Centorrino, 2018; Centorrino, Fève, and Florens, 2025). Here it pushes the wrong way. Composing with the inverse derivative operator raises the degree of ill-posedness from a to $a+1$ *and* lowers the source index from $s/(2a)$ to $(s-1)/\{2(a+1)\}$. At the frontier-clearing configuration $(s,a)=(3,\frac{3}{4})$ of the Monte Carlo that returns $\frac{4}{7}$ in place of 2, below the $\frac{5}{6}$ that (55) requires.

We leave recovering marginal effects from $\hat{\varphi}$ to future research.

We stop the iteration on a discrepancy principle at a level that is stage-dependent and deterministic, and both features are load-bearing.

Stage-dependent, because the data error of the stage-k equation is not $\hat{\delta}_n$ but $\hat{\delta}_n \ln n$ plus the linearisation remainder $h_u^{-2}\varepsilon_{k-1}^2$. A fixed level sits below the true data error in exactly the regime the recursion must pass through, and there the classical discrepancy guarantee does not apply.

Deterministic, for two reasons of different status. The first is this paper's own and is sharp: the realised residual cannot supply the level, because bounding the unobservable error from below through the residual would require a lower bound on $\|\hat{\mathcal{A}}v\|$ in terms of $\|v\|$. That $\|\hat{\mathcal{A}}v_\dagger\| \ll \|v_\dagger\|$ is available is not an accident of this construction but the definition of ill-posedness. The second is classical and is cited: no rule depending on the data alone can make a regularisation method convergent for a genuinely ill-posed problem (Bakushinskii, 1984).

That statement is weaker than it looks here. It is worst-case and deterministic, and in the statistical setting we inhabit it does not exclude data-driven selection. Lepskii-type balancing

over the regularisation index is purely data-driven and attains the optimal rate up to a logarithmic factor (see Dunker, 2021; Lepskii, 1990). We therefore prefer the schedule of (37), and are not forced into it.

It is preferred at a stated price. That schedule is *computable* from four inputs and it is not *known*: what c_{14} and c_{15} must dominate are analytic constants built from the kernel, from the design, and from the spectral pre-image radius R_* of (P-a.3).

The error is not symmetric in direction. Chosen too large, the rule stops earlier, the rate is unchanged in order and the constant degrades. Chosen too small, the implemented level again sits below the true data error at some stage and the discrepancy guarantee is void, leaving whatever the cap (36) returns. This is the one place the procedure carries a tuning constant the theory does not pin, and we say so here rather than leave it to be discovered.

Proposition 28 is the rate. It discharges A5(i) rather than assuming it, and it reaches neither A5(ii) nor A5(iii), which Theorem 29 carries as hypotheses at (T-4). Two further things we do not claim should be read with it. It is an upper bound and not a sharpness statement, and nothing there says δ_n cannot be improved. And the source index μ is *primitive*, at (P-a.3), together with R_*.

What pins μ is not the identity $\mu = s/(2a)$, and the reason is worth giving to a referee who asks. The passage from a Hilbert-scale source condition plus a link condition to the *spectral* condition the classical discrepancy theorem consumes is operator interpolation at exponent 2μ. That is available only at $2\mu \le 1$, while (55) forces $2\mu > \frac{5}{3}$.

The θ-step runs on the debiased moment ψ of (38): the differenced least-squares moment plus $\widetilde{W}$ times a weight evaluated at the residual. We compute every nuisance out of fold (Definition 24) and take the one-step (47) from the plug-in preliminary (45) (see Chernozhukov, Chetverikov, et al., 2018; Newey and Robins, 2018).

The preliminary inherits the first-stage rate, $\|\hat{\theta}^{(0)} - \theta_\dagger\| = O_P(n^{-1/2}) + O_P(\delta_n)$, and it inherits

it because $\theta_\dagger$ is identified by restriction (B) alone (the plug-in preliminary lemma of the supplement); no condition on the weight, on the propensity or on A2 enters it. Its variance estimator is the sample sandwich (48), which carries a transpose because $\hat{J}$ is not symmetric even though J is.

We run the φ-step at a fixed anchor $\theta_\circ$, licensed by Corollary 12 and Lemma 10 under A1b(iii) rather than assumed. What is θ-dependent is the conditioning, and the mechanism is worth naming: in (24) a conditional density is evaluated at a θ-dependent point, which moves the singular values of the operator and with them every constant Proposition 28 is read at. The population solution set is θ-free; the operator is not.

Debiasing enters through the choice of weight, and Proposition 20 prices all three nuisance directions. The adjoint equation its first clause turns on is A2. In the weight direction the derivative vanishes at every perturbation and not merely to first order, because ψ is bi-affine there.

In the propensity direction the equivalence is with N1, conditional centring given ΔZ. N1 can always be arranged by a straightforward re-anchoring, at no cost to $\partial_u L$ and none to $E[\psi]$, but it does not hold automatically, and Proposition 20(iii) is the statement of exactly what its failure costs. That equivalence is what demotes ϱ from a $\sqrt{n}$ nuisance to a product-rate one. Forming the weight never requires a weighting measure on $\mathcal{T}$ to be chosen: by Lemma L-2 of the supplement the estimator solves for L inside a dense class, so we compute no t-integral in forming it, which is also the argument against a continuum-GMM formulation (Carrasco and Florens, 2000).

We assume two ingredients of this estimator to exist with stated properties and construct neither in this paper; three further conditions we take as primitive rather than derive. Each is named in a hypothesis rather than buried in a regularity condition.

The two unbuilt ingredients are estimators. A5(iv) prices the representer estimator $\hat{\ell}$, and no construction attaining that rate is supplied. The preliminary $\hat{\varphi}^{(0)}$ has no supplier either,

the schedule (37) needing it at a declared deterministic envelope $\hat{\varepsilon}_0$ of polynomial order, for which consistency will not do.

The three primitives are conditions, not objects. Both μ and R_* are primitive, as above. Global identification of $\varphi_\dagger$ is assumed at A1b(iii) and not derived. And on the sieve branch of the first-order bias, the sieve display of Lemma L-7 in the supplement is assumed.

Obligations of a second kind are construction claims rather than assumptions and are carried explicitly by the hypotheses that consume them. The first is that $\hat{L}^{(-k)}$ *is* an anchored antiderivative $L^{\hat{b}^{(-k)}}_{\hat{\ell}^{(-k)}}$ of a bounded, u-continuous $\hat{\ell}^{(-k)}$, which is (R-a) of Theorem 29. The second is that the direction $\hat{L}^{(-k)} - L_\star$ is conditionally centred, which is (R-c), a disjunction whose first branch assumes N1′ and whose second *prices the failure* at $r_n \epsilon_n^L$ through the fourth term of the supplement's bias decomposition. Pricing a failure is the honest third option, and it is neither a discharge nor an evasion. The supplement's remark on open verification obligations is the single inventory.

Definition 22 (The estimator: iterated linearisation with sieve Landweber–Fridman)**.** Let $\hat{\varphi}^{(0)}$ be a preliminary estimator which is a *measurable random element of* $\mathcal{V}$ *with respect to the sample it is computed on* **and which lies in** $\mathcal{S}_n$, the sieve fixed at the foot of this paragraph; and let $\hat{\varepsilon}_0 \to 0$ be a *known* deterministic sequence with $P\big(\|\hat{\varphi}^{(0)} - \varphi_\dagger\|_{L^2(\pi)} \le \hat{\varepsilon}_0\big) \to 1$ and $P\big(\|\hat{\varphi}^{(0)} - \varphi_\dagger\|_{H^s} \le \mathsf{R}\big) \to 1$. Let $\mathcal{S}_n := \operatorname{span}(b_1, \dots, b_{L_x}) \cap \mathcal{V}$ with $b_1, \dots, b_{L_x}$ the B-spline basis of A8. Why the second display is a statement about a finite quantity, and why the sieve-membership clause is not idle, are in the supplement's discussion of the estimator.

Fix the reference density's estimate, and floor it. The adjoint (S:hatAadj), through which every step of (b) below is taken, carries $1/\pi(x)$ in front of the whole expression, and π is an unknown feature of P by Definition 5; the operator the analyst forms is (S:hatAadj) at an estimate $\hat{\pi}$. **Fix** $\varepsilon_\pi \in (0, 1]$ before the data are seen. **The construction convention** is that $\hat{\pi}$ takes values in $[\varepsilon_\pi, \varepsilon_\pi^{-1}]$ at every $x \in \mathcal{X}$; a density estimator that does not is replaced by the pointwise clipping $(\hat{\pi} \vee \varepsilon_\pi) \wedge \varepsilon_\pi^{-1}$, its metric projection in $L^2(\mathcal{X}, dx)$ onto *the set of*

functions with values in $[\varepsilon_\pi, \varepsilon_\pi^{-1}]$ *Lebesgue-almost everywhere on* $\mathcal{X}$. What the floor buys, what it costs, and the one thing it does **not** establish — that running the iteration at $\hat{\pi}$ rather than at π leaves (56) in place, an **open item** stated as one at Remark S:phirscope(7) — are in the supplement's discussion of the estimator.

Normalise the scale, ***in the pairing the iteration runs in.*** Let

$$\omega_n \;:=\; 2\,(2c_C)^{1/2}\,\mathsf{c}_2^2\,c_\Pi\,\max\left\{\|1/\pi\|_\infty^{1/2},\;\varepsilon_\pi^{-1/2}\right\} h_u^{-1}h_x^{-1/2} \tag{35}$$

where c_2 and c_C are the two kernel-envelope constants (16) of C3(ii) fixes, and c_Π is the range-measure constant of B1(iv.c). This is the bound of Proposition S:hatAprop(i) and (i′), deterministic and sure at every $\varphi \in \mathcal{V}$, read at the two costs *the maximum pays*: $\|1/\pi\|_\infty^{1/2}$ in $L^2(\pi)$ and $\varepsilon_\pi^{-1/2}$ in $L^2(\hat{\pi})$. (35) *does not move with* L. Run the iteration below on $\omega_n^{-1}\hat{\mathcal{A}}_\varphi$ and $\omega_n^{-1}\hat{T}_{h_x}$, so that $\|\omega_n^{-1}\hat{\mathcal{A}}_\varphi|_{\mathcal{S}_n}\| \le 1$ surely *as an operator out of* $L^2(\pi)$ ***and*** *out of* $L^2(\hat{\pi})$. By Lemma S:scaleinv this is a choice of relaxation parameter: it changes neither the iterates nor the stopping index and leaves the rate (56) unchanged, while inflating by ω_n^2 the **bound** (F-f) supplies on that index. The cap is sized accordingly:

$$\bar{m}_n \;\ge\; \omega_n^2\,\hat{\delta}_n^{-2}. \tag{36}$$

The stopping levels are computed before the data are touched. Fix $\mathsf{c}_{14}, \mathsf{c}_{15} < \infty$ and $\kappa := 2\mu/(2\mu+1)$, and define the deterministic *stagewise noise schedule*

$$\hat{\delta}_{n,k}^{\mathrm{eff}} \;:=\; \mathsf{c}_{14}\Big(\hat{\delta}_n \ln n \;+\; h_u^{-2}\,\hat{\varepsilon}_{k-1}^2\Big), \qquad \hat{\varepsilon}_k \;:=\; \mathsf{c}_{15}\Big(\hat{\delta}_{n,k}^{\mathrm{eff}}\Big)^{\kappa}, \qquad k \ge 1. \tag{37}$$

c_{14} and c_{15} must be *chosen*, because they are not identified by the data. They must dominate two analytic constants, (S:effnoise)'s c_{16} and (S:linlf)'s c_{17}, which Proposition 28 Steps 2 and 3 consume and neither of which is known to the analyst. What each is built from, which way the error bites, and why the radius the second carries is the *spectral pre-image* radius R_*

of (P-a.3) and **not** R, are in the supplement's discussion of the estimator. For $k = 1, \dots, \bar{k}$:

(a) form the linear equation in $v \in \mathcal{S}_n$, $\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} v = -\hat{T}_{h_x}\big(\hat{\varphi}^{(k-1)}, \theta\big)$;

(b) run Landweber–Fridman on it from $v_0 = 0$, at the relaxation (35), $v_{m+1} = v_m - \omega_n^{-2}\, \hat{\mathcal{A}}^*_{\hat{\varphi}^{(k-1)}} \big[\hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} v_m + \hat{T}_{h_x}(\hat{\varphi}^{(k-1)}, \theta)\big]$, projected onto $\mathcal{S}_n$, which by Lemma S:scaleinv(i) *is* the unrelaxed iteration for the normalised pair, *the adjoint being formed at the floored* $\hat{\pi}$ *fixed above and not at* π;

(c) stop at

$$\hat{m}^{(k)} \;:=\; \min\Big\{ m \le \bar{m}_n : \; \big\| \hat{\mathcal{A}}_{\hat{\varphi}^{(k-1)}} v_m + \hat{T}_{h_x}(\hat{\varphi}^{(k-1)}, \theta) \big\|_{L^2(\Pi)} \;\le\; \tau_{\mathrm{DP}}\, \hat{\delta}^{\mathrm{eff}}_{n,k} \Big\},$$

the discrepancy principle *at the stage-k level* (37), *with the convention that* $\hat{m}^{(k)} := \bar{m}_n$ *when the set is empty*;

(d) set $\hat{\varphi}^{(k)} := \hat{\varphi}^{(k-1)} + v_{\hat{m}^{(k)}}$.

Put $\hat{\varphi} := \hat{\varphi}^{(\bar{k})}$.

Two readings of clause (b)*, and they are two different random functions.* As written, (b) forms the adjoint at the floored $\hat{\pi}$: that is the **feasible** reading, the one an analyst can run, and $\hat{\varphi}$ names its output — here and in every other file. Write $\hat{\varphi}^{\pi}$ for the output of the *same* recipe with two ingredients replaced: $\hat{\mathcal{A}}^*_{\hat{\varphi}^{(k-1)}}$ read as (S:hatAadj) at the *population* π, and the projection in (b) the $L^2(\pi)$-orthogonal projection onto $\mathcal{S}_n$. That is the **infeasible** reading: no analyst can run it, π being an unknown feature of P by Definition 5. *The stage superscripts* $\hat{\varphi}^{(0)}, \dots, \hat{\varphi}^{(\bar{k})}$ *are* ***reading-relative***, denoting the stages of whichever reading the ambient statement fixes; so $\hat{\varphi} = \hat{\varphi}^{(\bar{k})}$ read here and $\hat{\varphi}^{\pi} = \hat{\varphi}^{(\bar{k})}$ read there, and no third family of symbols is introduced. (56) is a statement about $\hat{\varphi}^{\pi}$ and Definition 24(b) plugs in $\hat{\varphi}$; what is settled and what is open about the passage from the first to the second is Remark S:phirscope(7), in five parts of which two are open, and the convention's scope and typography are in the supplement's discussion of the estimator.

Write $\mathcal{I}_*$ for the fold the iteration is run on, $\mathcal{I}_{-k}$ under Definition 24(b), $\mathcal{I}_m$ under Definition S:hatTexp, and read every "measurable" below at $\sigma(\mathcal{I}_*)$.

The output is a measurable random element, and that is **verified** rather than required elsewhere — *verified at the* ***fixed*** *sieve* $\mathcal{S}_n$*, which is what the infeasible reading projects onto, and* ***pending*** *at the feasible one, whose sieve is random.* Induction on $k \le \bar{k}$, from a $\sigma(\mathcal{I}_*)$-measurable $\hat{\varphi}^{(0)}$, gives the claim for $\hat{\varphi}^{\pi} = \hat{\varphi}^{(\bar{k})}$, and *that is not the whole of what Definition 24*(c′.1) *asks.* The verification, the two features of (c) it consumes, the three stated results that consume it, and what it does **not** reach at the feasible, random sieve are in the supplement's discussion of the estimator. So are the two load-bearing features of (c) — the level is stage-dependent, and it is deterministic and supplied from outside — together with the status of Bakushinskii's veto, and the price the schedule charges, which is a *rate* $\hat{\varepsilon}_0$ for the preliminary and not mere consistency and which hypothesis (P-a.2) of Proposition 28 carries.

Every Landweber iteration in this construction is applied to a ***linear*** *equation*; the nonlinearity is carried by the outer index k and by nothing else. That is the entire difference from the route Remark S:lfnonlinear declines, and it is what makes a cone condition unnecessary.

Definition 23 (The debiased moment function, and the two matrices)**.** For a nuisance triple $\zeta = (\varphi, L, \varrho)$ with $\varphi \in \mathcal{V}$, $L : \mathbb{R} \times \mathbb{R}^p \to \mathbb{R}^{p \times (\mathsf{L}-1)}$ Borel and $\varrho : \mathbb{R}^p \to \{q \in \mathbb{R}^{\mathsf{L}-1} : q_l \ge 0 \text{ for every } l,\ \mathbf{1}'q \le 1\}$ Borel, write $\widetilde{W}_l(\varrho) := \mathbb{1}\{W = l\} - \varrho_l(\Delta Z)$ for $l = 1, \dots, \mathsf{L}-1$ and $\widetilde{W}(\varrho) := \big(\widetilde{W}_1(\varrho), \dots, \widetilde{W}_{\mathsf{L}-1}(\varrho)\big)'$, the contrast vector (1) formed at the trial propensity, and define the $\mathbb{R}^p$-valued *debiased moment function*

$$\psi(\theta, \zeta) \;:=\; \Delta Z\, \Delta U(\theta, \varphi) \;+\; L\Big(\Delta U(\theta, \varphi),\, \Delta Z\Big)\, \widetilde{W}(\varrho), \tag{38}$$

with $\Delta U(\theta, \varphi)$ as in (19). Write $\widetilde{W} := \widetilde{W}(\varrho)$ at the true propensity, which is (1) itself, $\zeta_\star := (\varphi_\dagger, L_\star, \varrho)$ as in Definition 19, and

$$\psi_\star \;:=\; \psi(\theta_\dagger, \zeta_\star) \;=\; \Delta Z\, \Delta U + L_\star(\Delta U, \Delta Z)\, \widetilde{W}. \tag{39}$$

Define

$$J \;:=\; E\Big[\Delta Z\,\Delta Z'\Big] \;+\; E\Big[\ell_\star(\Delta U,\Delta Z)\,\widetilde{W}\,\Delta Z'\Big], \qquad \Sigma \;:=\; E\Big[\psi_\star\psi_\star'\Big]. \tag{40}$$

Every product above is read at the type declaration of the instrument block: L, $L_\star$ and $\ell_\star$ carry one column per contrast, $\widetilde{W}(\varrho)$ is a column of $\mathsf{L}-1$ contrasts, so $L\widetilde{W}(\varrho)$ is $\mathbb{R}^p$-valued and the second summand of (40) is $(p\times 1)(1\times p)$, the shape J must have. *At* $\mathsf{L}=2$ *every display above is the printed one*: the simplex is the interval $[0,1]$, $\mathbb{1}\{W=1\}=W$ makes $\widetilde{W}(\varrho)=W-\varrho(\Delta Z)$ a scalar, L, $L_\star$ and $\ell_\star$ have one column, and $L\widetilde{W}=\widetilde{W}L$, $L_\star\widetilde{W}=\widetilde{W}L_\star$ and $\ell_\star\widetilde{W}=\widetilde{W}\ell_\star$ for that reason.

Notes on Definition 23. Why the second summand of (40) carries $\ell_\star$ and not $L_\star$ and is nevertheless the zero matrix, why J is not "a different matrix" from $E[\Delta Z\Delta Z']$, and how the typing of $E[\ell_\star\widetilde{W}\Delta Z']$ transfers to $\hat{J}$ are Remark S:momentread.

Definition 24 (Cross-fitting, and the fixed anchor of the φ-step)**.** Fix $\mathsf{F}\geq 2$, not depending on n, and a partition of $\{1,\dots,n\}$ into folds $\mathcal{I}_1,\dots,\mathcal{I}_{\mathsf{F}}$ with $n_k:=|\mathcal{I}_k|$ and $\lfloor n/\mathsf{F}\rfloor\leq n_k\leq\lceil n/\mathsf{F}\rceil$, chosen independently of the data. Write $\mathcal{I}_{-k}:=\{1,\dots,n\}\setminus\mathcal{I}_k$, O_i for the i-th observation vector of A0, and

$$\mathbb{E}_n[\varsigma]:=\frac{1}{n}\sum_{i=1}^{n}\varsigma(O_i), \qquad \mathbb{E}_{n,k}[\varsigma]:=\frac{1}{n_k}\sum_{i\in\mathcal{I}_k}\varsigma(O_i).$$

At $\mathsf{F}=2$ this is the split of Definition 2, with $(\mathcal{I}_k,\mathcal{I}_{-k})=(\mathcal{I}_m,\mathcal{I}_a)$.

Convention (P8)*, re-declared here.* (P8)'s *conditioning convention*, "every expectation and every O_P is conditional on the auxiliary fold", was stated **for the φ-step**, at $\sigma(\mathcal{I}_a)$, and is therefore **re-declared** rather than inherited: throughout the θ-step it is read at $\sigma(\mathcal{I}_{-k})$, with $\mathcal{I}_k$ in place of $\mathcal{I}_m$. A0 supplies the independence of $\mathcal{I}_k$ from $\mathcal{I}_{-k}$, which with clause (c′) below is what (P8)'s *freezing lemma* consumes. Which of Lemma S:pconv(i)–(iii) reaches which site — and in the θ-step it is **never the transfer rule** — is in the supplement's discussion of the estimator.

For each k the nuisances are computed on $\mathcal{I}_{-k}$:

(a) *Propensity.* $\hat{\varrho}^{(-k)} = (\hat{\varrho}_1^{(-k)}, \ldots, \hat{\varrho}_{\mathsf{L}-1}^{(-k)})'$ satisfies A4(iii) with n replaced by n_{-k} and is truncated to the ε_ϱ-interior simplex (11) as in Definition 2, so that the fold contrast vector

$$\widehat{\widetilde{W}}^{(-k)} := \left(\mathbb{1}\{W=1\} - \hat{\varrho}_1^{(-k)}(\Delta Z),\ \ldots,\ \mathbb{1}\{W=\mathsf{L}-1\} - \hat{\varrho}_{\mathsf{L}-1}^{(-k)}(\Delta Z)\right)',$$

written $\widehat{\widetilde{W}}_i^{(-k)}$ when evaluated at observation i and $\widehat{\widetilde{W}}_{i,l}^{(-k)}$ for its l-th entry there, on the indexing of Definition 2, satisfies $\varepsilon_\varrho \le \left|\widehat{\widetilde{W}}_{i,l}^{(-k)}\right| \le 1$ surely *at every* $l \in \{1,\ldots,\mathsf{L}-1\}$, this is Remark S:hatTread(2) read at the fold, and it is the **per-cell** bound and **not** a bound on $\|\widehat{\widetilde{W}}_i^{(-k)}\|$, which (S:estcontrnorm) **derives** from it rather than assumes. And $\mathsf{r}_k := \hat{\varrho}^{(-k)} - \varrho$ satisfies $\|\mathsf{r}_k\|_\infty \le 1 - 2\varepsilon_\varrho < 1$ **by A4(i)**, the supremum taken over z *and* over the $\mathsf{L}-1$ components. Why A4(i) is named, and why this display is **not** (S:estcontrnorm) — a sup-norm and a Euclidean statement about the same object, only the first of which reads an overlap condition — are in the supplement's discussion of the estimator.

(b) *Structural function.* $\hat{\varphi}^{(-k)}$ is the estimator of Definition 22, run on $\mathcal{I}_{-k}$ **at a fixed anchor** $\theta_\circ \in \mathbb{R}^p$ **chosen before the data are seen**, and $h_k := \hat{\varphi}^{(-k)} - \varphi_\dagger$. The anchor is what removes the circularity, and it is licensed by Corollary 12 and Lemma 10 under A1b(iii). *What is not θ-free is the conditioning of the inverse problem*, so Proposition 28's hypotheses (P-a)–(P-g), (55) and A7 in the form (13) are read *at* $\theta_\circ$, and a, μ, R_* and $\hat{\delta}_n$ are $\theta_\circ$-specific.

(c) *Debiasing weight.* $\hat{L}^{(-k)}$ is an admissible weight in the sense of Definition 19(b), $\hat{L}^{(-k)} = L_{\hat{\ell}^{(-k)}}^{\hat{b}^{(-k)}}$, generated by an estimator $\hat{\ell}^{(-k)}$ of $\ell_\star$ that is Borel, bounded, and continuous in u at every (u,z); $D_k := \hat{L}^{(-k)} - L_\star$. Hypothesis (R-a) of Theorem 29 strengthens that continuity to continuous differentiability, which (c′.2) uses and no other clause here does, so the sup-norm convention (42) is **not** unconditional. What

the three properties buy is in the supplement's discussion of the estimator.

(c′) Measurability of the five estimated nuisances, and the reading of their sup-norms. Clause (c) states a per-realisation "Borel" and nothing more, without which the hypotheses of (R-a) and the displays (43) are not well-posed probability statements; *and the right container is not the same for all five.* (c′.1) is a requirement on the estimators, with **one** exception: for $\hat{\varphi}^{(-k)}$ the property is **verified** rather than required, at the measurability paragraph of Definition 22, and *that verification is discharged at the fixed sieve* $\mathcal{S}_n$ *and* ***pending*** *at the feasible, random one.*

(c′.1) Measurability, and it is two different statements, because three of the five are only ever integrated and two are evaluated pointwise at arguments their container does not control.

The three integrated nuisances, as random elements of L^2 at a fixed representative. Each of $\hat{\varrho}^{(-k)}$, $\hat{b}^{(-k)}$ and $\hat{\varphi}^{(-k)}$ is a $\sigma(\mathcal{I}_{-k})$**-measurable random element** of, respectively, $L^2(P_{\Delta Z})$, $L^2(P_{\Delta Z})$ and $\mathcal{V}$, *a measurable map from the sample space into the space and not a family of functions indexed by the sample point*; *fix a jointly measurable representative for each, once, as A2c does for* $\ell^{\Pi}_{\star}$. *For the third this is a verification and not a requirement*, supplied by Definition 22's measurability paragraph, which is written at the fixed $\mathcal{S}_n$ — the sieve the *infeasible* reading $\hat{\varphi}^{\pi}$ projects onto, while clause (b) names the *feasible* output, for which **both** halves of the present requirement — measurability, by Remark S:phirscope(7a), and membership in $\mathcal{V}$, by Remark S:phirscope(7c) — are *pending*, and are printed as such rather than discharged by being carried as a hypothesis. Why, and the two grounds on which the choice of representative is immaterial for these three, are in the supplement's discussion of the estimator.

For $\hat{\varrho}^{(-k)}$ the choice is moreover **pinned**, and that is what clause (a)'s *sure* bound rests on: A4(iii), read at the fold by clause (a), requires product measurability of

$\hat{\varrho}$'s *own pointwise function*, the (41) shape, so for that object no equivalence class is formed and no representative is chosen at all.

The two weight objects, for which the L^2 container is not merely unnecessary but wrong. What is required of $\hat{\ell}^{(-k)}$, in addition to clause (c), is that its *own* pointwise function, at every realisation and every (u, z), be product measurable:

$$(\omega, u, z) \;\longmapsto\; \hat{\ell}^{(-k)}(\omega)(u, z) \quad \text{is} \quad \sigma(\mathcal{I}_{-k}) \otimes \mathfrak{B}\Big(\mathbb{R} \times \mathbb{R}^p\Big)\text{-measurable}, \tag{41}$$

and $\hat{L}^{(-k)} = L^{\hat{b}^{(-k)}}_{\hat{\ell}^{(-k)}}$ then *inherits* the same property from (31), *so* (41) *is imposed on one object and not on two*; under (R-a) the derivative inherits it too, which (c′.2) needs for its second envelope. *What is retained from the L^2 reading is membership*: $\hat{\ell}^{(-k)}$ and $\hat{L}^{(-k)}$ lie in $L^2(P_{\Delta U, \Delta Z})$ almost surely, the second *derived and not required*, and that is what makes the third display of (43) finite. (41) is also what (P8)'s freezing lemma consumes for the $\hat{\ell}^{(-k)}$ *ingredient* of its composite. Why the L^2 reading fails for these two and not for the three above, both inheritances, and why the third display has a truth value, are in the supplement's discussion of the estimator.

(c′.2) The sup-norms are essential suprema in z and genuine suprema in u. For the two envelopes (R-a) prices,

$$\Big\|\hat{\ell}^{(-k)}\Big\|_\infty \;:=\; \operatorname{ess\,sup}_z \Big\{ \sup_{u \in \mathbb{R}} \Big\|\hat{\ell}^{(-k)}(u, z)\Big\| \Big\}, \qquad \text{the essential supremum taken under } P_{\Delta Z}, \tag{42}$$

and likewise for $\|\partial_u \hat{\ell}^{(-k)}\|_\infty$. Both halves of that reading are needed and neither can be interchanged with the other. Hence $\omega \mapsto \|\hat{\ell}^{(-k)}\|_\infty$, and likewise $\omega \mapsto \|\partial_u \hat{\ell}^{(-k)}\|_\infty$, is $\sigma(\mathcal{I}_{-k})$-measurable as a map into $[0, +\infty]$, which is what (R-a) needs to price both envelopes as $O_P(1)$. The measurability of the inner supremum and of the **outer** essential supremum, and why the essential supremum must be

taken in z and **not** in (u, z), are in the supplement's discussion of the estimator.

(d) The fold reading of the maintained nuisance-rate clauses, declared once. A4(iii) is read at each fold in (a); the **four** clauses of A5 that price an *estimated* nuisance are read the same way, and every later citation of them is a citation of the corresponding display here:

$$\max_{k\le \mathsf{F}} \left\|h_k\right\|_{L^2(\pi)} = O_P(\delta_n), \qquad \max_{k\le \mathsf{F}} \left\|h_k\right\|_{H^s} = O_P(1), \qquad \max_{k\le \mathsf{F}} \left\|\hat{\ell}^{(-k)} - \ell_\star\right\|_{L^2(P)} = O_P(\epsilon_n), \tag{43}$$

being A5(ii)'s first half, A5(vi), and A5(iv)'s first half respectively, *the third in force precisely when (R-e.1) of Theorem 29 is the branch taken, since under (R-e.2) no rate for $\hat{\ell}^{(-k)}$ is assumed at all. The second display is finite at every realisation only* ***once*** $\hat{\varphi}^{(0)}$ ***is in*** $\mathcal{S}_n$*, a premise stated here because this is a* ***definition*** *and inherits no hypothesis list*: Definition 22(d) puts the path in $\mathcal{S}_n$ if and only if its *starting point* is there, (8) places $\mathcal{S}_n$ in H^s and H1 places $\varphi_\dagger$ there. *Every result that reads* (43) *supplies the premise*, (P-a.1) being in group (T-5) of Theorem 29. *The indexing is not cosmetic and it is not free either*: A5(ii), (iv) and (vi) are stated at a **single** estimated nuisance, and for the first two the cost is nil, Theorem 29 Step 0 *deriving* them from (T-5) with F fixed and $n_{-k} \asymp n$. *For the third it is a strengthening of A5(iv) in form*, of the same order for fixed F but a strengthening nonetheless, and it is therefore carried explicitly by hypothesis (R-e.1) of Theorem 29 rather than read into A5(iv). *The fourth clause is A5(vi′)*, which prices the *stagewise* Sobolev stability of the φ-step, $P(\max_{j\le \bar{k}} \|\hat{\varphi}^{(j)} - \varphi_\dagger\|_{H^s} \le \mathsf{R}) \to 1$ — $\bar{k}$ a *stage* index and not a fold index — and is read at each fold complement exactly as the other three are. What that costs, and why the radius R must nonetheless serve all F fold complements at one value, are in the supplement's discussion of the estimator.

Put $\hat{\zeta}^{(-k)} := (\hat{\varphi}^{(-k)}, \hat{L}^{(-k)}, \hat{\varrho}^{(-k)})$ and, for $\theta \in \mathbb{R}^p$,

$$\bar{\psi}_n(\theta) \;:=\; \frac{1}{n}\sum_{k=1}^{\mathsf{F}}\sum_{i\in\mathcal{I}_k}\psi\Big(\theta, \hat{\zeta}^{(-k)}\Big)(O_i), \qquad \hat{\psi}^{(-k)} := \psi\Big(\theta_\dagger, \hat{\zeta}^{(-k)}\Big). \tag{44}$$

Since F *is fixed,* $n_k/n \to \mathsf{F}^{-1}$ *and* $n_k \asymp n$; the constant F is carried explicitly in every bound rather than absorbed.

Definition 25 (The preliminary and the one-step estimator)**.** With the notation of Definition 24 define

$$\hat{\theta}^{(0)} \;:=\; \Big(\mathbb{E}_n\big[\Delta Z \Delta Z'\big]\Big)^{-1}\frac{1}{n}\sum_{k=1}^{\mathsf{F}}\sum_{i\in\mathcal{I}_k}\Delta Z_i\Big(\Delta Y_i - \Delta\hat{\varphi}_i^{(-k)}\Big), \tag{45}$$

on the event that $\mathbb{E}_n[\Delta Z \Delta Z']$ is nonsingular, and $\hat{\theta}^{(0)} := 0$ otherwise. Define the *cross-fitted Jacobian*, for $\theta \in \mathbb{R}^p$,

$$\hat{J}(\theta) \;:=\; \frac{1}{n}\sum_{k=1}^{\mathsf{F}}\sum_{i\in\mathcal{I}_k}\Big[\Delta Z_i \Delta Z_i' + \hat{\ell}^{(-k)}\Big(\Delta U_i(\theta, \hat{\varphi}^{(-k)}), \Delta Z_i\Big)\,\widehat{\widetilde{W}}_i^{(-k)}\,\Delta Z_i'\Big], \tag{46}$$

so that $\partial_\theta \bar{\psi}_n(\theta) = -\hat{J}(\theta)$ identically, and put $\hat{J} := \hat{J}(\hat{\theta}^{(0)})$. *The order of the three factors is what the identity forces*: by (38) the θ-derivative of $\hat{L}^{(-k)}(\Delta U_i(\theta,\cdot), \Delta Z_i)\,\widehat{\widetilde{W}}_i^{(-k)}$ is $-\hat{\ell}^{(-k)}(\cdot)\,\widehat{\widetilde{W}}_i^{(-k)}\,\Delta Z_i'$, which is $(p \times (\mathsf{L}-1))((\mathsf{L}-1)\times 1)(1\times p)$. At $\mathsf{L} = 2$ the three factors commute in the first two positions. The *one-step estimator* is

$$\hat{\theta} \;:=\; \hat{\theta}^{(0)} \;+\; \hat{J}^{-1}\,\bar{\psi}_n\Big(\hat{\theta}^{(0)}\Big) \tag{47}$$

on the event $\{\hat{J}$ nonsingular$\}$, and $\hat{\theta} := \hat{\theta}^{(0)}$ otherwise. The variance estimator is

$$\hat{\Sigma} \;:=\; \frac{1}{n}\sum_{k=1}^{\mathsf{F}}\sum_{i\in\mathcal{I}_k}\psi\Big(\hat{\theta}, \hat{\zeta}^{(-k)}\Big)(O_i)\;\psi\Big(\hat{\theta}, \hat{\zeta}^{(-k)}\Big)(O_i)', \qquad \hat{V} \;:=\; \hat{J}^{-1}\hat{\Sigma}\,\hat{J}^{-1\prime}, \tag{48}$$

the last **on the event** $\{\hat{J}$ **nonsingular**$\}$**, with** $\hat{V} := 0$ **otherwise**.

Remark 26 (How Definition 25's construction is read). Three clauses, each of which stood inside the definition. None of it is construction, and the estimator Definition 25 defines is unchanged by its removal.

(1) *That identity uses* ***one*** *derivative and no more*: $\partial_u \hat{L}^{(-k)} = \hat{\ell}^{(-k)}$, which Definition 19(b) supplies at every point under the three properties Definition 24(c) requires of $\hat{\ell}^{(-k)}$, together with $\theta \mapsto \Delta U_i(\theta, \cdot)$ being affine by (19) and the sum being finite. No second derivative is claimed here. $\partial_u^2 \hat{L}^{(-k)} = \partial_u \hat{\ell}^{(-k)}$ is used at Theorem 29 Step 1, for Taylor's theorem in θ, and it is hypothesised *there*, at (R-a), *a definition should not carry a differentiability claim that a theorem assumes. That step is not the only reader of the envelope* $\|\partial_u \hat{\ell}^{(-k)}\|_\infty$*, and no claim of exhaustiveness is made here*: Lemma S:biasexp Step 4 and Lemma S:sandwich Step 5 read it too, on the right of an inequality, as (R-a)'s own reading records. The draft this sentence replaces said "only" and named one of three, while that reading named two of the same three; the set is one and the two claims covered disjoint halves of it.

(2) *The fallback is not optional.* $\hat{\theta}^{(0)}$ is given a value off $\{\mathbb{E}_n[\Delta Z \Delta Z']$ nonsingular$\}$ and $\hat{\theta}$ is given one off $\{\hat{J}$ nonsingular$\}$; $\hat{V}$ inverts the *same* $\hat{J}$ and needs one too. That matters because Theorem 29's **second** conclusion, $\hat{V} \xrightarrow{p} J^{-1}\Sigma J^{-1\prime}$, and the studentised statement that follows it are statements about $\hat{V}$ at every n, and a convergence in probability is not well posed for a quantity that is undefined on an event of positive probability. *The choice of value is immaterial and the zero matrix is chosen because it is the one value that needs no further object*: by Lemma S:sandwich(i), $P(\hat{J}$ nonsingular$) \to 1$, so the event on which the fallback is active has probability tending to zero and no conclusion of Lemma S:sandwich(ii) or of Theorem 29 changes. *In a finite sample the fallback is not innocuous and should not be reported as a variance estimate*: $\hat{V} = 0$ makes the studentised statistic undefined, which is the honest report on that event and is the reason the value is stated rather than left to the reader.

(3) One step and not a root, and the ground is the **sample** criterion and not the population one. The realised map $\theta \mapsto \bar{\psi}_n(\theta)$ is *not* affine: by (38) θ enters $\hat{L}^{(-k)}$ through its first argument, and $\widehat{\widetilde{W}}^{(-k)}$ is not conditionally centred, so $\bar{\psi}_n(\theta) = 0$ need not have a unique root at any n. *The ground is* ***not*** *that no global rank condition is available*: by Lemma S:momzero(iv) and (S:psiaffine) the population moment $\theta \mapsto E[\psi(\theta, \zeta_\star)]$ is affine with gradient $-E[\Delta Z \Delta Z']$, so a global rank condition *is* available, it is A1b(iv), and $\theta_\dagger$ is its unique root on all of $\mathbb{R}^p$, which says nothing about the sample criterion at estimated nuisances. *What the block supplies is a preliminary with a rate* (Lemma S:prelimtheta), and a one-step from a preliminary with a rate is exactly the construction that converts a rank condition on the population Jacobian into a limit distribution without solving anything; the cross-fitted form is that of Chernozhukov, Chetverikov, et al. (2018), and the two classical anchors for the one-step itself are Newey and McFadden (1994, §3.4) and Vaart (1998, Ch. 5). *The Jacobian is evaluated at* $\hat{\theta}^{(0)}$ *and not at* $\hat{\theta}$, which is what makes (47) a closed-form estimator and what makes the algebra of Theorem 29 Step 1 exact rather than asymptotic.

4 Asymptotic properties

Three objects in what follows are constructed in the supplement and are recalled here so that the statement can be read where it is printed. $\mathcal{B}_{\text{sym}} := \frac{1}{2}(\mathcal{B} + \mathcal{B}^*)$ is the symmetric part of $\mathcal{B}$; $q := E[\partial_u \ell_\star \widetilde{W} \mid X_1, X_2] \in \mathbb{R}^p$, the $\mathbb{R}^{p \times (\mathsf{L}-1)}$-valued $\partial_u \ell_\star$ contracted against the contrast vector $\widetilde{W}$ of (1) and then conditioned; and, for an integrable g, $\sigma_g(x) := E[g \mid X_2 = x] f_{X_2}(x) + E[g \mid X_1 = x] f_{X_1}(x)$, taken coordinatewise for a vector-valued g, so that σ_q is that object at $g = q$ and is $\mathbb{R}^p$-valued. The multiplier is written out as $\vartheta' \sigma_q / \pi$ throughout and is given no abbreviation, because $\mathcal{B}_{\text{mult}}$ is multiplication *by* it and an abbreviation would collapse the distinction. At $\mathsf{L} = 2$ the contraction is a product of scalars in the second factor, $\partial_u \ell_\star \widetilde{W} = \widetilde{W} \partial_u \ell_\star$, and q, σ_q and $\vartheta' \sigma_q / \pi$ are the objects earlier rounds printed.

Proposition 27 (The second-order remainder on the ball: the sharp constant, and the exact δ^2 order under O-16(a))**.** *The supremum of the second-order remainder over a ball of radius δ in the structural direction is exactly $\frac{1}{2}\delta^2$ times the norm of the compressed symmetrised form, and it is bounded below by $\frac{1}{2}\delta^2\|\vartheta'\sigma_q/\pi\|_\infty$; so, provided the form does not degenerate, no bound of the form $o(\delta^2)$, and in particular no bound in the weak norm $\|\mathcal{A}_{\mathrm{cond}}\cdot\|_{L^2(\Pi)}$, holds uniformly on that ball. Suppose the hypotheses of Lemma* S:solset *hold, and for clause* (iv) *in addition* ***C2′(i)****. Fix $\vartheta\in\mathbb{R}^p$ with $\|\vartheta\|=1$, and let the* second-order remainder in the structural direction *at $h\in\mathcal{V}$ be*

$$\tfrac{1}{2}\,\vartheta'\partial^2_{\tau_1\tau_1}\Psi(0) \;=\; \tfrac{1}{2}\,E\Big[\vartheta'\,\partial_u\ell_\star\,\widetilde{W}\,(\Delta h)^2\Big] \;=\; \tfrac{1}{2}\,\Big\langle\mathcal{B}h,h\Big\rangle_{L^2(\pi)}, \tag{49}$$

which is Lemma S:secondord(i)*, read at the admissible direction pair $(D,\mathsf{r})=(0,0)$, contracted in ϑ and transferred to $L^2(\pi)$ by Lemma* S:solset(iii). Two non-degeneracy hypotheses are carried by name below, and this display is where they are defined*: at the fixed ϑ above,*

$$\text{O-16(a)}: \quad \exists\, h\in\mathcal{V} \;\; \textit{with} \;\; \vartheta' E\Big[q\,(\Delta h)^2\Big]\neq 0; \qquad\qquad \text{O-16(b)}: \quad \vartheta'\sigma_q\not\equiv 0.$$

Both are hypotheses and neither is a consequence of O-11*, which is Remark* S:sharpscope(3)–(4)*;* that (b) implies (a), and that the converse fails, are claims of clause (iii) below and of Remark S:sharpscope(3) respectively, not background assumed here. *Then, for every $\delta>0$:*

(i) The sharp constant.

$$\sup_{h\in\mathcal{V},\;\|h\|_{L^2(\pi)}\le\delta}\;\tfrac{1}{2}\,\Big|\Big\langle\mathcal{B}h,h\Big\rangle_{L^2(\pi)}\Big| \;=\; \tfrac{1}{2}\,\delta^2\,\Big\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\Big\|, \tag{50}$$

a supremum that need not be attained.

(ii) Bracketing. *The constant in* (50) *obeys*

$$\Big\|\vartheta'\sigma_q/\pi\Big\|_\infty \;\le\; \Big\|P_{\mathcal{V}}\mathcal{B}_{\mathrm{sym}}P_{\mathcal{V}}|_{\mathcal{V}}\Big\| \;\le\; \Big\|\mathcal{B}_{\mathrm{sym}}\Big\| \;\le\; \Big\|\mathcal{B}\Big\|, \tag{51}$$

where the first inequality holds under O-16(b) *and the outer two hold always.*

(iii) Positivity, which is a hypothesis and is stated as one. Under ***O-16*****(a)** *the constant in* (50) *is strictly positive, so* $\sup_{\|h\|\le\delta}\frac{1}{2}|\langle\mathcal{B}h,h\rangle| \asymp \delta^2$ *with a constant free of* δ*, and no bound of the form* $o(\delta^2)$ *as* $\delta\downarrow 0$ *holds uniformly over* $\{h\in\mathcal{V}:\|h\|_{L^2(\pi)}\le\delta\}$*. Under* ***O-16*****(b)** *the constant is at least* $\|\vartheta'\sigma_q/\pi\|_\infty > 0$*, and O-16*(b) *implies O-16*(a)*.*

(iv) No weak-norm bound. *Under O-16*(b) *there is no finite* λ *with*

$$\tfrac{1}{2}\Big|\big\langle\mathcal{B}h,h\big\rangle_{L^2(\pi)}\Big| \;\le\; \lambda\,\big\|\mathcal{A}_{\mathrm{cond}}h\big\|^2_{L^2(\Pi)} \qquad \text{for all } h\in\mathcal{V}; \tag{52}$$

nor with $\|\mathcal{A}_{\mathrm{cond}}h\|^2$ *replaced by* $\|h\|_{L^2(\pi)}\,\|\mathcal{A}^*_{\mathrm{cond}}\mathcal{A}_{\mathrm{cond}}h\|_{L^2(\pi)}$*.*

Notes on Proposition 27. How its hypothesis list is read is Remark S:sharpquadread; what it claims and the four things it does not are Remark S:sharpscope.

Proposition 28 (The rate for the φ-step)**.** *Suppose A0, A1, A1b(iii)–(iv), A4(ii)–(iii), A5(vi′), B1(i)–(v), C1(ii), C2′, C3 and C5′(i) hold. Suppose H1 holds,* for the Sobolev scale H^s that Definition S:srcball, A5(vi′), (P-c) and (P-g) are written in, and for A2's Picard form; the identity $\mu = s/(2a)$ is **not** assumed and is **not** used, μ being primitive at (P-a.3). *Suppose A8 holds in its* basis *clause,* $b_1,\dots,b_{L_x}$ *is a B-spline basis of degree exceeding* $s-\frac{1}{2}$ *on a quasi-uniform mesh over* $\mathcal{X}$*, and in its* upper end $L_x \ll n^{1/2}$*, but* ***not*** *in its lower end* $L_x \gg n^{1/(2s)}$*; and suppose* $p_K \ge s-\frac{1}{2}$*, which with* $p_K = 8$ *caps* $s\le\frac{17}{2}$ *and therefore, against* (54)*'s floor,* restricts $\alpha < \frac{68}{145}$*. And suppose*

(P-a) (preliminary, and the **spectral** source condition at the compression) *Three clauses, all three* envelopes holding with probability tending to one *rather than* O_P *statements, Remark* S:paenvelope *says why the distinction is load-bearing.*

(P-a.1) (preliminary, H^s, and $\sigma(\mathcal{I}_*)$-measurable) $\hat\varphi^{(0)}$ *is a* $\sigma(\mathcal{I}_*)$*-measurable random element of* $\mathcal{V}$*, in the sense and with the fold* $\mathcal{I}_*$ *of Definition 22;* $\hat\varphi^{(0)}\in\mathcal{S}_n$*; and* $P\big(\|\varphi_\dagger - \hat\varphi^{(0)}\|_{H^s}\le \mathsf{R}\big)\to 1$*, with* R *the* same *deterministic radius as in Definition* S:srcball*,*

A5(vi′), (P-c) and (P-g). A5(vi′), (P-c) and (P-g) are named here as instances and this clause maintains no list*: Definition* S:srcball *carries the enumeration — seven sites, this clause itself the third of them — and this points at it.* The distinction is not clerical here, because this clause is a **hypothesis** *and a reader must act on it: a radius chosen to clear only the sites named above need not clear* (S:entropy) *or Lemma* S:unifop *Step 4, which are the remaining two, and the existential is over* ***one*** R *meeting all seven at once.*

(P-a.2) (preliminary, rate) $P\big(\|\varphi_\dagger - \hat{\varphi}^{(0)}\|_{L^2(\pi)} \le \hat{\varepsilon}_0\big) \to 1$ *for a known deterministic* $\hat{\varepsilon}_0 = O(n^{-\varsigma_0})$ *whose exponent clears the repelling fixed point of the stagewise recursion,*

$$\varsigma_0 \;>\; \frac{\kappa\,\alpha}{4\,(2\kappa-1)}, \qquad \kappa = \frac{2\mu}{2\mu+1}, \quad \alpha = \frac{r}{2r+d_Z}$$

(P-a.3) (source condition, stated **spectrally** and **at the compression**) *There are* $\mu > 0$ *and a finite* R_**, neither depending on* n*, such that with*

$$\hat{\mathcal{P}}_\varphi^{\mathcal{S}} \;:=\; P_{\mathcal{S}_n}\,\hat{\mathcal{A}}_\varphi^*\hat{\mathcal{A}}_\varphi\big|_{\mathcal{S}_n} \;:\; \mathcal{S}_n \to \mathcal{S}_n \tag{53}$$

the compression *of* $\hat{\mathcal{A}}_\varphi^*\hat{\mathcal{A}}_\varphi$ *to* $\mathcal{S}_n$*,*

$$P\Big(\;\bar{v}_\dagger^{(k)} = \big(\hat{\mathcal{P}}^{\mathcal{S}}_{\hat{\varphi}^{(k-1)}}\big)^{\mu}\varsigma^{(k)} \;\; \textit{with} \;\; \big\|\varsigma^{(k)}\big\|_{L^2(\pi)} \le \mathsf{R}_* \;\; \textit{for every } k \le \bar{k}\;\Big) \;\longrightarrow\; 1,$$

where $\bar{v}_\dagger^{(k)} := P_{\mathcal{S}_n}\big(\varphi_\dagger - \hat{\varphi}^{(k-1)}\big)$ *and* $\bar{k}$ *is the stage count of Definition 22,* which the proof's own construction puts at $\bar{k} \ge 2$, so this clause is **never** a one-stage condition and is always read at a $\bar{v}_\dagger^{(k)}$ formed at a realised iterate as well as at one formed at the preliminary.

(P-b) (no hypothesis). *There is* ***no*** *hypothesis (P-b): the list runs* (P-a), (P-c)–(P-g)*;*

(P-c) (sieve approximation, a **lower** end for L_x) $\mathsf{R}\,L_x^{-s} = O\big(h_u\,\hat{\delta}_n\big)$*;*

(P-d) (uniformity, an *upper* end for L_x) (S:entropysieve)*, so that Lemma* S:unifop *clause (b) is available.* (P-c) and (P-d) are the two ends of one window and must be read together*: with* $\hat{\delta}_n = n^{-\alpha}$ *and* $h_u \asymp \hat{\delta}_n^{1/8}$ *they are* $L_x \gg n^{9\alpha/(8s)}$ *and* $L_x = o\big(n^{1-2\alpha}/\ln n\big)$*, so the window is non-empty if and only if*

$$s \;>\; \frac{9\alpha}{8\,(1-2\alpha)}; \tag{54}$$

(P-e) (stopping) *A7 in the amended form* (13)*, with* $\tau_{\mathrm{DP}} > 2$*;*

(P-f) (the linearisation remainder is absorbed) $\delta_n^2 = O\big(h_u^2\,\hat{\delta}_n \ln n\big)$*, a hypothesis in which a bandwidth appears explicitly, as (P-c) and (P-g) also do, and no claim of exhaustiveness is made in naming it so;*

(P-g) (x-smoothing bias) $\mathsf{R}\,h_x^{s-1/2} = O\big(h_u\,\hat{\delta}_n\big)$*, an* upper *bound on* h_x*, which C5′ does not carry;*

and suppose finally

$$\mu \;>\; \tfrac{5}{6}. \tag{55}$$

Then there is a finite $\bar{k} \geq 2$*, not depending on* n *and computable from* $(\mu, \varsigma_0, \alpha)$*, such that the* ***infeasible*** *reading* $\hat{\varphi}^{\pi}$ *of Definition 22 — clause* (b) *of that definition with the adjoint formed at the population* π *and the projection* $L^2(\pi)$*-orthogonal onto* $\mathcal{S}_n$*, which is the object every step of the proof below runs, and* ***not*** *the estimator* $\hat{\varphi}$ *that clause instructs, whose adjoint is formed at the floored* $\hat{\pi}$ *— satisfies*

$$\big\|\hat{\varphi}^{\pi} - \varphi_{\dagger}\big\|_{L^2(\pi)} \;=\; O_P(\delta_n), \qquad \delta_n \;:=\; \big(\hat{\delta}_n \ln n\big)^{2\mu/(2\mu+1)}, \tag{56}$$

and $\delta_n = o_P(n^{-1/4})$ *whenever* (7) *holds.* What is **not** claimed is that $\hat{\varphi}$ satisfies (56)*, and no hypothesis of this proposition reaches that passage: it is the open item Remark* S:phirscope(7)*, whose parts* (7c) *and* (7d) *are open.* In particular this proposition does **not** discharge A5(ii),

which is stated at $\hat{\varphi}$; Theorem 29 carries that clause on its own line, at (T-4)*, and carries it as a hypothesis.*

Notes on Proposition 28. What its hypotheses carry, where the threshold (55) comes from and why (P-a.3) is the load-bearing one are Remark S:phirhyp; why (P-a.3) is spectral and at the compression is Remark S:pasrc; why (P-a)'s clauses are w.p.$\to 1$ envelopes is Remark S:paenvelope; the nonlinear route it declines is Remark S:lfnonlinear; and what it does *not* claim is Remark S:phirscope. *The last of those is not a pointer to fine print*: the conclusion above is scoped to the infeasible reading $\hat{\varphi}^{\pi}$, and Remark S:phirscope(7) is where the passage to the estimator Definition 24(b) actually plugs in is divided into five parts — two settled outright, one settled with an undischarged rider, two open.

Theorem 29 (Asymptotic normality of the cross-fitted one-step estimator)**.** *The one-step estimator is $\sqrt{n}$-consistent and asymptotically normal with the sandwich variance, under mild ill-posedness and a propensity smooth enough to put the structural rate strictly inside $n^{-1/4}$. Suppose*

(T-1) (sampling, model, identification) *A0; A1; A1b(i), A1b(iii), A1b(iv);*

(T-2) (the representer and the weight) *A2; A2′(i.a), A2′(i.b), A2′(iii); N1 at $L_\star$,* which is $p(\mathsf{L}-1)$ scalar conditions and **not** a condition on a contraction of the $\mathsf{L}-1$ columns *(Corollary 21),* and which, together with (R-a), pins the anchor A2′(iii) leaves free*:* (R-a) *forces $\hat{b}^{(-k)} \to b_\star$, so $b_\star$ must be the N1 anchor of Remark* S:anchor *and the construction must target it;*

(T-3) (index set, densities, propensity) *B1(i), (ii), (iv); C1(ii); C2′(i); A4(i), A4(ii), A4(iii);*

(T-4) (rates) *A5(i) and A5(ii) in the form* (S:deltalog)*; A5(iii); and A5(vi) with $s > \frac{1}{2}$, read at each fold as in Definition 24(d).* A5(ii) is **assumed** here and is not delivered by (T-5)*: it is stated at $\hat{\varphi}$, the estimator Definition 24*(b) *plugs in, while Proposition 28 concludes at the infeasible $\hat{\varphi}^{\pi}$, and the passage between them is Remark* S:phirscope(7)*,*

two of whose five parts are open. This theorem is therefore conditional on that clause, and says so on its own hypothesis line rather than in a remark*;*

(T-5) (the φ-step) *the hypotheses of Proposition 28, read at the anchor $\theta_\circ$ of Definition 24(b), H1; A5(vi′); A8's basis and upper-end clauses; $p_K \geq s - \frac{1}{2}$, which against (54)'s floor gives $\alpha < \frac{68}{145}$; (P-a.1)–(P-a.3), (P-c)–(P-g); and (55), $\mu > \frac{5}{6}$ strictly, so that (56) holds on each $\mathcal{I}_{-k}$* — at the **infeasible** reading $\hat{\varphi}^{\pi}$ of Definition 22 that (T-4) names, and not at the $\hat{\varphi}^{(-k)}$ of Definition 24(b)*; so this group supplies A5(i) and the fold reading of A5(vi′) and does **not** supply A5(ii);*

and suppose in addition

(R-a) (the weight is built as an anchored antiderivative, with a bounded derivative) *for each k, $\hat{L}^{(-k)} = L^{\hat{b}^{(-k)}}_{\hat{\ell}^{(-k)}}$ in the sense of Definition 19(b), with $\hat{\ell}^{(-k)}$ continuously differentiable in u at every point and bounded,*

$$\max_{k\leq \mathsf{F}} \left\{ \left\| \hat{\ell}^{(-k)} \right\|_\infty + \left\| \partial_u \hat{\ell}^{(-k)} \right\|_\infty \right\} = O_P(1), \qquad \max_{k\leq \mathsf{F}} E \left\| \hat{b}^{(-k)}(\Delta Z) \right\|^4 = O_P(1),$$

and $\max_{k\leq \mathsf{F}} \|\hat{L}^{(-k)} - L_\star\|_{L^2(P)} = o_P(1)$*;*

(R-b) (fourth moments, and exactly two of them) $E\|\Delta Z\|^4 < \infty$ *and* $E[\Delta U^4] < \infty$*;*

(R-c) (the $\varrho \times L$ cross term) ***either*** (c.1) $D_k = \hat{L}^{(-k)} - L_\star$ *satisfies N1′ for every k,* ***or*** (c.2) $\max_k \|D_k\|_{L^2(P)} = O_P(\epsilon_n^L)$ *with $r_n\, \epsilon_n^L = o(n^{-1/2})$;*

(R-d) (no hypothesis). *There is **no** hypothesis (R-d): the list runs (R-a), (R-b), (R-c), (R-e), on the* (P-b) *pattern above;*

(R-e) (the first-order bias) ***either*** (e.1) *A5(iv) read at each fold, i.e. the third display of (43) together with $\delta_n \epsilon_n = o_P(n^{-1/2})$,* ***or*** (e.2) *the hypotheses of Lemma* S:balance(iii)*, the sieve solve sets term* (II) *of* (S:balance3) *to zero, A8 holds with a B-spline basis of degree exceeding $s - \frac{1}{2}$, A5(vi) holds and* (S:nuhat) *holds, together with* $\max_k \|\hat{\ell}^{(-k)} - \ell_\star\|_{L^2(P)} =$

$o_P(1)$, ***and*** *with*

$$s \;>\; \frac{1}{2\,(1-2\alpha)}. \tag{57}$$

Then, with $\hat\theta$ *of* (47) *and* J, Σ *of* (40),

$$\begin{aligned}\sqrt{n}\left(\hat\theta - \theta_\dagger\right) \;&=\; J^{-1}\,\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_\star(O_i) \;+\; o_P(1)\\ &\xrightarrow{d}\; \textit{the centred Gaussian law on } \mathbb{R}^p \textit{ with covariance matrix } J^{-1}\Sigma J^{-1\prime},\end{aligned} \tag{58}$$

and $\hat V \xrightarrow{p} J^{-1}\Sigma J^{-1\prime}$, *so that for* ***every*** $\vartheta \in \mathbb{R}^p$ *with* $\|\vartheta\| = 1$,

$$\sqrt{n}\,\vartheta'\left(\hat\theta - \theta_\dagger\right)\Big/\left(\vartheta'\hat V\vartheta\right)^{1/2} \;\xrightarrow{d}\; \textit{the standard Gaussian law on } \mathbb{R},$$

the ratio being read on $\{\vartheta'\hat V\vartheta > 0\}$, *an event of probability tending to one.*

Notes on Theorem 29. What each hypothesis group does and does not carry is Remark S:rootngroups; how the conclusion is read is Remark S:rootnread; what the theorem covers and the six things it does not are Remark S:rootnscope; the clause-by-clause usage audit is Remark S:rootnhyp; the logarithm in (56) is Remark S:rootnlog.

Corollary 30 (T2′: the rate floor, without (7))**.** *The same decomposition delivers a rate whether or not the frontier holds;* on the sieve branch, and only there, every exponent in that rate is a function of the primitives. *Suppose (T-1)–(T-3), (T-5), (R-a), (R-b) hold, together with A5(i), A5(vi) with* $s > \frac{1}{2}$, $\delta_n \to 0$, $\epsilon_n \to 0$, $r_n \to 0$, *and (R-c.2), (R-e.1) with the two products left* unrestricted. *Then*

$$\left\|\hat\theta - \theta_\dagger\right\| \;=\; O_P\!\left(n^{-1/2} \;+\; \delta_n\epsilon_n \;+\; \delta_n^2 \;+\; r_n\delta_n \;+\; r_n\epsilon_n^L\right). \tag{59}$$

Under (R-e.2) in place of (R-e.1) the term $\delta_n\epsilon_n$ *is replaced by* $O_P(n^{-1/2}\delta_n) + O_P(\mathsf{R}\,L_x^{-s})$, *and no rate for* $\hat\ell$ *enters* (59) *at all. Finally,* (58) *is the special case of* (59) *in which every term after the first is* $o(n^{-1/2})$.

Notes on Corollary 30. How much of (59) is in primitives, and which branch buys which answer, is Remark S:ratefloorprim.

Theorem 29 is our main result, and what carries it is an *exact* decomposition rather than an expansion. The bias decomposition of the supplement writes the conditional bias of the debiased moment at the estimated nuisances as the balancing pairing $-\langle \nu_{g_k}/\pi, h_k \rangle$ plus exactly three remainders, one quadratic in the structural error and two products of two estimation errors. There is no Taylor remainder in the weight or the propensity directions, because the moment is bi-affine there. What makes each of the three $o_P(n^{-1/2})$ is the frontier (7).

The empirical-process term costs nothing at scale $n^{-1/2}$, and that is a positioning commitment rather than a technical convenience. Lemma L-8 of the supplement obtains $\sqrt{n}\sum_{k=1}^{\mathsf{F}} \frac{n_k}{n}(\mathbb{E}_{n,k} - E)[\hat{\psi}^{(-k)} - \psi_\star] = o_P(1)$ from conditional Chebyshev at a frozen nuisance, so the only input is $L^2(P)$-consistency of the estimated moment: no entropy condition, no Donsker property, no stochastic equicontinuity at the estimated point. We make that concession once, in the φ-step, where we take a uniform law with a rate over a slowly growing ball at scale $\hat{\delta}_n \gg n^{-1/2}$. A penalised sieve minimum-distance treatment of θ would need the harder statement at the harder scale.

The sandwich is simpler than its construction suggests. At the $J = E[\Delta Z \Delta Z']$ of §2, the variance splits exactly, by the same tower step applied to the cross term: $\Sigma = E[\Delta Z \Delta Z' \Delta U^2] + E[L_\star \widetilde{W} \widetilde{W}' L_\star']$. So the sandwich is the variance the infeasible plug-in (45) would have at a *known* $\varphi_\dagger$, plus a positive semidefinite term that does not vanish under overlap.

That the second term does not vanish is reached differently at $\mathsf{L} \geq 3$, and the binary route is unavailable there rather than merely weaker. At $\mathsf{L} = 2$, $\widetilde{W}^2 \geq \varepsilon_\varrho^2$ surely under A4(i), which bounds the term below by $\varepsilon_\varrho^2 E[L_\star L_\star']$. At $\mathsf{L} \geq 3$ that argument fails outright: $\widetilde{W}\widetilde{W}'$ has rank one, so no sure inequality $\widetilde{W}\widetilde{W}' \succeq cI$ holds at any $c > 0$, and $L_\star \widetilde{W}\widetilde{W}' L_\star'$ is singular at every sample point once $p > 1$. The route that works at every L is the conditional one,

and it is what (6) was written for: under A1, $E[\widetilde{W}\widetilde{W}' \mid \Delta U, \Delta Z] = \Omega(\Delta Z)$, and A4(i) gives $\Omega(\Delta Z) \succeq (\varepsilon_\varrho/\mathsf{L})I$.

What the debiasing weight buys is not variance but the removal of the first-order bias that the plug-in carries at exact order $\|h_k\|$. That is the standard debiasing trade (see Chernozhukov, Escanciano, et al., 2022; Ichimura and Newey, 2022), and it is not a defect.

We do not compare the estimator to a semiparametric efficiency bound. The bound Poirier (2017) computes under independence restrictions is for a model without a structural inverse (see also Ai and Chen, 2012; Bickel et al., 1993), and computing one here, where the nuisance solves an ill-posed nonlinear operator equation, is a step we leave to future research.

No correction for the estimation of ϱ appears in the influence function, and its absence is a consequence rather than an omission. By Proposition 20(iii) the ϱ-derivative of the moment vanishes exactly under N1, so there is nothing for a correction to remove; Lemma L-11 of the supplement states this as a property of the limit and not as an approximation to it.

The positivity the studentised statement needs is *implied by the theorem's hypotheses rather than assumed*. Notice that its conclusion is quantified over every ϑ with $\|\vartheta\| = 1$, not over the directions on which the sandwich happens to be positive. By part (v) of the moment-validity lemma of the supplement, $\Sigma \succ 0$ follows from A1b(iv) and C2′(i), both of which the theorem already carries, since $E[\Delta Z \Delta Z']$ nonsingular gives $P(\vartheta'\Delta Z \neq 0) > 0$ and a conditional density for ΔU gives $P(\Delta U = 0) = 0$.

A proviso that a theorem's hypotheses entail belongs in its conclusion and not on the user's checklist.

The scope is mild ill-posedness, and severe ill-posedness is a place the theorem does not reach rather than a gap in the proof. What (7) requires is that δ_n be a *polynomial* rate faster than $n^{-1/4}$, which under H1's $\sigma_j \asymp j^{-a}$ is what a source condition delivers. Under exponentially decaying singular values the rate attainable from *any* source condition is logarithmic in n, so

A5(ii) fails at every r and the theorem is unavailable.

H1's scale supplies a shorter reading of that scope, and it needs qualifying. The reading pairs mild ill-posedness with $s > a + \frac{1}{2}$, hence $\mu > \frac{1}{2} + 1/(4a)$, and at $a \leq \frac{3}{4}$ that inequality is taken to deliver the source-index restriction the theorem carries.

The inequalities in that reading are statements about $\mu_{\mathrm{H1}} = s/(2a)$, the index of A2 at the *population* operator on all of $\mathcal{V}$ at $\theta_\dagger$, whereas (55) and (7) carry (P-a.3)'s primitive μ at the finite-rank compression of the realised operator at $\theta_\circ$. The bridge between them is the operator interpolation that the supplement's remark on (P-a.3) shows unavailable at every $2\mu > 1$, and the commutativity proviso that would restore it is stated and declined.

Read as an interpretation it is right; read as a scope statement it both excludes designs the theorem covers and claims designs it does not.

The rate condition cannot be relaxed by bounding the remainder more carefully, *and that is a weaker claim than necessity.* Proposition 27 makes the direct bound order-sharp under O-16(a), so no argument that merely improves it will weaken (7).

The sharpness statement invites readings that the supplement's scope remark blocks. It does not say that the realised remainder at $h_k = \hat{\varphi}^{(-k)} - \varphi_\dagger$ fails to be $o_P(\|h_k\|^2)$: a regularisation path confined to directions on which the form is small could do better, and nothing here excludes that. It does not say that $\delta_n = o(n^{-1/4})$ is *necessary* for $\sqrt{n}$-estimability of $\theta_\dagger$ by some estimator. And it does not say the moment class is exhausted.

What relaxing the frontier would require is a different selection, one with a smaller σ_q and compact part, which is open. The alternative is a restriction on the *direction* of the first-stage error, which is an extra hypothesis and not a sharper argument. The sharpness statement carries a hypothesis of its own, O-16, which asserts that the form does not degenerate and which does not follow from Corollary 14. Here q can vanish by cancellation inside the conditional expectation, and $q \not\equiv 0$ alone is not enough either, an antisymmetric $q\, f_{X_1,X_2}$

making the form vanish against the symmetric $(\Delta h)^2$.

We acknowledge that near the frontier the theorem's asymptotics do not bind at any sample size anyone will run, and we would rather say so than let a referee find it.

Write $\hat{\delta}_n = n^{-\alpha}$ and $\kappa = 2\mu/(2\mu+1)$. The θ-remainder is $O_P(n^{-1/2} + \sqrt{n}\,\delta_n^2)$ and its second term is $\sqrt{n}\,\delta_n^2 = n^{-\beta}(\ln n)^{2\kappa}$ with $\beta = 2\kappa\alpha - \frac{1}{2}$, which is strictly positive by (7). In $t = \ln n$ that is $e^{-\beta t}t^{2\kappa}$, which is *increasing* for $t < 2\kappa/\beta$, so the remainder *rises* in n up to $n^* = \exp\{2\kappa/\beta\}$ and only then falls. The Monte Carlo experiment at the configuration this theory names, which the supplement carries, runs at $\mu = 2$, hence $\kappa = \frac{4}{5}$, and at $\alpha = 0.32$, so that $\kappa\alpha = 0.256 > \frac{1}{4}$ and the design sits *inside* the frontier. There one gets $\beta = 0.012$ and $n^* = e^{400/3} \approx 10^{58}$.

That threshold is a fact about where the design sits and not about the theorem, and the same formula prices designs that sit elsewhere. It is not an artefact of a badly chosen parameterisation: $\beta \downarrow 0$ as $\kappa\alpha \downarrow \frac{1}{4}$, so $n^* \uparrow \infty$ at the frontier and the 10^{58} is a property of (7) itself. But n^* falls away from the frontier as steeply as it climbs towards it, and it does so well inside the admissible range rather than at its edge.

Take $d_Z = 1$. A first stage with $r = 5$ derivatives gives $\alpha = \frac{5}{11}$, hence $\beta = \frac{5}{22}$ and $n^* = e^{176/25} \approx 1{,}140$ at $\mu = 2$, and $n^* = e^{352/61} \approx 320$ at $\mu = 4$; at $r = 12$ and $\mu = 10$, $n^* = e^{400/87} \approx 99$. These are ordinary smooth designs with a scalar conditioning variable, satisfying every hypothesis the theorem carries, and at them the turn happens at sample sizes anyone will run. *So the concession is that the asymptotics do not bind near the frontier, and that the configuration that experiment runs is near it.* It is not that they bind nowhere.

Nothing in this weakens (58). The $o_P(1)$ there is a statement about the limit and it is proved. What changes is how finite-sample evidence should be read. Below n^* the object that describes the estimator is Corollary 30, in which δ_n^2, $\delta_n\epsilon_n$, $r_n\delta_n$ and $r_n\epsilon_n^L$ appear as *terms* of the rate and not as remainders sent to zero.

That corollary is honest about its own exponents in the same way. On the branch that prices $\hat{\ell}$ directly, two of them, ϵ_n and ϵ_n^L, are assumed and not derived, because neither has a supplier in the maintained block. Only on the sieve branch for the first-order bias, together with N1′ at each estimated direction, do both unsupplied symbols disappear, and the primitives the surviving exponents are functions of are (μ, r, d_Z, s).

So the correct reading of (7) is that it is a *qualitative* frontier: crossing it makes the remainder vanish eventually and says nothing about where.

Finally, the theorem is *pointwise in* P, and this is the quantifier the finite-sample caveats all sit inside. Every hypothesis is imposed at a single data-generating law, and μ, R_*, a, s, ε_ϱ, r, $\bar{c}_\pi$, the representer's budgets and the singular system of $\mathcal{A}_{\text{cond}}$ are all P-specific constants.

The consequences are real limitations rather than formalities. The $o_P(1)$ in (58) is not uniform over any neighbourhood of P, so the theorem does not deliver confidence sets whose coverage is honest over a class, and no such class is defined here. The 10^{58} of the previous paragraph is that gap displayed numerically: within any class containing laws arbitrarily close to the frontier the convergence cannot be uniform. And $\lambda_{\min}(\Sigma) > 0$ holds at each P and may be arbitrarily small along a sequence of laws, so the positivity established above is pointwise too.

A uniform statement would need a class on which those constants are bounded, a uniform version of (56) at a random compressed operator, and a Lindeberg condition uniform in P. We attempt none of the three.

5 Monte Carlo

We run the shipped estimator on a multinomial design, on the code Section 6 estimates with and with every nuisance fitted from the sample. The instrument is drawn at four cells and read at $\mathsf{L} \in \{2, 3, 4\}$ off one draw, so the cardinality varies inside a replication and the three readings are paired. Three laws are drawn from that design: a balanced one; a skewed one,

which moves the distribution of the differenced controls and leaves the propensity correctly specified; and a third that misspecifies $\hat{\varrho}$, the nuisance A4(iii) prices. The estimator cross-fits at $\mathsf{F} = 5$ folds, so the operator is built on $\lceil 0.4n \rceil$ design points. Everything below is produced by `scripts/simulations/sim_merged_cardinality.R`.

The grid is $n \in \{400, 800, 1{,}600\}$ and it is chosen to bracket the panel of Section 6, which has $n = 983$ counties and so sits between its second and third points. *That is the experiment and not a limitation of it.* Three further experiments run the same estimator against population objects computed in closed form, with the nuisances injected at controlled rates rather than fitted, and they reach $n = 181{,}019$ at 5,000 replications in every cell; the supplement carries all three, in its first four sections. The first of them, at the configuration the theory names, finds the estimator matching its limit law at large n only: the simulated standard deviation clears a two-standard-error bar against its asymptotic value at $n = 181{,}019$ at one of two equally admissible constants and at neither below $n = 64{,}000$. The second, run at a configuration with more than twenty times the frontier margin and a propensity no smoother, meets a pre-committed coverage criterion from $n = 354$ upward, which locates that verdict in the design rather than in Theorem 29. What none of the three can answer is how the shipped code behaves at a sample size a reader has, because none of them runs the shipped code. *Nor do the three suites merge into a single cardinality contrast*: they differ in the design, in the estimator and in the nuisance regime, and their sample-size grids do not intersect at any point, so a table whose columns read $\mathsf{L} = 2$ and $\mathsf{L} = 4$ but whose rows came from two of them would attribute three differences to the cardinality. The supplement sets the three beside each other under a rule that keeps one design, one estimator and one nuisance regime to a row (Table S29 of the supplement).

One precision statement governs everything below. The suite was run at 120 replications and returns between 112 and 120 usable ones, the shortfall being the draws on which the boundary kernel drove $\hat{\pi}$ non-positive at a design point, so that the adjoint could not be formed and

the replication was dropped rather than returned degraded; the count is a property of the cell and is identical across the three cardinalities in all nine of them, which is what leaves the paired contrast defined. At a coverage rate near 0.95 the Monte Carlo standard error is about 0.02, so two of them resolve to about ± 0.040 and a gap smaller than that is not evidence of anything. *That is the smallest resolution in the suite and it is calibrated at one point*: at the misspecified constant, where coverage runs from 0.277 to 0.795, the printed standard errors are 0.038 to 0.046. Every reading below is priced at the standard error its own cell prints. What the dropped draws could have done is bounded rather than merely disclosed, and each table carries the bound in its own header row: at most 0.024 on the balanced design and at most 0.063 on the misspecified one. The selection is real and its direction is known, a draw surviving when the nonparametric stage behaved, so the printed rates are the optimistic reading of that interval.

On the balanced design the estimator behaves as the theorem asks at every cardinality. The bias at $n = 1{,}600$ is inside two of its own Monte Carlo standard errors at all nine cells, the largest being the constant at four cells at -0.0200 (0.0125). The quantity Theorem 29 makes constant in n, $\sqrt{n}$ times the simulated standard deviation, falls across the grid at seven of the nine cells and is larger at $n = 1{,}600$ than at $n = 400$ at two, both of them the constant, by 19.19 and 5.35 per cent; *neither move is resolved from flat*, at 1.50 and 0.45 of the standard error Panel B prints, which is computed from the fourth moment rather than from the Gaussian formula because the realised kurtosis reaches 10.8 over the suite. Coverage of the nominal 95% interval at $n = 1{,}600$ sits between 0.866 and 0.975 at all nine cells (Table 1).

That range is true and it conceals the shape of its own low end. Three of the nine cells sit more than two Monte Carlo standard errors below nominal and they are not scattered: all three are the constant, at 0.882, 0.866 and 0.866 down the three cardinalities. The shortfall does not shrink in n either. At four cells the constant runs 0.863, 0.842 and 0.866 across

the three sample sizes, which is flat and 2.7 standard errors under nominal at the largest of them. What kind of failure that is, the interval answers and the limit law does not. The mean estimated standard error over the simulated standard deviation is below one in 77 of the suite's 81 cells and below 0.75 in 18 of them; take the coverage a centred normal carrying that ratio would deliver and subtract the realised coverage, and the difference is inside ± 0.070 at all 54 well-specified cells, at a root mean square of 0.024 (Table S32 of the supplement). *The entire well-specified coverage shortfall is the interval's scale and not the limit law.*

The third design is the one that breaks, and it breaks in the propensity. Its constant's bias does not fall in n at four cells, running -0.3040, -0.3242 and -0.3830, and $\sqrt{n}$ times its standard deviation *rises*, 6.031, 6.960 and 8.605, where the theorem makes it constant; that move is 3.40 of its own standard error and so, unlike the two at the balanced design, it *is* resolved from flat. Coverage of the constant at $n = 1{,}600$ is 0.697 at two cells, 0.294 at three and 0.277 at four, and the paired bias difference against the binary arm is -0.2287 (0.0226) and -0.2765 (0.0241), which are 10.1 and 11.5 standard errors and are not a resolution artefact. The two slope coefficients are untouched, covering between 0.891 and 0.950 at every cardinality at that sample size (Table 2). The damage is therefore confined to the coordinate the propensity enters, and it grows with the number of cells at the two larger sample sizes. *At* $n = 400$ *it does not*: coverage there runs 0.795, 0.661 and 0.732 and the bias -0.1813, -0.3266 and -0.3040, so four cells show less damage than three on both measures. Both reversals are inside the bound $\mathrm{SE} \le \mathrm{SE}_3 + \mathrm{SE}_4$, at 0.82 and 0.38 of it, and $n = 400$ under this design is the thinnest cell in the suite, at 112 usable replications.

The stopping rule fails in the same place and in the same direction, and its constant is where the failure is visible. At the two well-specified designs the discrepancy rule selects an iterate inside its budget in most replications at every cardinality, at 1.000, 0.917 and 0.875 on the balanced design at $n = 1{,}600$ and 1.000, 0.792 and 0.667 on the skewed one. At the misspecified design it selects one in 0.875 of replications at two cells, 0.250 at three and *none*

Table 1: The shipped engine at three cardinalities: the balanced design

Block III — multinomial design, shipped engine, nuisances estimated, *Balanced design*
`sim_multinomial_dgp.R` with `lf_panel.R`; $\mathsf{L} \in \{2, 3, 4\}$ off one draw
B used of 120 drawn: 117 (n=400), 120 (n=800), 119 (n=1,600); equal at all L
Dropped draws move any coverage by at most 0.024; worst 0.966 → [0.942, 0.967]

Design	n	Coefficient	$\mathsf{L} = 2$	$\mathsf{L} = 3$	$\mathsf{L} = 4$
Panel A: bias, with its Monte Carlo standard error. Truth (0, 1.0, −0.5).					
Balanced	400	Constant	0.0875 (0.0319)	-0.0178 (0.0258)	-0.0804 (0.0239)
		z_1	0.0090 (0.0270)	-0.0145 (0.0167)	-0.0141 (0.0128)
		z_2	0.0240 (0.0262)	0.0140 (0.0170)	-0.0010 (0.0147)
	800	Constant	0.0074 (0.0194)	-0.0305 (0.0191)	-0.0429 (0.0173)
		z_1	-0.0015 (0.0103)	-0.0262 (0.0098)	-0.0267 (0.0089)
		z_2	-0.0014 (0.0103)	-0.0005 (0.0092)	0.0001 (0.0088)
	1,600	Constant	0.0139 (0.0129)	0.0144 (0.0152)	-0.0200 (0.0125)
		z_1	0.0031 (0.0068)	0.0001 (0.0072)	-0.0038 (0.0059)
		z_2	0.0054 (0.0078)	0.0012 (0.0062)	0.0021 (0.0054)
Panel B: $\sqrt{n}\times$ s.d., which the theorem makes constant in n, with its s.e.					
Balanced	400	Constant	6.896 (0.494)	5.571 (0.416)	5.178 (0.488)
		z_1	5.848 (0.845)	3.621 (0.474)	2.766 (0.265)
		z_2	5.659 (0.535)	3.686 (0.245)	3.175 (0.254)
	800	Constant	6.024 (0.490)	5.920 (0.466)	5.356 (0.331)
		z_1	3.182 (0.252)	3.021 (0.192)	2.772 (0.193)
		z_2	3.186 (0.188)	2.850 (0.186)	2.724 (0.209)
	1,600	Constant	5.621 (0.445)	6.640 (0.582)	5.455 (0.368)
		z_1	2.959 (0.173)	3.134 (0.272)	2.564 (0.170)
		z_2	3.414 (0.216)	2.709 (0.126)	2.372 (0.152)
Panel C: coverage of the nominal 95% interval, with its Monte Carlo standard error.					
Balanced	400	Constant	0.855 (0.033)	0.889 (0.029)	0.863 (0.032)
		z_1	0.855 (0.033)	0.966 (0.017)	0.957 (0.019)
		z_2	0.803 (0.037)	0.863 (0.032)	0.940 (0.022)
	800	Constant	0.942 (0.021)	0.908 (0.026)	0.842 (0.033)
		z_1	0.958 (0.018)	0.917 (0.025)	0.942 (0.021)
		z_2	0.975 (0.014)	0.950 (0.020)	0.925 (0.024)
	1,600	Constant	0.882 (0.030)	0.866 (0.031)	0.866 (0.031)
		z_1	0.975 (0.014)	0.916 (0.025)	0.958 (0.018)
		z_2	0.916 (0.025)	0.975 (0.014)	0.958 (0.018)

Notes: Block III of Table S29 of the supplement: a multinomial design run through the shipped engine with every nuisance estimated from the sample, at $\mathsf{L} \in \{2, 3, 4\}$ off one draw. *The realised replication count is in this table's own header row and is not repeated here*, so that the two cannot drift apart; the shortfall against the 120 drawn is the draws on which the boundary family drove $\hat{\pi}$ non-positive, so that the adjoint could not be formed. Rows are the three coefficients of $\theta_\dagger$, whose true values are (0, 1.0, −0.5). Panel A is the bias with its Monte Carlo standard error; Panel B is $\sqrt{n}$ times the simulated standard deviation, which Theorem 29 makes constant in n, so a column flat down the three sample sizes is the theorem's prediction; Panel C is coverage of a nominal 95% interval with its Monte Carlo standard error. *At those counts a coverage rate resolves to about* ±0.04, so a gap smaller than that is not evidence; Table S33 of the supplement carries the paired contrast, which is the sharper comparison. No population variance exists for this block, so no standard-deviation ratio against an asymptotic value is reported for it. Source: `scripts/simulations/sim_merged_cardinality.R`; table `paper/tables/sim/sim_merged_rootn_balanced.tex`.

Table 2: The shipped engine at three cardinalities: a misspecified propensity

Block III — multinomial design, shipped engine, nuisances estimated, *Misspecified $\hat{\varrho}$ design*
`sim_multinomial_dgp.R` with `lf_panel.R`; $\mathsf{L} \in \{2, 3, 4\}$ off one draw
B used of 120 drawn: 112 (n=400), 118 (n=800), 119 (n=1,600); equal at all L
Dropped draws move any coverage by at most 0.063; worst 0.946 → [0.883, 0.950]

Design	n	Coefficient	$\mathsf{L} = 2$	$\mathsf{L} = 3$	$\mathsf{L} = 4$
Panel A: bias, with its Monte Carlo standard error. Truth (0, 1.0, −0.5).					
Misspecified $\hat{\varrho}$	400	Constant	-0.1813 (0.0355)	-0.3266 (0.0316)	-0.3040 (0.0285)
		z_1	-0.0168 (0.0208)	0.0238 (0.0156)	0.0254 (0.0143)
		z_2	-0.0385 (0.0212)	-0.0268 (0.0164)	-0.0040 (0.0154)
	800	Constant	-0.1550 (0.0219)	-0.2916 (0.0220)	-0.3242 (0.0227)
		z_1	0.0104 (0.0139)	0.0224 (0.0107)	0.0326 (0.0107)
		z_2	-0.0049 (0.0141)	-0.0069 (0.0123)	-0.0120 (0.0101)
	1,600	Constant	-0.1066 (0.0163)	-0.3353 (0.0182)	-0.3830 (0.0197)
		z_1	-0.0025 (0.0078)	0.0095 (0.0071)	0.0114 (0.0065)
		z_2	0.0075 (0.0082)	-0.0063 (0.0069)	-0.0014 (0.0064)
Panel B: $\sqrt{n}\times$ s.d., which the theorem makes constant in n, with its s.e.					
Misspecified $\hat{\varrho}$	400	Constant	7.519 (0.641)	6.698 (0.521)	6.031 (0.442)
		z_1	4.395 (0.525)	3.305 (0.205)	3.034 (0.189)
		z_2	4.480 (0.421)	3.474 (0.265)	3.265 (0.209)
	800	Constant	6.734 (0.463)	6.753 (0.558)	6.960 (0.473)
		z_1	4.282 (0.348)	3.302 (0.218)	3.284 (0.217)
		z_2	4.330 (0.397)	3.782 (0.281)	3.096 (0.219)
	1,600	Constant	7.128 (0.743)	7.943 (0.596)	8.605 (0.614)
		z_1	3.390 (0.221)	3.076 (0.263)	2.848 (0.192)
		z_2	3.571 (0.266)	3.021 (0.208)	2.779 (0.180)
Panel C: coverage of the nominal 95% *interval, with its Monte Carlo standard error.*					
Misspecified $\hat{\varrho}$	400	Constant	0.795 (0.038)	0.661 (0.045)	0.732 (0.042)
		z_1	0.929 (0.024)	0.938 (0.023)	0.929 (0.024)
		z_2	0.902 (0.028)	0.946 (0.021)	0.946 (0.021)
	800	Constant	0.771 (0.039)	0.525 (0.046)	0.424 (0.045)
		z_1	0.907 (0.027)	0.924 (0.024)	0.898 (0.028)
		z_2	0.890 (0.029)	0.873 (0.031)	0.924 (0.024)
	1,600	Constant	0.697 (0.042)	0.294 (0.042)	0.277 (0.041)
		z_1	0.950 (0.020)	0.941 (0.022)	0.933 (0.023)
		z_2	0.899 (0.028)	0.891 (0.029)	0.908 (0.027)

Notes: As Table 1, at the design that misspecifies $\hat{\varrho}$ and therefore violates (T-4). *This is the arm that is supposed to fail*, and it is included so that a match at the other two designs is a non-rejection by a procedure shown able to reject. What the table adds beyond the failure is where it lands and how it moves: the damage is confined to the constant, the coordinate the propensity enters, and it grows with the number of cells at $n = 800$ and $n = 1{,}600$ but not at $n = 400$, where four cells sit above three on both bias and coverage by less than $\mathrm{SE}_3 + \mathrm{SE}_4$; the two slope coefficients cover between 0.873 and 0.950 at every cell. Panels and resolution are as in Table 1, and the realised replication count is in this table's own header row; its $n = 400$ cell is the thinnest in the suite. Source: `scripts/simulations/sim_merged_cardinality.R`; table `paper/tables/sim/sim_merged_rootn_misspec.tex`.

at all at four, and a rule that never fires returns the cap, which is the configuration Section 6's bootstrap runs under and says so (Table 3). *The threshold is not pinned down by the data, and we do not claim otherwise.* The same table's first block sweeps it on the abstract design, where moving c_{14} by a single factor of three takes the probability that the rule fires at all from 0.077 to 0.760 at $n = 500$ under a zero preliminary and from 0.027 to 0.603 under a rate-accurate one. Outside that neighbourhood the rule is degenerate, returning the cap in every replication of all twelve cells at $\mathsf{c}_{14} \leq 10^{-3}$ and one step in every replication of all twelve at $\mathsf{c}_{14} \geq 10^{-1}$, so it stops in the interior only inside a window one to seven per cent of the four decades swept, and that window moves with n (Table S5 of the supplement). What we can say is that the cap is not an independent cause of the misspecified design's failure. Run at $\tau_{\mathrm{DP}} = 0$, where the rule never fires and the returned iterate *is* the cap, moving $\bar{m}$ from 15 to 100 at $n = 1{,}600$ and $\mathsf{L} = 4$ raises the misspecified design's integrated squared error from 0.195 to 0.488, at Monte Carlo standard errors of 0.010 and 0.067, while it lowers the balanced design's from 0.091 to 0.036. Iterating longer helps where the propensity is right and hurts where it is wrong.

One hypothesis of Theorem 29 fails on every design here, and we report it rather than absorb it. N1 asks that $E[L(\Delta U, \Delta Z) \mid \Delta Z]$ vanish, which Corollary 21 types as $p(\mathsf{L} - 1)$ scalar conditions and not p. Measured in population on the weight the engine solves for, the entrywise defect over $\max_{j,l} \|L_{jl}\|$ runs from 0.531 to 0.720 across the nine design-by-cardinality cells at $n_{\mathrm{solve}} = 640$, against a floor of 7.6×10^{-14} returned by the same code with the conditional dependence switched off (Table S30 of the supplement). *This is not a defect in Theorem 29.* It sits between the theory and the implementation: the engine builds ℓ in a class over the residual alone and anchors it by subtracting a constant, and on a location-scale family with a continuum of conditional means the zero weight is the only member of that class that satisfies N1. The re-anchoring of Section 4 would restore it at the same ℓ; the engine does not build it. Two things this does not establish. It says nothing about Section 6's data, where the defect is estimated on a panel whose conditional law is unknown, and

Table 3: The one statistic both φ-step arms measure

Design and preliminary	Threshold	Budget	n	Pr(rule selects an iterate) $\mathsf{L}=2$	$\mathsf{L}=3$	$\mathsf{L}=4$
Block I — named configuration, abstract operator, nuisances injected at controlled rates $(\mu, s, a, r, d_Z, p_K) = (2, 3, \frac{3}{4}, \frac{8}{3}, 3, 8)$, $p = 4$; $\mathsf{L} = 2$ in this block						
zero preliminary	10^{-2}	39,528,911,316	500	0.077	—	—
		185,035,802,711	2,000	0.250	—	—
		875,713,042,937	8,000	0.653	—	—
	3×10^{-2}	39,528,911,316	500	0.760	—	—
		185,035,802,711	2,000	0.990	—	—
		875,713,042,937	8,000	1.000	—	—
rate preliminary	10^{-2}	39,528,911,316	500	0.027	—	—
		185,035,802,711	2,000	0.147	—	—
		875,713,042,937	8,000	0.220	—	—
	3×10^{-2}	39,528,911,316	500	0.603	—	—
		185,035,802,711	2,000	0.820	—	—
		875,713,042,937	8,000	0.933	—	—
Block III — multinomial design, shipped engine, nuisances estimated `sim_multinomial_dgp.R` with `lf_panel.R`; $\mathsf{L} \in \{2, 3, 4\}$ off one draw						
Balanced	$\tau_{\mathrm{DP}} = 0.7$	100	400	0.958 [23/24]	1.000 [24/24]	1.000 [24/24]
			800	1.000 [24/24]	0.917 [22/24]	0.750 [18/24]
			1,600	1.000 [24/24]	0.917 [22/24]	0.875 [21/24]
Skewed	$\tau_{\mathrm{DP}} = 0.7$	100	400	0.958 [23/24]	0.917 [22/24]	0.917 [22/24]
			800	0.958 [23/24]	0.917 [22/24]	0.833 [20/24]
			1,600	1.000 [24/24]	0.792 [19/24]	0.667 [16/24]
Misspecified $\hat{\varrho}$	$\tau_{\mathrm{DP}} = 0.7$	100	400	0.958 [23/24]	0.833 [20/24]	0.125 [3/24]
			800	0.958 [23/24]	0.500 [12/24]	0.042 [1/24]
			1,600	0.875 [21/24]	0.250 [6/24]	0.000 [0/24]

Notes: The probability that the discrepancy rule selects an iterate at or before its budget, which in Block I is one minus the probability that the rule returned the budget without ever meeting its threshold, and in Block III is the share of paths with a realised stopping index. *The two arms do not parameterise the threshold the same way and their budgets differ by more than eight decades*, which is why both are printed as columns: Block I sweeps c_{14} against the cap (36), of order 10^{10} to 10^{11}, and Block III fixes $\tau_{\mathrm{DP}} = 0.7$ against a cap of 100. Reading a Block I row against a Block III row without reading those two columns is reading two different rules. Only the two swept values at which Block I's branch frequencies are non-degenerate are printed, 10^{-2} and 3×10^{-2}; outside them the answer is constant, at Pr(cap) = 1 in all twelve cells for $\mathsf{c}_{14} \leq 10^{-3}$ and $\Pr(\hat{m}^{(1)}{=}1) = 1$ in all twelve for $\mathsf{c}_{14} \geq 10^{-1}$. *That degeneracy is not something Table S5 of the supplement prints.* Its Panel A reports median stopped iterations, and a median does not establish what every replication did; its Panel B prints branch frequencies at $\mathsf{c}_{14} = 10^{-2}$ alone. What that table does print bearing on it is Panel B's last column, the interval of c_{14} in which the rule stops in the interior; the opening pages of this section read those intervals off it and state the same degeneracy in the same terms. Both statements rest on the sweep behind the two tables, `quality_reports/sim/sim_thm_rootn_phistep.rds`, and not on a printed cell. Block III entries carry the count of replications in brackets, out of 24 paths per cell. Source: `scripts/simulations/sim_merged_cardinality.R`; table `paper/tables/sim/sim_merged_stopping.tex`.

it does not show that the theory's own $\ell_\star$ fails N1 here. By Proposition 20(iii) a live defect leaves the moment's ϱ-derivative non-vanishing, so $\hat{\varrho}$ enters at first order and the sandwich omits a term that does not shrink; *we report that as consistent with the standard-error ratio above and not as demonstrated by it*, the correction itself not being computed here.

Three readings this section does not support. It does not establish that the shipped engine and the abstract construction agree, because no cell here shares a sample size with any of the three experiments in the supplement. It does not establish that the cardinality is harmless at the well-specified designs, because 112 to 120 replications cannot resolve a coverage gap below about ± 0.04 and the paired contrasts bound the bias difference rather than the coverage one (Table S33 of the supplement). And it does not establish that the N1 defect is what the standard-error ratio measures: the two quantities are set beside each other rather than fitted to one another.

6 Empirical application

Hedonic price regressions read the value of a local amenity off the cross-section of property values (see Rosen, 1974; Bishop et al., 2020), and regulatory designations are a standard source of exogenous variation in them (see Greenstone and Gallagher, 2008). Air quality is the amenity with the longest such literature. The regressor of interest is almost always entered linearly, and we relax that here, on a two-period county panel whose instrument is discrete because the regulation that generates it is a rule on thresholds. The section illustrates the estimator of Section 3 and claims nothing beyond the illustration; what it does not establish is set out in the last subsection rather than left to be discovered there.

The data are the county panel Chay and Greenstone (2005) assembled to measure how the 1970 Clean Air Act Amendments capitalised into property values. The file holds 989 counties. One candidate control, per-capita expenditure on pollution abatement, is missing in all 989 rows and is dropped as an empty column rather than as item non-response, which leaves

22 usable controls and not 23; six more are missing the local-fiscal block, which leaves an analysis sample of $n = 983$ (Table S34 of the supplement).

Write Y_{ij} for the log of county i's median property value in $j \in \{1970, 1980\}$, X_{ij} for its mean concentration of total suspended particulates in $\mu\text{g/m}^3$, W_i for its 1975 nonattainment designation, and ΔZ_i for an intercept together with d_Z differenced county characteristics, at $d_Z = 5$ throughout. That is the differenced model of Section 2 term by term: a continuous endogenous scalar entering through $\varphi_\dagger$, a discrete instrument, a differenced control vector, and a county effect the differencing removes. Descriptive statistics are in Table S35 of the supplement, where just under a third of the sample is designated, at a mean of 0.303. The intercept is not decoration and A4(ii) requires it: read against A1b(iv), that clause forces ΔZ to carry exactly one constant coordinate, which in a differenced equation is the period effect.

The object of economic interest here is $\varphi_\dagger$ and not $\theta_\dagger$, which inverts Theorem 29, whose $\sqrt{n}$ statement is about $\theta_\dagger$: the structural function is the hedonic price of air quality, while $\theta_\dagger$ collects coefficients on demographic and fiscal controls. Which of the two roles carries the question is a property of the application rather than of the model.

6.1 The instrument

The designation rule fires on either of two triggers: a county is designated if its annual geometric mean exceeds the standard, or if its second-highest daily reading does. We read W as the pair of triggers rather than as their disjunction, which gives a four-valued instrument: 685 counties trigger neither, 114 the bad-day rule alone, 66 the annual mean alone, and 118 both, the last three being the cell sizes Table 5 prints. The four cells are ordered in baseline particulates and are not ordered in the decadal change, so they are not a ladder in the regressor; both orderings come from the same trigger constructor on the same sample and are printed in no table. The refinement discards nothing, since the binary reading is the indicator of the first cell's complement.

The reason for reading the designation at $\mathsf{L} = 4$ is methodological and not empirical. Four cells are what the rule says, so the binary indicator is a coarsening of the datum rather than the datum itself, and under A1 the three contrasts the four cells generate are further restrictions on the same $\varphi_\dagger$, by Proposition 17. What the extra cells do not buy should be said in the same breath: not identification, since a coarsening imposes fewer restrictions; not testability, which is present already at $\mathsf{L} = 2$; and not instrument strength, since splitting an instrument creates no variation. What they buy is the decomposition Proposition 17 describes, in which a piece far from zero at the common fit is a cell where the four-valued reading conflicts with $\hat{\varphi}$.

All three cells exceed their own reference, at $p = 0.039$ for the bad-day cell, 0.006 for the annual-mean cell and 0.003 for the joint cell, against a null that redraws W from the fitted propensity (Table 5). The reading runs one way only, since a small p localises a conflict to that cell while a large one exonerates nothing: consistency across cells is necessary and not sufficient for A1 by Proposition 17(iii), and the false-negative rate of the attribution is unpriced. The table repeats the calculation at the cross-fitted propensity, and which of the two Proposition 17's diagnostic should use is not settled here. We report the three cells and draw no ordering from them.

The instrument moves the regressor. Nonattainment lowers the decadal change in mean particulates by 8.37 to 9.11 $\mu\text{g/m}^3$ across the four control sets, and does so at the 1 per cent level at every one of them (Table 4, Panel A). We read that table on the state-clustered standard error, because designation was made at the level of the Air Quality Control Region and A0 assumes independent sampling. Clustered, F runs from 10.6 with no controls to 13.1 at the widest set, so the first stage clears the conventional threshold of ten at every control set, with little room at the narrowest. Linear instrumental variables give -5.519×10^{-3} log points per $\mu\text{g/m}^3$, which at the 1970 mean of 66.0 $\mu\text{g/m}^3$ is an elasticity of -0.364, against the -0.20 to -0.35 Chay and Greenstone (2005) report on this design; it sits just past the

more negative end of that range and reproduces the sign and the order of magnitude rather than falling inside it.

Table 4: The instrument: first stage, reduced form, and the linear benchmark

	$d_Z = 0$	$d_Z = 5$	$d_Z = 10$	$d_Z = 22$
Panel A: first stage, dependent variable ΔX ($\mu g/m^3$)				
W	-8.627***	-8.369***	-8.755***	-9.108***
robust SE	(1.742)	(1.697)	(1.896)	(1.917)
clustered SE	[2.655]	[2.321]	[2.470]	[2.519]
F, robust	24.5	24.3	21.3	22.6
F, clustered	10.6	13.0	12.6	13.1
Panel B: reduced form, dependent variable ΔY				
W	0.0354	0.0462***	0.0368***	0.0209*
robust SE	(0.0121)	(0.0086)	(0.0091)	(0.0087)
clustered SE	[0.0252]	[0.0123]	[0.0126]	[0.0116]
Panel C: linear IV, coefficient on ΔX ($\times 10^3$)				
ΔX	-4.099	-5.519**	-4.206**	-2.298
robust SE	(1.769)	(1.619)	(1.431)	(1.091)
clustered SE	[3.640]	[2.296]	[1.885]	[1.406]
Observations	983	983	983	983

Notes: Chay–Greenstone county panel, $T = 2$, first differenced; $n = 983$ counties in every column. Columns vary the control set, d_Z counting the differenced controls entered: none, five core hedonic demand shifters, those plus five demographic variables, and all twenty-two usable. Tables S36 and S38 of the supplement name the twenty-two and tabulate ten. W is the imputed 1975 nonattainment designation, here coarsened to binary; ΔX the 1970–1980 change in mean total suspended particulates, μg/m^3; ΔY the change in log median property value. Robust standard errors in parentheses, state-clustered in square brackets; stars use the clustered errors, which this section reads. Panel C instruments ΔY on ΔX with W, just identified. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: `scripts/application/app_tsp_01_diagnostics.R`; table `paper/tables/app_tsp_firststage.tex`.

A4(i) holds in an in-sample fit and binds in the fit the estimator uses. The multinomial propensity, projected onto the truncation set of Definition 2 at $\varepsilon_\varrho = 0.01$, has in-sample minimum $\min_l \hat{\varrho}_l = 0.0115$; over the 29,500 cross-fitted evaluations one estimation run makes, the projection binds on 0.997 per cent of them, at a pre-projection minimum of 4.3×10^{-5}, and at the binary coarsening on none (Table 6). Where it binds, A1 holds at $\hat{\varrho}$ and fails at the truncated propensity substituted for it, so those evaluations are outside the assumption;

Table 5: The localisation diagnostic, cell by cell

Cell	Trigger	n	$\|\hat{M}_l\|$	Null median	$\#\{\geq\}$	p	p_{cf}
1	bad-day only	114	6.75×10^{-3}	3.00×10^{-3}	78	0.039	0.042
2	annual-mean only	66	6.91×10^{-3}	2.30×10^{-3}	12	0.006	0.012
3	both	118	9.59×10^{-3}	3.02×10^{-3}	6	0.003	0.004
Sensitivity to the Bierens t-grid, null draws held fixed							
seed 20260805			$p = 0.0395,\ 0.0065,\ 0.0035$				
seed 1			$p = 0.0395,\ 0.0065,\ 0.0035$				
seed 20260901			$p = 0.0395,\ 0.0075,\ 0.0035$				
seed 4242			$p = 0.0395,\ 0.0075,\ 0.0035$				

Notes: Chay–Greenstone county panel, $n = 983$, at the headline control set $d_Z = 5$ and $\varepsilon_\varrho = 0.01$, the instrument at its four trigger cells. Rows are the three non-base cells; the base cell (685 counties, neither rule) carries no contrast. $\|\hat{M}_l\|$ is the $L^2(\Pi_u \otimes \Pi_t)$ norm of the sample analogue of (21) at cell l and the common fit $(\hat{\varphi}, \hat{\theta})$, which A1 sends to zero up to sampling error. *The reference distribution redraws W from the fitted multinomial* $\hat{\varrho}(\Delta Z)$, imposing A1 with the propensity held fixed; a permutation would destroy the conditioning too. "$\#\{\geq\}$" counts null draws at or above the observed norm out of $B = 2{,}000$, $p = (\#\{\geq\} + 1)/(B + 1)$, and p_{cf} uses the cross-fitted propensity on the same draws. No multiplicity adjustment is applied. The last block re-draws the t-grid at three further seeds, free under B1(iv). Source: `scripts/application/app_tsp_08_localise.R` and `app_tsp_common.R`; table `paper/tables/app_tsp_localise.tex`; run object `quality_reports/application/app_tsp_localise.rds`.

the theory prices that as a bias term rather than excluding it. The implied covariance clears (6) with room, at $\lambda_{\min}(\hat{\Omega}) = 0.0113$ against $\varepsilon_\varrho / \mathsf{L} = 0.0025$, so no cells need merging, which matters because merging strengthens A1b(iii) and gives back the restrictions the refinement was taken for.

The designation in hand is not the designation that was made. Between 1972 and 1978 nonattainment status was assigned at the level of the Air Quality Control Region, and the county-level variable in this file is an imputation from monitor readings. Cropper et al. (2023) quantify the discrepancy: the estimated reduction in total suspended particulates attenuates from 10.2 and 9.1 μg/m^3 across their two panels under the official designation to 6.0 and 7.7 under the county-level definition this section adopts. The imputed variable is admissible here under non-differential misclassification, so that the probability of a mislabelling does not depend on ΔU given ΔZ. Under it W remains valid and is merely less relevant; under the alternative the estimator is inconsistent rather than imprecise, so no sample size repairs it.

We assume the condition and do not test it.

6.2 The estimator on this panel

One property of the estimator changed between the diagnostics above and the estimates below, and it is a hypothesis the theory consumes rather than a tuning choice. The x-direction kernel is the supplement's boundary family at $p_K = 8$, which enforces every truncated moment (15) asks for, and with it the clause $\int_{\mathcal{X}} \hat{\nu}\, dx = 0$ that places the adjoint's range inside $\mathcal{V}$. The engine it replaces satisfied neither (Table S47 of the supplement).

Table 6 reports $\hat{\theta}$ at the headline control set, at $\mathsf{L} = 4$ and at the binary coarsening, on one cross-fitting stream and at a common truncation level and iteration cap, so the two columns vary the cardinality alone. The two agree on sign and on order of magnitude at every coordinate, and we put the agreement no more strongly than that: measured in units of the four-cell column's own clustered standard error, no coordinate moves by as much as 0.53 of a standard error and four of the six move by less than a quarter of one. That is the expected direction of no effect rather than a null result: under A1 the surplus contrasts are further restrictions on the same $\varphi_\dagger$, so they buy localisation and not a different answer, and a large shift would have been evidence against A1 or against the binary estimate. They do not buy precision either, the clustered standard error being larger in the four-cell column at per capita income.

Holding the cap fixed across the two columns is what makes them a one-factor contrast, and that is not a second-order detail: a third arm at the $\bar{m} = 15$ the binary cardinality would itself have set moves that column further than the cardinality moves it, by 0.1930 at its largest coordinate, and the two contrasts are opposite in sign at five of the six coordinates, so a design that varied both at once would net them against each other (Table S39 of the supplement).

The standard errors in both columns are conditional on a hypothesis the estimator does not

Table 6: The slope vector at four instrument cells and at the binary coarsening

	$\varepsilon_\varrho = 0.0100$, $\bar{m} = 60$	
	$\mathsf{L} = 4$	$\mathsf{L} = 2$
Constant	-0.0025	-0.0156
	(0.0560)	(0.0568)
Per capita income	0.7112	0.6057
	(0.2027)	(0.1862)
Population	0.3102	0.3433
	(0.0754)	(0.0838)
Unemployment rate	-0.9905	-1.1148
	(0.5922)	(0.6238)
Manufacturing employment	0.3918	0.4198
	(0.2398)	(0.2751)
High school	0.8743	0.9008
	(0.3188)	(0.3396)
Truncation binds (%)	0.997	0.000
Cross-fitted evaluations	29500	29500
$\min_l \hat{\varrho}_l$ pre-projection	4.3×10^{-5}	1.9×10^{-2}
Mean rel. balance	0.0012	0.0013

Notes: Chay–Greenstone county panel, $n = 983$, at the headline control set $d_Z = 5$; columns are the instrument's four trigger cells and their binary coarsening. Both are the cross-fitted one-step estimator of (47), with state-clustered standard errors in parentheses and no significance stars. *θ has no ΔTSP coordinate*: the regressor enters through $\varphi_\dagger$ alone, so the rows are the intercept and five differenced controls, per capita income, population, the unemployment rate, manufacturing employment and the share with a high-school education. *The two columns also share a cross-fitting stream*, so cardinality alone varies; the ladder of Table S38 of the supplement differs in all three and is comparable with neither column. The last four rows are what A4(i) costs, the projection being onto (11). Both run the boundary-adjusted x-kernel of the text; its ablation is `scripts/application/app_tsp_09_boundary.R` and `paper/tables/app_tsp_boundary.tex`. Source: `scripts/application/app_tsp_08_localise.R`; table `paper/tables/app_tsp_localise_theta.tex`.

impose. Corollary 21 makes N1 a condition on a $p \times (\mathsf{L}-1)$ matrix, hence $p(\mathsf{L}-1)$ scalar conditions rather than p, and the anchor this estimator removes is a constant per (j,l) rather than a function of ΔZ, so it imposes no version of N1 beyond the unconditional one. What the supplement measures is the smallest defect any constant anchor can reach, and at $\mathsf{L}=4$ it is not resolvable from zero, as it is not at either binary arm (Table S41 of the supplement). That is not evidence of absence but a statement about the power of the check at $n=983$, and no moment-level diagnostic could have substituted for it, since $E[\widetilde{W}_l \hat{L}_{jl} \mid \Delta Z]=0$ holds automatically under A1 whatever the anchor is. Inference here is conditional on N1, and by Proposition 20(iii) its failure would cost the propensity direction of the orthogonality, at the full rate r_n and undamped.

Figure 1 draws $\hat{\varphi}$ at the headline control set. The curve drawn is the four-cell fit, and the binary-coarsening curve does not sit on top of it: at the common cap the sup gap between the two is 0.0327, or 85 per cent of the four-cell curve's own range of 0.0383, against a root-mean-square gap of 0.0136, so the two separate at some points of the support and not on average. Letting the cap move with the cardinality instead would put that sup gap at 0.0081 and license the sentence that the two cardinalities agree. That number measures nothing: it nets two effects that nearly cancel, the cap alone moving the binary curve by 0.0380.

The curve is not monotone, and the sign the hedonic reading asks for is delivered on part of the support and not on the rest. It rises at 32 of its 59 grid steps and falls at 27, in four monotone stretches: it descends to -0.0174 at $48.7\,\mu\text{g/m}^3$, the 28th percentile; climbs to $+0.0150$ at 68.5, the 71st; falls to $+0.0001$ at 86.4, the 89th; and rises again over the last four steps. Its whole range is 0.0349, and its largest sustained movement, at $+1.64 \times 10^{-3}$ log points per $\mu\text{g/m}^3$ between the trough and the peak, runs the wrong way for the hedonic reading.

The linear slope of the preceding subsection delivers 0.380 log points over the $68.8\,\mu\text{g/m}^3$ separating the 5th and the 95th percentiles of X, and the nonparametric estimate's whole

range is under a tenth of that. It is nowhere as steep either: its steepest single descending step is -3.28×10^{-3}, or 59.5 per cent of the linear slope. *The two estimates therefore disagree about the level of the gradient and not only about its shape.* Every sentence of this reading is a statement about the point estimate: the constant slope is not a feature of $\hat{\varphi}$ anywhere on the support, and the sign it imposes fails on the two rising stretches, which carry 48.8 per cent of the pooled distribution of X between them. We do not call that a negative result, because the envelope of the next subsection is wide enough that nothing here rules a constant slope out.

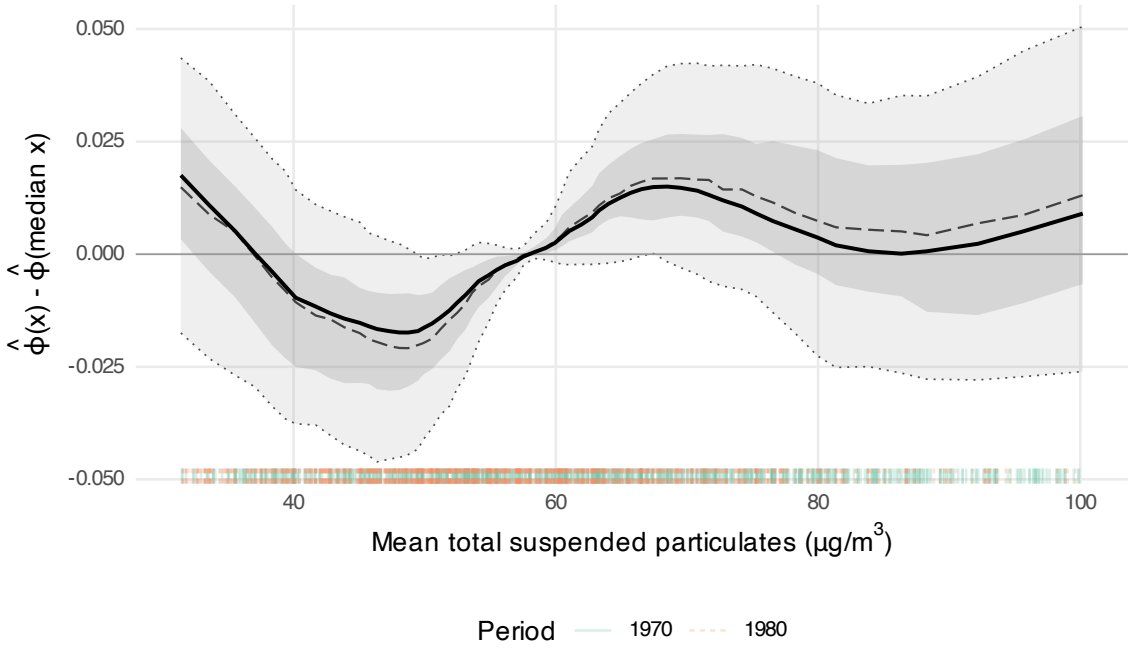


Figure 1: The estimated hedonic price function and its variability

Notes: $\hat{\varphi}(x) - \hat{\varphi}(x_{0.5})$ against mean total suspended particulates, at $d_Z = 5$, on the 983-county Chay–Greenstone panel, the instrument at its four trigger cells. Levels are log points of median property value, normalised at the pooled median of X, 58.0 μg/m^3: the figure carries shape, not level. The solid line is the point estimate and the dashed line the median of 99 bootstrap replications, which it tracks to within 0.0049 log points. The shaded regions are the twenty-fifth-to-seventy-fifth and fifth-to-ninety-fifth percentiles of those replications, the outer dotted. *They illustrate variability and are not confidence sets; no coverage is claimed for them and none is computed*, as in Centorrino, Fève, and Florens (2025). The rug marks the marginal distribution of X in each period, thin in 1980 above 80 μg/m^3 where few counties support the curve. Source: `scripts/application/app_tsp_02_ladder.R` and `app_tsp_05_phi_band.R`; figure `paper/figures/app_tsp_phi.pdf`.

6.3 The bootstrap

Valid uniform inference on $\hat{\varphi}$ does not exist in this model, and the obstruction is a property of the estimator rather than of this application. Uniform bands for ill-posed problems are built on a source condition written in the spectral calculus of T^*T and on completeness of the instrument, and neither is available here: the operator equation of Section 2 is nonlinear in φ, so there is no adjoint in which to write the condition, and a W with four values delivers completeness no more than a two-valued one does (see Severini and Tripathi, 2006; Babii and Florens, 2025). Pointwise sets do exist under nonlinear ill-posedness (see Chen and Pouzo, 2015), but they are indexed by a sieve dimension where the regularisation index of Definition 22 is a Landweber iteration count. We therefore follow the companion paper (Centorrino, Fève, and Florens, 2025), which estimates the same operator equation from a discrete instrument under the same regularisation: a point estimate, a bootstrap band drawn to show variability, and one sentence saying what the band is not. *The envelope in Figure 1 illustrates the variability of $\hat{\varphi}$ and is not a confidence set. No coverage is claimed for it and none is computed.*

The 99 replications resample counties with replacement and re-run the regularisation in each, holding $\hat{\theta}$ fixed, and the envelope they give is wide against the curve it surrounds: its median width over the sixty-point grid is 1.33 times everything the curve does. Three further schemes ask whether that width is an artefact of the resampling or a property of the estimator, since a resample of n counties leaves roughly 37 per cent of the draw as duplicates while halving the information: Exp(1) multiplier weights, which have the same first two moments and the same effective sample size but duplicate nothing, and subsamples of 621 and of 492 distinct counties. *That comparison is not evaluable, and we report it as such*: the three criteria that would have separated the two mechanisms are all statements about the realised stopping index, and that index equals the cap in every replication of every scheme, because the discrepancy rule of Section 3 is run at $\tau_{\mathrm{DP}} = 0$ here and the patience is set equal to the cap. What the schemes

do show is agreement on the size of the band, the multiplier arm being 8.1 per cent narrower over the grid as a whole and wider at the ends of the support. The five schemes are set out in Table S44 of the supplement, the multiplier envelope in Table S45 of the supplement, and the eleven levels this figure draws, together with the bootstrap's convergence record, in Table S43 of the supplement.

6.4 What the illustration shows

The estimator runs on this panel and returns the two objects it is built to return: a slope vector with a clustered standard error, at either cardinality of the instrument, and a structural curve whose shape contradicts the constant slope the linear specification imposes. That is the whole of what the section claims.

At $n = 983$ it claims nothing about the asymptotics, and there are two reasons for that rather than one. Evaluated at $(\mu, r) = (2, \frac{8}{3})$, the frontier-clearing configuration the Monte Carlo names, the frontier (7) is met at no control set that also satisfies A4(ii): it fails once any control is entered, and the one rung at which it holds is the intercept-only rung, which A4(ii) excludes because at $\alpha = \frac{1}{2}$ the sieve-entropy condition admits no integer sieve dimension (Table S49 of the supplement). The second is the cardinality arm of the Monte Carlo, which finds the limit law's own remainder failing to vanish at $\mathsf{L} = 4$ on the paper's named design (Table S22 of the supplement), in a propensity direction whose price this panel cannot bound because the N1 defect above is not resolvable from zero. The object that describes this estimator at this sample size is therefore Corollary 30 and not Theorem 29. *The application illustrates the procedure on a design the method was built for; it does not demonstrate that the asymptotics bind at this sample size*, and we would rather say so than let the section be read as though they did.

7 Conclusions

In this paper, we estimate the slope vector of a first-differenced partially linear panel model whose nonparametric argument is a continuous endogenous regressor and whose only instrument takes finitely many values. Conditional independence between the differenced error and the instrument replaces the conditional mean, a Bierens family converts that restriction into unconditional moments, sieve Landweber–Fridman regularisation solves the nonlinear equation that results, and a cross-fitted one-step on a Neyman-orthogonal moment returns a slope vector and a standard error.

Five results carry the paper.

Identification of $\theta_\dagger$ runs through the mean restriction alone and needs no source condition and no rank condition beyond nonsingularity of $E[\Delta Z \Delta Z']$. A moment free of the structural function exists exactly when the two periods' tilted marginals of the endogenous regressor coincide, so the partialling-out route is closed here as a property of the differenced design and not of the estimator. Orthogonality in the structural direction holds if and only if the representer solves the adjoint equation, which is the range condition of the linear theory read at a Fréchet derivative. Under A1 every set of instrument cells admits a common structural function, so a conflict between two cells refutes A1 and attributes the refutation to the cells that carry it, and that statement carries no identification hypothesis at any subset of cells. And the cross-fitted one-step is $\sqrt{n}$-consistent and asymptotically normal with a consistent sample sandwich, whenever the first-stage rate is faster than $n^{-1/4}$.

What the extra cells buy is that attribution and nothing else. They do not buy identification, a finer instrument making the identification hypothesis weaker rather than stronger; they do not buy testability, which is present already at $\mathsf{L} = 2$; and they do not buy instrument strength, since splitting an instrument creates no variation.

Efficiency is not among them.

The sandwich is the variance the infeasible plug-in would carry at a known $\varphi_\dagger$, plus a positive semidefinite term that does not vanish under overlap, and no semiparametric efficiency bound is computed for this model. What the debiasing weight buys is the removal of a first-order bias and not a smaller variance.

We assume two ingredients of the estimator to exist and construct neither. The representer estimator $\hat{\ell}$ is priced by A5(iv) at a rate ϵ_n with $\delta_n \epsilon_n = o_P(n^{-1/2})$, and no construction attaining it is supplied. The preliminary $\hat{\varphi}^{(0)}$ is needed at a declared deterministic envelope $\hat{\varepsilon}_0$ of polynomial order, for which consistency will not do, and it has no supplier either. Three further conditions are primitive rather than derived: μ and R_*, global identification of $\varphi_\dagger$ at A1b(iii), and the sieve display of Lemma L-7 on the sieve branch of the first-order bias.

The frontier is qualitative, and the Monte Carlo prices what that costs. Crossing it makes the second-order remainder vanish eventually and says nothing about where. At the configuration the theory names the turn falls at $n^* \approx 10^{58}$, the simulated standard deviation clears a two-standard-error bar only at $n = 181{,}019$ in DGP A and only at the smaller of two equally admissible constants, and nominal 90% intervals under-cover at 29 of the 32 cells; that experiment is in the supplement, and Table S1 of the supplement carries the standard-error units the sentence reads. What describes a panel anyone will run is the rate floor of Section 4 and not the $\sqrt{n}$ statement. That reading is about the configuration and not about Theorem 29: at a design whose frontier margin is 0.1318 in place of 0.0060, and whose propensity score is no smoother, the same estimator meets the pre-committed coverage criterion from $n = 354$ upward (Table S13 of the supplement) and clears the standard-deviation ratio's two-standard-error bar from $n = 8{,}000$ at the same constant (Table S19 of the supplement).

The stopping constant is the second open item. It is computable and not identified, and at this configuration the criterion it thresholds falls by 11 to 43 per cent over the whole iteration budget, so the stop is settled almost entirely by a number the theory does not pin. Whether

a Lepskii-balanced rule should replace it turns on the preliminary: given the discrepancy rule its own oracle constant, Lepskii is ahead in all three cells started from zero and behind in all three started from a rate-accurate one.

The third is severe ill-posedness, which the theorem does not reach. Under exponentially decaying singular values the rate attainable from any source condition is logarithmic, so the frontier fails at every smoothness of the propensity score.

We run the estimator in Section 6 on the county panel of Chay and Greenstone (2005), where the object is the hedonic price of air quality and the instrument is a nonattainment designation read as the pair of triggers that produces it, at $\mathsf{L} = 4$. At $n = 983$ the frontier fails at every control set an applied literature would accept (Table S49 of the supplement), so the object that describes the estimator there is the rate floor and not Theorem 29. That section illustrates the procedure on a design the method was built for, and shows neither that the asymptotics bind at this sample size nor anything about the Clean Air Act. We take no position on A1 from it.

Future work should focus on constructing the two missing ingredients, since a representer estimator with a rate and a preliminary with a declared envelope would discharge the only maintained hypotheses that name an object without building it. Recovering marginal effects from $\hat{\varphi}$ is a second step, and the derivative parameterisation is not the route, since it lowers the source index below what the frontier requires. A semiparametric efficiency bound for a model whose nuisance solves an ill-posed nonlinear operator equation is a third, and a statement uniform in P is a fourth. The latter would need a class on which the P-specific constants are bounded, a uniform first-stage rate at a random compressed operator, and a Lindeberg condition uniform in P.

Valid inference on $\hat{\varphi}$ is a fifth, and Section 6 locates it. Uniform bands in ill-posed models are built on a source condition written at an adjoint and on completeness of the instrument, and a nonlinear equation with a discrete instrument supplies neither (see Babii, 2016; Severini

and Tripathi, 2006; Babii and Florens, 2025). Chen and Pouzo (2015) do reach nonlinearly ill-posed conditional moment models, but index their sets by a deterministic sieve dimension, while the regularisation index of Definition 22 is the random stopping time of a discrepancy rule. The mathematics stops in the same place, Monard, Nickl, and Paternain (2021) delivering Bernstein–von Mises for nonlinear forward maps at smooth linear functionals only. Both literatures give valid inference for the regular $\sqrt{n}$ part and neither goes past it, so Theorem 29 sits on that boundary rather than short of it.

8 Supplementary material

Proofs of all results are in the online supplement. The supplement also carries the statements this article refers to by name and does not print, the interpretation of each numbered condition, and the smoothness budget in one table. It repeats the statements printed here, in the same order and under the same numbers, so that a proof and the result it proves can be read together.

Two pointers appear in the statements above and both name a target in the supplement rather than numbering one. `Lemma S:sandwich` is the environment whose label is `lem:sandwich`, and `(S:smdiff)` is the display whose label is `eq:smdiff`. Each is the name the supplement gives the object, so it is findable there by the heading it carries.

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